Label granularity in supervised pretraining of jet foundation models

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Abstract

Jet foundation models are pretrained to separate many fine classes of simulated jets, and analyses build discriminants by summing the scores of those classes. Whether the label granularity, the fineness of the pretraining classes, changes what the network learns has not been tested with the training jets held fixed. We pretrain the Sophon transformer several times on the same jets, in the same order and from the same initial weights, and change only the label set. The label sets form a hierarchy from 188 to 17 classes in which each merge removes a named set of physics distinctions. Random partitions of the classes, some of which differ only in a boundary between quark flavors, compare label sets at a fixed number of classes. Linear classifiers on the fixed pretrained features recover what the labels separate: merging decays that differ only in lepton or quark flavor costs the most, and at a fixed number of classes the cost depends on which classes are merged. After fine-tuning, the 17-class labels still lower the background rejection for several signal classes of JetClass, a separate simulation with every input present. On the public top-tagging and quark/gluon benchmarks, which lack some inputs, the label sets differ far less. An auxiliary jet-mass regression makes the pretrained features carry the mass and reduces the sculpting of the background mass distribution, at a small cost to the classification. A pretraining label set should keep every distinction that downstream discriminants sum over, and the boundaries between flavors.

Keywords: **[keywords: the PI chooses]**

1 Introduction

A top quark, a W, Z or Higgs boson, or a new resonance produced with large momentum at the Large Hadron Collider [1] decays into quarks or leptons whose showers merge into a single jet. Analyses identify the particle from the structure of that jet [2, 3, 4, 5]. Three properties of the jet carry the identification. The first is the number of prongs: two-prong jets, for example, contain two separate energy clusters from the two quarks of a heavy-particle decay. The second is the quark flavor, which a b or c hadron reveals through a displaced vertex, where tracks meet away from the collision point after the hadron travels a few millimeters [6, Sec. 4.1]. The third is the jet mass. The taggers that do this are deep networks that read every measured particle of the jet, trained on simulated jets to separate the

signal decays from jets from light quarks and gluons, the quantum chromodynamics (QCD) background [7, 8, 9, 10]. Increasingly, a single network is pretrained to separate many fine classes of simulated jets and then serves many analyses, each of which sums the fine classes into the signal and background it needs [11, 12]. The Sophon model is pretrained on the 188 classes of the JetClass-II dataset [11, 13], and the CMS tagger GloParT grew from 37 classes in its first version to 374 in its third [12]. How fine to make the label set, the classes the network is trained to separate, is a choice every such model makes.

At the output of the network, finer labels cost nothing in principle. For an ideal classifier, the score of a class split into exclusive subclasses equals the sum of the subclass scores, and adding classes leaves the ratios between the scores of the original classes unchanged [11, Sec. III, Properties 1 and 2]. A model trained on fine classes therefore serves every coarser definition of the signal. Both properties concern the output of an ideal classifier. They say nothing about what the backbone, the network below the classification head, learns to carry, and the backbone is the part that is reused when the model is adapted to a new task. Whether the label set shapes the backbone is untested, because the published comparisons changed the training jets together with the labels. Sophon's 42-class model identifies $X \to b\bar{b}$ decays less well than the 188-class model, but it was trained only on two-prong decays and QCD jets, with fewer jets per epoch [11, Sec. III, App. B.2]; its classes are also not the 43-class level studied here. In the thesis that introduced GloParT, adding two-prong and top-quark classes to a ParticleNet-based classifier of $H \to WW^*$ decays raised the area under the receiver operating characteristic curve (AUC) against QCD jets from 0.9555 to 0.9593. The author reads this as a minor change, and the added classes came with their own training jets [14, Sec. 11.4.1].[1]

Image classification has studied the same choice, and its answers depend on how the classes are grouped and on how the features are used. Pretraining on coarse groups of the ImageNet classes costs little in detection after fine-tuning [17]. On a taxonomy of species, finer labels lower the error of a fine-tuned network on the coarse classes up to about 10^3 classes. Class identities assigned at random to individual images do worse than the coarse labels at every granularity [18]. The way the pretrained network is used matters as well. A linear probe, a logistic regression trained on frozen features (the backbone's output with its weights held fixed), measures what a linear classifier can read from them. A larger label set of web hashtags gives a linear probe a clear lead, yet under fine-tuning, which continues training all pretrained weights on the downstream task with a new output layer, the lead nearly disappears [19]. Jets have not been studied this way, although simulation makes them the cleaner case: every jet carries its exact decay, so the same jets can be labeled at every level of a hierarchy.

We pretrain Sophon's network on the same JetClass-II jets, in the same order and from the same initial weights, and change only the label set and with it the width of the output layer (figure 1). The label sets are four levels of a class hierarchy, each pretrained three times. Each level merges classes of the level above, so every coarse class is a union of finer ones, and each merge removes a named set of distinctions.

[1] Studies that vary the composition of the pretraining data change the labels together with the jets: Uslu et al. [15] change which JetClass-II classes enter pretraining at a fixed total number of jets, and SuperJEPA [16] pretrains on a subset of the JetClass classes and on all of them with different numbers of jets.

The 162-class level merges the QCD classes, which Sophon divides by their numbers of b, c and s quarks. The 43-class level also merges decays with different numbers of b or c quarks, gluons with light quarks, and electrons with muons. The 17-class level merges b, c and light quarks as well and keeps only the number and type of the prongs. Two controls fix the number of classes at 17. Random partitions are label sets with the same number of classes, formed by assigning the resonant classes to groups at random and chosen to cut across the 17-class groups; they test whether the cost of a coarse label set depends on which classes are merged. Random partitions with controlled flavor boundaries differ from one another only in their flavor boundary, a class boundary between decays that differ only in quark flavor, and test whether one such boundary changes what the frozen features carry. We evaluate every model with linear probes and with MLP probes, small multilayer perceptrons (MLP) trained on the same frozen features, first on the JetClass-II classes themselves. The downstream tasks, on which we also fine-tune, are the JetClass dataset [8], top tagging [20] and quark/gluon discrimination in Pythia and Herwig samples [21, 22, 23]; Sophon's $X \to b\bar{b}$ and $X \to bs$ benchmark completes them [11].

Frozen features keep what the labels ask the network to separate (figure 2). Merging the QCD classes leaves what a linear probe reads from the resonant classes unchanged and costs only on the QCD classes themselves. The merge that removes quark flavor is the costliest step of the hierarchy. Class by class, the largest costs fall on decays that differ only in lepton flavor (an electron or a muon, directly or through a τ lepton). Summed over classes, the largest cost falls on decays that differ only in quark flavor (figure 3). At the same number of classes, the random partitions lose less than the 17-class level, so the cost is set by which classes are merged and not by how many remain. Any flavor boundary keeps flavor readable, and a boundary placed between the two decays of a pair separates that pair better still (figure 4).

After fine-tuning, the 17-class labels still lower the background rejection for several signal classes of JetClass, where every input is present (table 2). The public top-tagging and quark/gluon benchmarks lack some of the inputs the network reads, which we set to zero, and on them the label sets differ far less after fine-tuning (tables 3 and 4). Linear probes do poorly on these benchmarks. Setting the inputs they lack to zero on JetClass-II removes more of the pretrained features' accuracy than it removes from a randomly initialized network. We compare three substitutions for the missing inputs to find where the loss arises (figure 5). Finally, the 162- and 17-class models are also pretrained with an auxiliary regression of the jet mass, trained alongside the classification. The regression makes the frozen features carry the jet mass and reduces sculpting, the distortion of the background mass distribution by a selection on the classifier score, at a small cost to the classification (figures 7 and 8, table 5).

Section 2 places the study among work on pretraining objectives, label granularity, auxiliary targets and transfer. Section 3 describes the data, the label sets and the evaluation, section 4 gives the results task by task, and section 5 discusses them and states the limitations.

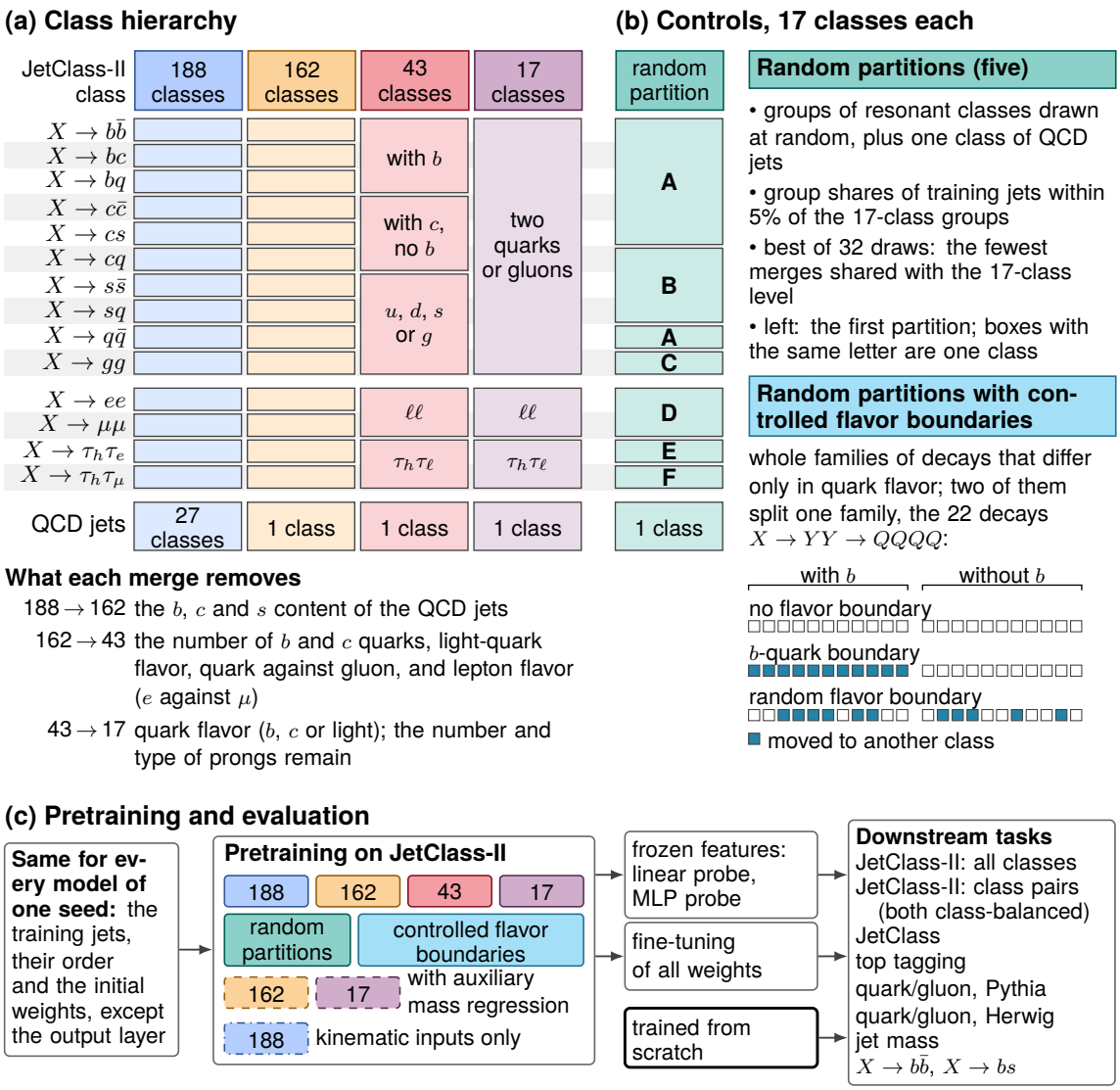


Figure 1: Design of the study. (a) The class hierarchy on the decays of a heavy resonance X and on the jets from quantum chromodynamics (QCD); q is a u or d quark, ℓ an electron or a muon, and τ_h, τ_e, τ_μ a τ lepton decaying to hadrons, an electron or a muon. Each box is one class of the label set above its column, except the QCD box of the 188-class column (Sophon's label set [11]), which stands for 27 classes. Each level merges classes of the level to its left; the lines below name what each merge removes. (b) The controls: the first random partition, where boxes with the same letter are one class, and the random partitions with controlled flavor boundaries on the decays $X \to YY \to QQQQ$ (one square per decay; filled squares are moved to a second class). (c) Models of one seed share the training jets, their order and the initial weights, except the output layer, the mass regression target, and the first layer and initial weights of the model with kinematic inputs only (the particle four-momenta). The hierarchy levels have three seeds each, the models with the auxiliary mass regression two, the random partitions with controlled flavor boundaries two and each random partition one. Frozen features, the output of the backbone (the network below the output layer) with its weights held fixed, feed linear probes (logistic regressions) and multilayer perceptron (MLP) probes; fine-tuning trains all weights. Colors identify the configurations in every figure; dashed and dash-dotted borders mark the two variants.

2 Related work

Pretraining objectives for jets. Supervised classification of simulated jets is the reference objective for jet foundation models [8, 11], and the alternatives are measured against it. The self-supervised families include masked particle modeling [24, 25], generative pretraining [26], contrastive learning on resimulated jets [27] and joint-embedding predictive architectures [28, 29]. Other models combine classification with a second objective: OmniLearn and OmniLearned with generation [30, 31], SuperJEPA with joint-embedding prediction [16]. One study compares the objectives on the same pretraining jets, all from JetClass. It finds classification best or tied when the full top-tagging training set is used for fine-tuning, with masked-particle pretraining ahead only for its smallest model fine-tuned on 10^4 jets, and concludes that the best objective is not a property of the model alone [32, Sec. IV.A]. A comparison on JetClass-II evaluates the objectives by fine-tuning alone, without linear probes [33]. Each of these studies keeps the classes of its supervised objective fixed, or changes them together with the jets. The label set of the objective that does best when labeled jets are plentiful has therefore not been varied on its own.

Label granularity in vision. The nearest precedent for our random partitions regroups whole CIFAR-10 classes at random into two coarse classes. Training on the fine labels beats training on the coarse ones for every grouping, by up to 2.78 percentage points of accuracy on the coarse task [34, Table VI], and in the same study the gain from fine labels on CIFAR-100 becomes negligible once the coarse task has many classes [34, Table VIII]. Both experiments train and test on one task, so neither measures transfer. Whether fine labels help also depends on how the classes are grouped. For cats and dogs against cars and trucks, training on the four fine classes never helps, while for a grouping that mixes animals and vehicles it consistently does [35, Fig. 3]. Class identities assigned at random to individual images, rather than to whole classes, do worse than every level of a species taxonomy [18]. With a linear probe on frozen features, pretraining on a larger set of web hashtags leads by about 7 percentage points of accuracy on the ImageNet task with the most classes, and under full fine-tuning the lead nearly disappears [19].[2] With the labels fixed, losses that separate the training classes more give frozen features from which a linear probe transfers worse, while after fine-tuning the losses are indistinguishable [36]. Our random partitions differ from these controls in three ways. They keep the number of classes of a physical level, 17 groups; they match each group's share of the training jets to that level; and they are pretraining label sets evaluated on fixed downstream tasks, not targets in their own right.

Auxiliary physics targets. Production taggers add auxiliary targets to the classification, with effects that point both ways. The ATLAS flavor tagger GN2 has two auxiliary objectives, which predict the origin of each track and which tracks share a vertex. Removing both lowers its c- and light-jet rejection by up to 30%, and keeping either one recovers most of the loss [9, Discussion]. GloParT adds two log-cosh regressions of the jet mass to its classification loss, with a weight tuned to the value our mass models also use. Its author argues that because the terms do not interact,

[2] After full fine-tuning, the larger label set leads on that task by a margin comparable to the difference the same study finds between two pretrainings on different random samples of images. The smaller label set, whose hashtags match the ImageNet classes, leads on the standard ImageNet task, and it was also pretrained on fewer images [19, Fig. 1, Sec. 3.1].

the minimum of the combined loss is optimal for both tasks [14, Sec. 11.2.1]; no measurement accompanies the argument. A CMS tagger of boosted Higgs bosons decaying to pairs of W bosons regresses the mass in GeV, so its weight is on a different scale from ours [10]. ATLAS measured a cost instead: a network for boosted Higgs bosons trained for classification and mass regression together was about 10% less efficient at classification than a dedicated classifier, and ATLAS regresses the mass in a separate network [37, Sec. 6.5]. These studies measure the tagger's outputs. None measures what an auxiliary target does to the frozen features while the labels stay fixed, which our 162- and 17-class models, pretrained with and without the mass regression, do.

Transfer from frozen features. Sophon used its label set downstream in two ways [11, Sec. III]. Its output scores form an $X \to b\bar{b}$ discriminant directly. A small network trained on its frozen features, with its first layer taken from the classification head, identifies $X \to bs$ decays, which no Sophon class contains. At low background efficiency this network does better than the Particle Transformer and ParticleNet trained on the task [11, Fig. 2, App. B.5]. Vigl et al. compared frozen and fine-tuned use of a Particle Transformer backbone pretrained on $X \to b\bar{b}$ jets in a search for $G \to HH \to b\bar{b}b\bar{b}$, and fine-tuning was better for every architecture they tried [38]. Neither study varied the label set with the training jets fixed, and Sophon reports no variation between pretraining runs. We repeat Sophon's benchmark at every level of the hierarchy.

3 Methods

3.1 Data, model and pretraining

We pretrain on JetClass-II [11, 13], the dataset of the published Sophon model, with its selection and its reweighting, so that every label set learns from the jet distribution Sophon learned from. JetClass-II holds about 139 million simulated jets, clustered with the anti-k_T algorithm [39] with $R = 0.8$. They are labeled with the 161 resonant and 27 QCD classes of JetClass-II. In a resonant jet a spin-0 resonance X decays either directly into a pair of quarks, gluons or leptons, or into a pair of intermediate resonances Y that decay in turn. The masses of X, up to 500 GeV, and of Y are sampled for each event, so a class is defined by its decay products and not by a mass. Resonances are generated with MadGraph5_aMC@NLO [40] and QCD multijet events with Pythia 8.3 [41], which also simulates the decays, the parton shower and hadronization. Delphes [42] simulates a CMS-like detector with an average of 50 additional proton–proton interactions per event, whose particles PUPPI [43] suppresses.

We keep jets with $200 < p_\mathrm{T} < 2{,}500$ GeV and $20 < m_\mathrm{SD} < 500$ GeV. Here m_SD is the soft-drop mass, the jet mass after soft, wide-angle radiation is removed, which peaks near the mass of the decaying particle. It is computed with the soft-drop procedure [44] at $z_\mathrm{cut} = 0.1$ and $\beta = 0$, the values CMS uses [45]. Every JetClass-II jet also satisfies $|\eta| < 2.5$ [11, App. A]. These requirements leave **[a committed count of the selected pretraining jets]** training jets. Training jets are drawn with Sophon's reweighting, which gives each of Sophon's reweighting categories the same distribution in $(p_\mathrm{T}, m_\mathrm{SD})$ in order to avoid sculpting the background mass distribution [11, Sec. II]. The categories are fixed by the JetClass-II classes and not

by the label set, so which jets are drawn, and how often, is the same for every label set. Every linear probe and every fine-tuning on JetClass-II uses jets from files that no pretraining run read.

The network is Sophon's [11], built on the Particle Transformer (ParT) architecture [8]. It has 8 particle-attention blocks, whose attention carries a bias computed from pairwise features of the particles, 2 class-attention blocks, an embedding dimension of 128, and a classification head with one hidden layer of 512 ReLU units. The number of particle-attention blocks follows the released code [46]: the Sophon paper states 6 blocks, but its 2.3 million parameters correspond to 8. As in Sophon, the four-momenta of the particles are rescaled so that every jet has the same transverse momentum. The frozen features, the output of the backbone, are the 128-dimensional class token after the class-attention blocks and the final layer normalization, which is the input of the head. The initial weights follow PyTorch's default initialization rather than the one in Sophon's code. Absolute performance therefore need not match Sophon's model, while every comparison in this paper is between models initialized the same way.

Every configuration is pretrained with the same settings, so that within a seed the label set is the only difference between two models (figure 1(c)). Models with the same seed share their initial weights and see the same training examples in the same order. The order of the training jets depends only on the seed and the epoch, and every layer starts from the same weights except the output layer, whose width is the number of classes. Each model sees 8.192×10^8 training examples, 80 epochs of 10,240,000 jets as in Sophon's released configuration [46]. It is trained with the Ranger optimizer [47, 48] and a learning rate that is constant at first and then decays exponentially; appendix B.1 gives the values. The checkpoint rule is shared too: each model is evaluated at the first maximum of its validation accuracy within epochs 70–79, where the learning rate is lowest. For every run, the code release records a hash of the jet identifiers read in each epoch, against which the training order of any two models can be checked.

3.2 Label sets

The label set is the one variable of the study. The four label sets we pretrain form the class hierarchy of figure 1(a), and each merge removes a named set of physics distinctions. The 188-class label set is Sophon's. It has one class for each resonant decay and divides the QCD jets into 27 classes by their numbers of b, c and s quarks. The 162-class level merges these QCD classes into one and keeps every resonant class, so it removes only the quark content of QCD jets. The 43-class level labels each decay by its topology, the pattern of intermediate resonances and prongs, and by whether it contains a b quark, else a c quark, else only light quarks or gluons. It removes three kinds of distinction at once: the number of b or c quarks, gluon against light quark, and electron against muon, also for τ leptons that decay to either. Grouping gluons with light quarks is our choice; Sophon's classes keep them apart. The 17-class level removes quark flavor. It keeps, for each decay, its number of prongs and the type of each: quark or gluon, electron or muon, τ lepton decaying to hadrons or to a lepton, or neutrino.

The levels remove flavor before structure, so the coarsest still describes the decay topology an analysis selects on. At every level a class holds only QCD jets, only

Table 1: Discriminants analyses build by summing the output scores of a pretrained tagger, against the number of classes in the label set (columns, finest first): Sophon's and the boosted $|V_{cb}|$ measurement's, and a discriminant that weights each QCD subclass separately, as GN3X evaluates its rejection per QCD subclass. •: the discriminant can be built exactly from that label set's outputs, because a merged class's score is the sum of the scores of the classes it merges; —: it cannot. This is a property of the label set, not of a trained model.

discriminant	source	188	162	43	17
$X \to b\bar{b}$ vs QCD	[11], Eq. (4)	•	•	—	—
$X \to c\bar{s}$ vs QCD (Sophon's A_1)	[11], Eqs. (5)–(6)	•	•	—	—
three quarks, exactly one b, vs QCD (Sophon's A_3)	[11], Eqs. (5)–(6)	•	•	—	—
$W' \to W\phi \to WWW$ event discriminant	[11], Eqs. (7)–(9)	•	•	—	—
D_{bc}, boosted $\|V_{cb}\|$ measurement	[49], Eq. (1)	•	•	—	—
QCD subclasses weighted separately (GN3X-style)	[37], Sec. 6	•	—	—	—

decays into quarks and gluons, or only decays with leptons. The step from 162 to 43 classes is the only one that removes more than one kind of distinction, so no result in this paper attributes an effect of that step to one of its three.

Merging classes removes analysis discriminants outright, not only information. Analyses build discriminants by summing the output scores of classes, and for an ideal classifier the score of a merged class is the sum of the scores of the classes it merges [11, Eq. (2)]. A discriminant can therefore be formed exactly from a label set only when every set of classes it sums is a union of that label set's classes, and table 1 lists the outcome for published discriminants. Sophon's $X \to b\bar{b}$ discriminant [11, Eq. (4)] can be formed at 188 and 162 classes but not at 43, where $X \to b\bar{b}$ shares a class with $X \to bc$ and $X \to bq$. Sophon's other discriminants [11, Eqs. (5)–(9)] exist only at 188 and 162 classes. So does the discriminant D_{bc} of the boosted $|V_{cb}|$ measurement, whose denominator contains $X \to bq$ and $X \to cs$ [49, Eq. (1)]. A discriminant that weights each QCD subclass separately, as GN3X evaluates its rejection for each QCD subclass [37], exists only at 188 classes.

Random partitions test whether, at a fixed number of classes and fixed loss weights, the cost of a coarse label set depends on which classes are merged. A random partition is a label set with as many classes as the 17-class level, formed by assigning the resonant classes to groups at random; with one class for all QCD jets the groups make 17 classes. Each group holds the share of the reweighted training jets of one 17-class group to within 5%, so it carries the same weight in the loss. A random partition thus has the class count and the loss weights of the 17-class level but not its physics (figure 1(b)). The partitions are selected rather than typical draws. Each of 32 draws per partition is refined by a search that merges as few of the class pairs merged by the 17-class level as the share rule allows, and the draw with the fewest such pairs is kept. The partitions are therefore built to cut across the 17-class groups. They mix decays with different numbers of prongs and keep many flavor boundaries. Every conclusion we draw from them is restricted to partitions built this way. We pretrain five partitions, one run each; appendix A gives their construction and their groups.

Random partitions with controlled flavor boundaries test whether a single flavor boundary, at a fixed number of classes and fixed group shares, accounts for the

cost of the flavor merge. The three partitions differ only in their flavor boundaries. In the random partition with no flavor boundary, every group is a union of whole families of decays that differ only in quark flavor, with the group shares of the 17-class groups. The random partition with a b-quark boundary starts from it and moves into another group the 11 of the 22 four-quark decays $X \to YY \to QQQQ$ that contain a b quark. To keep the shares exact, 9 decays of other forms move the other way in whole families. Its one flavor boundary thus separates the four-quark decays with a b quark from those without. The random partition with a random flavor boundary makes the same compensating move but moves 11 four-quark decays chosen at random, with group shares within 1% of those of the b-quark partition. Each is pretrained with two seeds. Together with the random partitions they are the controls; every control has 17 classes and is read against the 17-class level.

An auxiliary mass regression tests whether a physics target beyond the labels changes what the frozen features carry, and what it costs the classification. The 162- and 17-class models are also pretrained, with two seeds each, with the auxiliary mass regression. An extra output regresses

$$y = \ln\bigl(m_{\text{gen}}/m_{\text{SD}}\bigr), \tag{1}$$

where m_{gen} is the soft-drop mass of the matched generator-level jet and m_{SD} that of the reconstructed jet; only jets with a generator-level match enter the regression term. Its log-cosh loss is added to the classification loss with the weight $\lambda = 5$ at both label sets. This is the value GloParT tuned for its two regression terms on generator-level masses [14, Eq. (11.7)]; the target is ours, and λ is not tuned on validation. The published CMS tagger also uses a log-cosh loss but regresses the mass in GeV, with weight 0.05 [10], a value not comparable with ours. At a fixed weight, the share of the training loss taken by the regression term differs between the label sets because the classification loss is smaller with fewer classes. Averaged over training, it is 13.78 and 13.87% in the 162-class runs and 28.65 and 28.67% in the 17-class runs. The two label sets are therefore compared at a fixed weight, not at a matched share.

The 188-class model with kinematic inputs only is a diagnostic of the input mismatch on top tagging, not a method we propose. It is the 188-class model trained on the particle four-momenta only, that is on the 7 of the 17 input features that the top-tagging data provide, as ParT pretrains a model on kinematic features for top tagging [8]. Its pairwise features, its training jets and their order, and its training settings are those of the other 188-class models. Its initial weights are its own, because its first layer is narrower. The kinematic-only model, pretrained once, answers two questions. After fine-tuning, it measures how much top-tagging rejection the missing inputs cost, against the 188-class model fine-tuned with those inputs set to zero, which shares its rescaling, data and settings. With frozen features and nothing missing, it shows whether a linear probe on top tagging is limited by the frozen features or by the missing inputs.

3.3 Evaluation

Four classification tasks each ask one question. JetClass-II at its 188 classes asks which distinctions the frozen features keep. Its classes are the 188-class model's own labels, so this task favors finer label sets by construction; we also report it

over the resonant and the QCD classes separately, since only the 188-class level divides the QCD jets. JetClass [8] has every input present but other classes and a different simulation without pileup. It is the in-distribution downstream test and asks whether a label-set effect survives on new classes and a new simulation when no input is missing. Top tagging [20] and quark/gluon discrimination [21, 22] are the public benchmarks, with the companion Herwig sample [50, 23] as a change of generator. They lack some inputs, and they ask what the label set changes where analyses compare taggers. The jet mass and the tagging of $X \to b\bar{b}$ and $X \to bs$ complete the evaluation. Dataset sizes, selections and splits are given with each result.

The network reads 17 features per particle, and the public benchmarks lack some of them: the top-tagging data provide only the four-momenta, from which 7 of the features follow, and the quark/gluon data lack the track-displacement variables, ParT's "exp" scenario [8, Table 2] (sections 4.3 and 4.4 list them). Features a benchmark does not provide are set to zero for every label set, so top tagging and quark/gluon discrimination compare label sets on identical inputs, and fine-tuning with these zeros is the benchmark comparison. Zero follows JetParticle-JEPA, which sets unused inputs to zero without retraining [29]; that model saw zeros in pretraining through input dropout, and ours did not. OmniLearn argues against pretraining one model for each dataset [30, Sec. II.A]. ParT instead pretrains for the inputs of each benchmark [8], which we follow only through the kinematic-only model.

A linear probe cannot adapt to an input that no pretraining jet contains, as a zeroed particle is (section 4.3), so for linear probes on frozen features we compare three substitutions for the missing input features: zero, the training mean and random draws. The training mean is the running mean of the network's input normalization, the centroid of the training particles. The random draws are four draws of a real JetClass-II particle of a randomly drawn species, with the features averaged over the draws. Zero is the controlled comparison. The training mean asks whether a value at the center of the training distribution recovers what zero loses, and the random draws ask whether a value that a real particle can take recovers it. One substitution per task is applied to every model, the randomly initialized network included: the one whose linear probes have the lowest validation cross-entropy summed over all models. It is **[fill fits]** for top tagging and **[fill fits]** for quark/gluon discrimination, whose choice the Herwig jets share, and figure 5 shows the three substitutions side by side.

Linear probes are our main evaluation of what a label set puts into the frozen features, as linear evaluation is for transferable features in vision [36] and a frozen backbone with a linear head is for masked particle modeling [24]. On JetClass-II the linear probes are trained and tested on class-balanced samples from held-out files, with equal numbers of jets of each class where the data allow, so that rare classes weigh as much as common ones. The training sets are nested; their sizes and the fit settings are in appendix B.3. A linear probe reads only what a linear function of the features can. An MLP probe, a small multilayer perceptron with one hidden layer of 256 units trained on the same frozen features, therefore stands beside every linear probe that finds no difference, and tells whether the information is absent or only not linearly readable. The randomly initialized network is the saved initial weights of a 188-class model before any training, one for each seed. Evaluated in the same way, it gives the floor: what a linear probe on frozen features gets

from the architecture alone. Its BatchNorm statistics are recomputed on 200,000 training jets, so that it is compared with the pretrained networks under the same normalization of the inputs; appendix D.2 repeats the pretrained networks' linear probes with recomputed statistics.

Binary linear probes on pairs of JetClass-II classes ask whether a label set keeps a distinction it merges, or one that a downstream discriminant needs. In two pairs, $X \to b\bar{b}$ against $X \to c\bar{c}$ and $X \to YY \to bbqq$ against $X \to YY \to ccqq$, the decays differ only in quark flavor and are merged at 17 classes. The third, $X \to bc$ against $X \to bq$ and $X \to cs$ together, sets the numerator class of D_{bc} against the resonant classes of its denominator [49, Eq. (1)]; $X \to bc$ and $X \to bq$ share a class from 43 classes on. Each pair is merged in some random partitions and kept apart in others, which separates the effect of merging a pair from that of the class count. The three pairs of QCD classes that only the 188-class level separates are in appendix E.3, and the sizes of the pair samples in appendix B.3.

Fine-tuning asks what label-set effect remains once every weight adapts. Each pretrained model is fine-tuned once, with its whole head, hidden layer included, initialized anew. The same network trained on the downstream task alone, from random initial weights, is the reference for what pretraining adds over the downstream data; it is trained with three seeds. On the full training sets of top tagging and quark/gluon discrimination and on the largest JetClass training set, fine-tuning uses ParT's fine-tuning settings, and the network trained from scratch uses ParT's from-scratch settings [8]. The smaller training sets use one rule, identical for pretrained and from-scratch networks: a fixed maximum number of steps, early stopping on the validation set and a learning rate chosen on validation (appendix B.4). Missing inputs are set to zero in every fine-tuning. On JetClass-II the levels of the hierarchy and the models with the auxiliary mass regression are fine-tuned, while the controls are evaluated with linear probes only. One fine-tuning per pretraining run spends the compute on pretraining runs, whose variation this study must measure. Separating the variation of pretraining from that of fine-tuning would need several fine-tuning runs per pretraining run, as in MultiBERTs [51, App. D].

Each task is scored in the form its literature uses. On top tagging and quark/gluon discrimination the tables give the accuracy, the AUC and the background rejection $1/\epsilon_B$ at signal efficiencies of 50% and 30%, written R_{50} and R_{30}, where ϵ_B is the efficiency for the background. The text compares label sets by R_{30}, the background a tagger removes at a working point an analysis uses, because values of the AUC close to one hide differences that rejection shows. With pretraining, for example, ParT's top-tagging AUC rises only from 0.9858 to 0.9877, while its R_{30} rises from 1602 to 2766 [8]. On JetClass the tables follow ParT [8, Table 1]. They give the accuracy, the AUC averaged over all pairs of classes as in ParT, and the rejection of each signal class against the QCD background at the signal efficiency ParT uses for it. The per-class rejection shows whether a label set helps or hurts specific signals, in a form comparable with ParT. The AUC averaged over classes, each class against the rest, is given beside it in appendix E.2. On JetClass-II the tables give the accuracy and the AUC averaged over classes, each against the rest, over all 188 classes and over the resonant and the QCD classes, the form in which the separation of many classes is usually reported.

Between pretraining configurations on JetClass-II, label sets are compared by the ratio of $1 - \text{AUC}$, the fraction of signal–background pairs the classifier orders

wrongly:

$$\rho_{A/B} = \frac{1 - \mathrm{AUC}_A}{1 - \mathrm{AUC}_B}. \tag{2}$$

Here A and B are two configurations scored on the same task and the same test jets, with the AUC averaged over classes for the full task and the binary AUC for a pair of classes. A ratio above one means that A orders more pairs wrongly than B. The ratio answers by what factor a configuration changes the errors on the pretraining classes, where the AUCs lie close to one and differ by little in absolute terms. Rejection at a fixed signal efficiency does not serve here, because for some pairs of classes the best models pass almost no background jets of the test sample at the working point. The ratio of $1 - \mathrm{AUC}$ appears only in this role: never on JetClass, top tagging or quark/gluon discrimination, and never as a table headline.

No published jet foundation model reports results for every class it is pretrained on. Sophon and GloParT sum classes into discriminants [11, 12], and OmniLearned evaluates its classes only through a summed anomaly score [31]. We show all 188 JetClass-II classes in one heatmap of each class's ratio of $1 - \mathrm{AUC}$, the class against the rest, to the 188-class model (figure 3). The classes are grouped by their 17-class parent and the color scale is logarithmic, after the per-class heatmap of JetParticle-JEPA [29].

Three further descriptors each answer one question. The first is the relative drop in the accuracy of a JetClass-II linear probe when the inputs a benchmark lacks are set to zero in its training and test jets. It measures how much of what a linear probe reads depends on those inputs. JetParticle-JEPA reports the same relative drop when it removes inputs at inference [29]; here the linear probe is also trained without them. The second is the data-efficiency equivalence of a fine-tuned model: the number of training jets with which the network trained from scratch reaches the same metric, found by linear interpolation in the logarithm of the number of jets between neighboring training sizes. It asks how many from-scratch training jets fine-tuning at a given size is worth. The third is the share of the gain from pretraining that a linear probe keeps: its accuracy minus that of training from scratch, as a fraction of the accuracy of fine-tuning minus that of training from scratch, at the same number of training jets. It asks which part of the gain survives with the backbone frozen.

The jet-mass resolution asks how close a mass estimate m_{est} comes to the generator-level mass m_{gen} of equation (1), at every jet p_{T} and mass. For the response $r = m_{\mathrm{est}}/m_{\mathrm{gen}}$ we use the definition of ATLAS [52, Sec. 6],

$$\sigma_r = \frac{q_+ - q_-}{2\,\tilde{r}}, \tag{3}$$

where $\tilde{r}$ is the median response and q_- and q_+ bound the central 68% of the distribution of r; the median response is reported beside it. PELICAN uses the same quantile range of the relative residual without the division by the median [53, Sec. 7.3], and CMS reports the position and the width of the mass peak [10, 12]. The target m_{gen} is the same for resonant and QCD jets. We bin in the reconstructed jet p_{T}, where ATLAS bins in the true p_{T}. The jets are JetClass-II test jets with a generator-level match, divided into training, validation and evaluation parts for the regressions. Three estimates are compared on the same evaluation jets: the soft-drop mass itself, the regressed mass $m_{\mathrm{SD}}\, e^{\hat{y}}$, where $\hat{y}$ is the output of the auxiliary

mass regression, and a regression from frozen features. The last is an MLP probe trained to predict y after the training median of y in each JetClass-II class is subtracted, so that a better classifier is not credited as a better mass regression; it asks whether the frozen features themselves carry the jet mass. A multilayer perceptron is used because the mass is not a linear function of the features, and the linear (ridge) regression is in appendix E.5.

Mass sculpting asks whether selecting on the summed score of the resonant classes makes the background mass distribution peak where a signal would. As CMS does [10, Sec. 6.2], we select on $1 - p_{\mathrm{QCD}}$, at QCD efficiencies of 10% and 1%. We then measure the Jensen–Shannon distance, the square root of the base-2 Jensen–Shannon divergence, between the soft-drop mass distributions of the selected QCD test jets and of all QCD test jets; figure 8 also traces it over a grid of QCD efficiencies. A random selection of as many QCD jets gives a distance above zero from the size of the sample alone. The expected distance of such a selection, drawn without replacement, is therefore the floor against which each distance is read: it asks whether a distance is larger than a random selection of the same size would give.

We repeat Sophon's resonance-tagging benchmark [11, Sec. III] at every label set, for a spin-0 resonance X of mass **[samples not in]** GeV decaying to $b\bar{b}$ or to bs. In the 188- and 162-class models, where $X \to b\bar{b}$ is its own class, it is tagged by Sophon's direct discriminant [11, Eq. (4)]. This asks whether the pretrained output scores alone tag it without a new head. At every label set it is also tagged by a transfer head: a multilayer perceptron on the frozen features whose first layer starts from the pretrained head's hidden layer, trained with Sophon's settings [11, App. B.5]. $X \to bs$, a class in no label set, is tagged by the transfer head and by the same network trained from scratch with Sophon's settings for it [11, App. B.3], with three seeds. The score of the output class that contains $X \to bq$, the nearest pretrained class, is a control that asks whether a gain is in the new head or already in the nearest output. The background is the QCD jets of the JetClass-II test files, with no trigger and no jets from W or Z bosons or top quarks. Jets are counted in Sophon's soft-drop mass window, $150 < m_{\mathrm{SD}} < 230$ GeV for a resonance of 200 GeV, scaled by the ratio of the resonance masses to **[samples not in]**–**[samples not in]** GeV. We report $\epsilon_S/\sqrt{\epsilon_B}$, which is proportional to the discovery significance for a small signal [54] and so gives the reach of a selection without assuming event counts, and the rejection $1/\epsilon_B$ at $\epsilon_S = 30\%$. Sophon reports the discovery significance at an integrated luminosity of 40 fb^{-1} on the full Standard Model background after a trigger, which needs the event counts of every background process that our samples do not provide, so the two are not comparable in value. Each fit records whether it uses a fixed or a variable resonance mass; Sophon trains its transfer head on variable masses and evaluates it at 200 GeV.

3.4 Uncertainties

Every uncertainty is the standard deviation over the independently pretrained networks of a configuration, each followed by one linear probe or one fine-tuning. Each level of the class hierarchy has three pretraining runs. The models with the auxiliary mass regression have two each, the random partitions with controlled flavor boundaries two each, and each random partition and the kinematic-only model one; the

mean and standard deviation of the random partitions are taken over the five partitions. The spread therefore includes the variation between pretraining runs, which most published foundation-model results leave out by repeating only the downstream training on one pretrained network; Sophon reports no variation between pretraining runs [11]. Three runs are given as the mean and standard deviation, rounded following the Particle Data Group [55], two runs as both values and one run as its value. With three runs a standard deviation is itself uncertain by about 50%. The network trained from scratch varies its training seed, and the randomly initialized network its initial weights.

Ratios are formed between models with the same seed, which share their training jets and initial weights, and then averaged over seeds; a control is paired with the models of the seed it was pretrained with. The task's metric is $1-\mathrm{AUC}$ on JetClass-II, R_{30} on top tagging and quark/gluon discrimination, and the rejection of each signal class on JetClass. A paired ratio of it is a difference when it departs from one by more than twice its standard deviation and by more than its test-set uncertainty, which a bootstrap of the test jets gives (appendix D.3). JetClass rejections that rest on too few passing background jets, marked with a dagger in table 2, are not used for claims. For a configuration with two runs, a ratio is a difference only when both runs lie further from one than twice the standard deviation of the same ratio between levels of the class hierarchy. The finite test set adds less to the uncertainty than the variation between pretraining runs (appendix D.3). Training ran on several GPU types, and table 9 gives each run's types and GPU-hours. The three 188-class models trained on one GPU type each, one of them on a different type from the other two. The label-set ratios of that seed lie within 1.4% of the mean of the other two seeds, no more than the variation between seeds, so the GPU type does not move a result beyond that variation.

4 Results

Each subsection compares the four levels of the class hierarchy first and the controls second, the controls against the 17-class level whose class count they share. Linear probes on frozen features carry the comparison on the pretraining classes, with the MLP probe beside every linear result that shows no difference; fine-tuning carries it on the downstream benchmarks. On JetClass-II we compare label sets by ratios of $1-\mathrm{AUC}$, and on the downstream tasks by each task's background rejection. The subsections move outward from the pretraining classes. JetClass-II and JetClass come first, then the two public benchmarks, top tagging and quark/gluon discrimination; fine-tuning with few training jets, the jet mass and resonance tagging follow.

4.1 JetClass-II: the pretraining classes

The first question is which distinctions the frozen features keep on the 188 classes the networks were pretrained on, scored by linear probes on the class-balanced samples of section 3.3 (sizes in the caption of figure 2 and in table 17). Because this task favors finer label sets by construction, the claims below rest on comparisons that this bias cannot produce: the 162-class model, whose labels keep every resonant class of the 188-class model, and the controls, which have as many classes as the 17-class level.

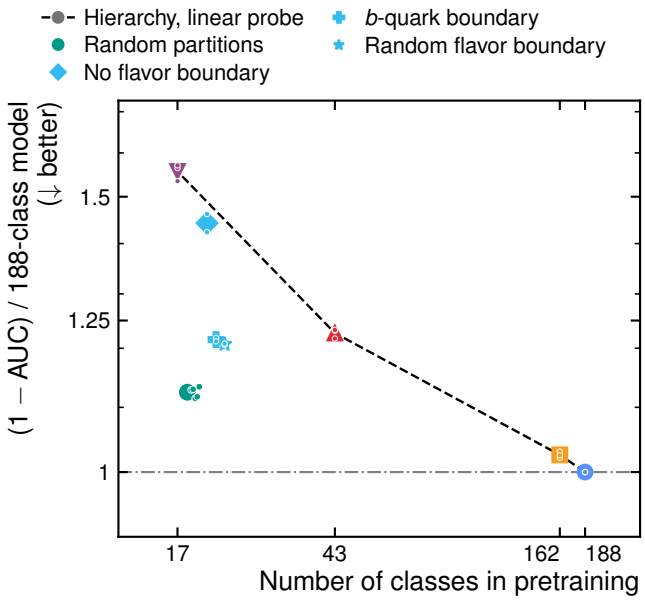


Figure 2: Linear probes on frozen features against the number of classes in pretraining. The vertical axis is the ratio of $1 - \mathrm{AUC}$ (one minus the area under the receiver operating characteristic curve, averaged over the 188 JetClass-II classes, each against the rest) to that of the 188-class model of the same seed; both axes are logarithmic, lower is better, and the dash-dotted line marks equality. The dashed line joins the four levels of the class hierarchy; the solid line joins the same levels after fine-tuning on the largest class-balanced training set. Large markers are means with bars of one standard deviation over three pretraining runs, small points single runs. At 17 classes, beside the 17-class level they are read against, sit the five random partitions (one run each; the larger marker is their mean and standard deviation) and the partitions with no, a b-quark and a random flavor boundary (two runs each). The test set holds up to 5,000 jets per class and all jets of the 11 classes with fewer. This is the 188-class model's own pretraining task, so finer label sets are favored by construction. For scale, the area under the curve is $0.9801 \pm <0.0001$ for the 188-class model and 0.8660 and 0.8630 **[2 of 3]** for the randomly initialized network with its normalization statistics recomputed.

A linear probe separates the 188 classes worse at every coarser level of the hierarchy, and the largest step is the merge of quark flavor between 43 and 17 classes (figure 2). On frozen features of the 43-class model it makes 23% more errors than on those of the 188-class model, and on the 17-class model 56% more errors. The step from 43 to 17 classes costs more than the step from 162 to 43 classes, although it merges far fewer classes and removes one kind of distinction where the other removes three: the number of b or c quarks, gluons against light quarks, and electrons against muons. The pretraining loss rewards exactly the distinctions in the label set, so a distinction it stops rewarding is no longer kept in a form a linear classifier can read, and how much that costs depends on what the distinction is, not on how many classes the merge removes.

The 162-class model matches the 188-class model on the resonant classes, under the linear probe (equal within 1%) and under the MLP probe (equal within 1%; table 12), so the equality is not a limit of what a linear classifier can read. Its

cost on JetClass-II sits entirely in the QCD classes it merges (the bottom block of figure 3; table 17). The merge from 188 to 162 classes moves no boundary between resonant classes: it removes only the division of QCD jets by their numbers of b, c and s quarks, which the loss then stops rewarding. Of the discriminants in table 1, this merge removes only one, a score against QCD jets split by their heavy-flavor content, as in the QCD subclasses that GN3X distinguishes [37]. Only an analysis that needs that subdivision pays for the merge.

At a fixed number of classes, which classes are merged decides the cost. A random partition with 17 classes makes 28% fewer errors than the 17-class level of the hierarchy, and 8.2% fewer errors than the 43-class level, which has more than twice as many classes (figure 2). A random partition keeps the class count and the share of training jets of every 17-class group but not the physics that defines it, and it keeps many of the flavor boundaries the 17-class level removes. The partitions with controlled flavor boundaries, at the same class count, lie between the 17-class level and the random partitions, with the partition that has no flavor boundary nearest the 17-class level. Because the partitions were selected to cut across the 17-class groups (section 3.2), they show that the class count alone does not set the cost on the pretraining classes, not that any grouping into 17 classes would do as well.

Fine-tuning tests whether the gap survives once every weight adapts to the 188 classes. After fine-tuning on the largest class-balanced training set, the 17-class model makes **[class-balanced JetClass-II fine-tuning]** than the 188-class model, where a linear probe on its frozen features makes 56% more errors (figure 2, solid line). Fine-tuning trains on all 188 classes, so every distinction the coarse labels removed is rewarded again, and what remains is the cost of starting from features that lack them, at this number of training jets. The partitions are not fine-tuned on JetClass-II, so this comparison is between levels of the hierarchy only.

Per class, the largest costs fall on decays that differ from another class only in lepton flavor (figure 3). From the 43-class level on, a decay with electrons shares its class with the same decay with muons, and the decays with a τ lepton that decays to an electron share theirs with those whose τ lepton decays to a muon. At that level the class with the largest ratio to the 188-class model is $X \to YY \to bb\mu\mu$, and the classes merged with a lepton-flavor counterpart are costly as a group (median 2.0 times the errors). They are few, and they stay among the costliest at 17 classes. Such a pair of decays differs in the inputs mainly through the particle-type flags of its leptons, so once the labels merge the two, nothing in the loss asks the backbone to carry which flag was set. Because the 188-class model separates these classes from the rest with few errors (table 27), a modest absolute loss appears as a large ratio.

Summed over classes, quark flavor carries most of the cost. At 17 classes, 60 of the 188 classes at least double their $1 - \mathrm{AUC}$, and the classes that share their group with a decay differing only in quark flavor account for 87% of the total, taken as the sum over the resonant classes of the logarithm of each class's ratio. Each of these classes loses less than the lepton-flavor classes, but together they are most of the resonant classes: most decays in JetClass-II come in variants that differ only in their quark flavors, and the 17-class labels merge every such set of variants.

The 17-class level alone cannot tell whether a class loses because it is merged with a flavor counterpart or because of the kind of decay it is, since it merges every

such class. The random partitions separate the two, because each merges some flavor counterparts and keeps others apart. Across the five partitions, the classes merged with a counterpart in quark flavor make equal within 1% at the median than those kept apart from all of theirs; for lepton flavor the same comparison gives equal within 3%. The cost follows the merge rather than the kind of decay: a class loses the separation that a merged label no longer rewards.

Decays that differ only in quark flavor show the cost of the merge most directly (figure 4). On the pair $X \to b\bar{b}$ against $X \to c\bar{c}$, frozen features of the 17-class model give a linear probe 4.2 times the errors of the 188-class model. The 43-class model, whose labels still keep the two decays apart, makes only 10% more errors. At 17 classes the two decays share a class and no class boundary anywhere separates b from c quarks, so the loss no longer rewards the differences in lifetime and decay multiplicity between b and c hadrons, which reach the network through the displacement and particle-type inputs. The 43-class labels separate decays with b quarks, decays with c quarks and decays with neither within every decay topology, and the pair stays readable.

At a fixed number of classes, a flavor boundary anywhere in the label set keeps quark flavor readable, even for a pair it does not cut. The three partitions with controlled flavor boundaries have 17 classes and the same group shares, and two pretraining runs each, so each comparison below quotes the value less favorable to a difference. On $X \to YY \to bbqq$ against $X \to YY \to ccqq$, the partition with a random flavor boundary, which keeps every pair of four-quark decays related by exchanging b and c quarks in one group, has at most 0.36 times the errors of the partition with no flavor boundary. The partition with a b-quark boundary, the only one of the three that cuts this pair, does better still: the partition with a random flavor boundary makes at least 50% more errors than it.

Since all three have the same number of classes, the flavor boundary and not the class count sets this cost. A boundary between four-quark decays with and without b quarks rewards the network for telling b from c quarks wherever it draws the line, which likely keeps the inputs that carry that difference in use for every pair; a boundary on the pair itself adds the separation specific to it.

The output layer can lose a discriminant while the frozen features keep most of what it needs. The boosted $|V_{cb}|$ measurement uses the scores of $X \to bq$ and $X \to cs$ in the denominator of its discriminant D_{bc} [49], so these classes must be told apart from $X \to bc$. The 162-class features separate them as well as the 188-class features do, under the linear and the MLP probe (figure 4c). At 43 classes, where $X \to bc$ and $X \to bq$ share a class and D_{bc} can no longer be formed (table 1), a linear probe makes only 15% more errors; at 17 classes it makes 95% more errors. The 43-class labels still separate decays with a b quark from decays with only a c quark, so the features keep what separates $X \to bc$ from $X \to cs$ and lose only the number of b quarks that separates it from $X \to bq$. The 17-class labels remove flavor altogether.

4.2 JetClass: the in-distribution downstream test

JetClass [8] holds nine classes of signal decays and a background of QCD jets; its jets carry all 17 particle inputs the networks were pretrained on and come from a simulation without pileup. Each pretrained model is fine-tuned once on a JetClass

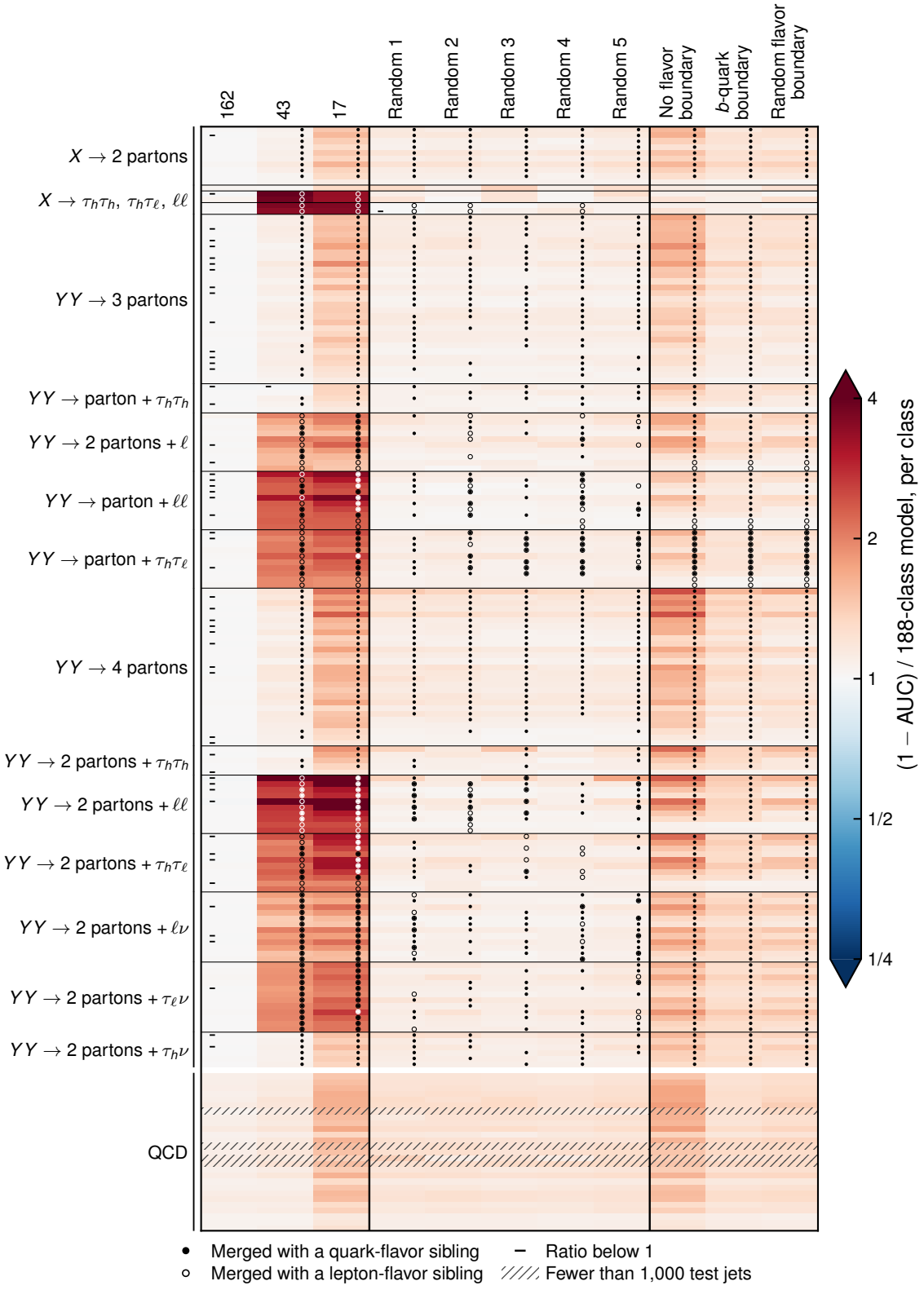


Figure 3: Where the cost of a coarser label set sits, class by class. Each row is one of the 188 JetClass-II classes, grouped by its 17-class group (brackets on the left), with the classes of QCD jets (quarks and gluons that do not come from a resonance decay) in a separate block at the bottom. Each column is a configuration: the 162-, 43- and 17-class levels of the hierarchy, the five random partitions, and the three partitions with controlled flavor boundaries (none, a b-quark boundary, a random flavor boundary). The color is the class's $1 - \mathrm{AUC}$ (one minus the area under the receiver operating characteristic curve of the class against the rest) under a linear probe on frozen features, divided by the same quantity for the 188-class model pretrained with that seed and averaged over the runs of the configuration (three per level, two per partition with a controlled flavor boundary, one per random partition).

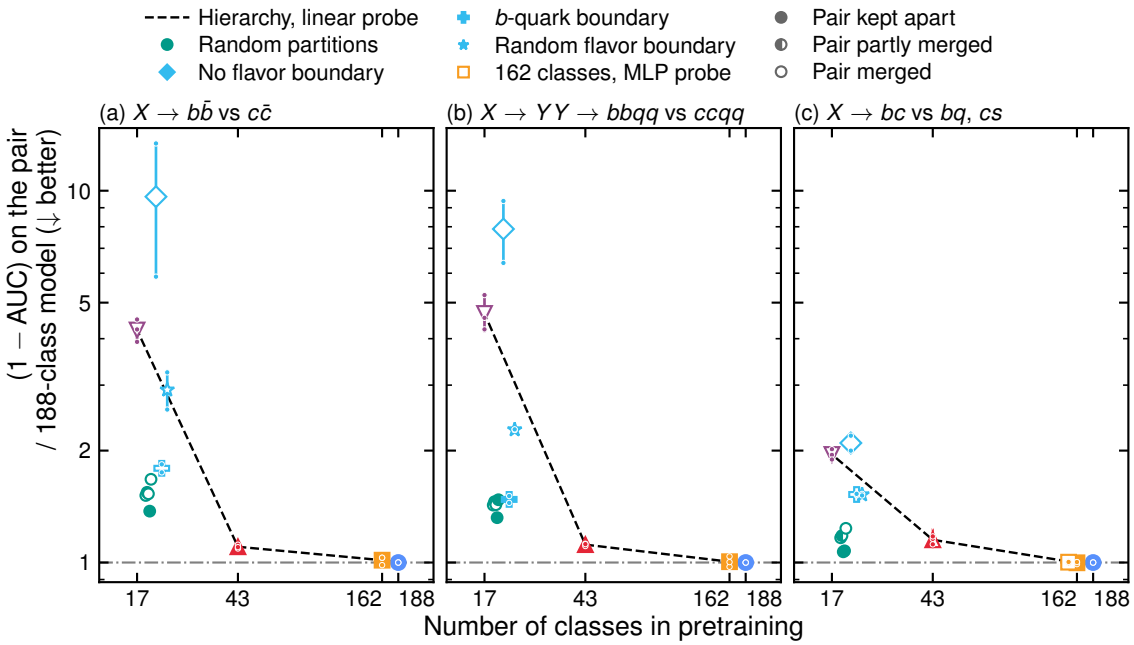


Figure 4: Linear probes on pairs of JetClass-II classes against the number of classes in pretraining: (a) $X \to b\bar{b}$ against $X \to c\bar{c}$; (b) $X \to YY \to bbqq$ against $X \to YY \to ccqq$; (c) $X \to bc$ against $X \to bq$ and $X \to cs$, the classes in the denominator of the discriminant of the boosted $|V_{cb}|$ measurement. Axes, markers, the hierarchy line and the pairing by seed are as in figure 2, for a binary linear probe on the pair. At 17 classes sit the five random partitions (one run each, no mean) and the three partitions with controlled flavor boundaries (two runs each). A marker is filled where the label set keeps the pair apart, open where it merges the pair, and filled on the left only where it merges part of it. In (c), the open square beside the 162-class point is the multilayer perceptron probe, divided by the multilayer perceptron probe of the 188-class model; the 162-class model matches the 188-class model under the linear (equal within 1%) and the multilayer perceptron probe (equal within 1%). In (b), the partition with no flavor boundary makes at least 51% more errors than the 17-class level, and the random partitions that merge the pair were all pretrained with one seed and those that keep it apart with the other, so their difference is confounded with the seed. Every value is in table 26.

training subset and compared with the same network trained from scratch, in the form of ParT's Table 1 [8]: accuracy, the AUC averaged over all pairs of classes, and the rejection of the background at ParT's working point for each signal class (table 2). The caption of table 2 gives our training and test sizes, which differ from those of the published rows.

After fine-tuning, the 162- and 43-class models match the 188-class model on JetClass: neither differs from it in any signal class by the rule of section 3.4 (table 2). The 43-class labels give up the number of b and c quarks, gluons against light quarks and electrons against muons, and none of these losses shows here. JetClass's signal classes differ from one another and from the background by their number of prongs and by whether they contain b or c quarks, and the 43-class labels keep both. The largest departure, in the Higgs decay to four quarks, stays within twice its standard deviation over the runs (table 22).

Table 2: JetClass, in the form of ParT's Table 1: accuracy, AUC averaged over all pairs of classes (one against one, ParT's convention; on JetClass's equal class sizes it equals the one-against-rest average, checked), and for each signal class the rejection of QCD jets at 50% signal efficiency (99% for $H \to \ell\nu qq'$, 99.5% for $t \to b\ell\nu$) on $p_c/(p_c + p_{\mathrm{QCD}})$. Published rows: trained from scratch on 2 million jets and scored on ParT's 20-million-jet test set (ParT on 100 million jets reaches an accuracy of 0.861 and an AUC of 0.9877); ours fine-tuned on one million jets and scored on two million (the AUC on 500,000 of them). Mean ± standard deviation over the three pretraining runs. From scratch: three seeds. Bold: the best value of each column, and every value within one standard deviation of it. †: fewer than 20 QCD jets pass in at least one run, so that run's rejection has a counting uncertainty above 22%; these columns carry no claim.

				rejection of QCD								
	jets	accuracy	AUC	$H \to b\bar{b}$ R_{50}	$H \to c\bar{c}$ R_{50}	$H \to gg$ R_{50}	$H \to 4q$ R_{50}	$H \to \ell\nu qq'$ R_{99}	$t \to bqq'$ R_{50}	$t \to b\ell\nu$ $R_{99.5}$	$W \to qq'$ R_{50}	$Z \to q\bar{q}$ R_{50}
ParticleNet [8]	2 M	0.828	0.9820	5,540	1,681	90	662	1,654	4,049	4,673	260	215
ParT [8]	2 M	0.836	0.9834	5,587	1,982	93	761	1,609	6,061	4,474	307	236
this work (Sophon's network, fine-tuned)												
from scratch	1 M	0.8269 ± 0.0010	0.9821 ± 0.0001	$5{,}090 \pm 200$	$1{,}540 \pm 50$	92.9 ± 1.0	680 ± 50	$1{,}539 \pm 24$	$5{,}300 \pm 600$	$4{,}150 \pm 270$	282 ± 4	227.0 ± 2.8
188 classes	1 M	$\mathbf{0.8491 \pm 0.0005}$	$\mathbf{0.9859 \pm 0.0001}$	$6{,}520 \pm 120$	$3{,}015 \pm 26$	$\mathbf{105.9 \pm 3.3}$	$\mathbf{1{,}205 \pm 33}$	$\mathbf{3{,}390 \pm 270}$	$\mathbf{16{,}900 \pm 2{,}200}^{\dagger}$	$\mathbf{11{,}400 \pm 1{,}000}^{\dagger}$	$\mathbf{413 \pm 13}$	$\mathbf{323 \pm 5}$
162 classes	1 M	$\mathbf{0.8495 \pm 0.0002}$	$\mathbf{0.9859 \pm <0.0001}$	$\mathbf{6{,}600 \pm 600}$	$\mathbf{3{,}110 \pm 150}$	104.3 ± 1.6	$\mathbf{1{,}180 \pm 50}$	$\mathbf{3{,}600 \pm 400}$	$\mathbf{14{,}700 \pm 3{,}200}^{\dagger}$	$\mathbf{10{,}400 \pm 1{,}200}^{\dagger}$	$\mathbf{424 \pm 13}$	318.7 ± 2.4
43 classes	1 M	0.8480 ± 0.0006	0.9857 ± 0.0001	$\mathbf{6{,}900 \pm 700}$	$\mathbf{2{,}880 \pm 230}$	$\mathbf{106.6 \pm 2.5}$	$1{,}070 \pm 50$	$3{,}210 \pm 150$	$15{,}100 \pm 1{,}400^{\dagger}$	$10{,}400 \pm 800^{\dagger}$	410 ± 9	$\mathbf{320 \pm 5}$
17 classes	1 M	0.8434 ± 0.0001	0.9849 ± 0.0001	$\mathbf{6{,}600 \pm 600}$	$2{,}680 \pm 130$	103.7 ± 2.4	$1{,}017 \pm 22$	$3{,}100 \pm 400$	$11{,}800 \pm 2{,}000^{\dagger}$	$8{,}800 \pm 1{,}100$	382 ± 10	299.3 ± 2.6

At 17 classes the rejection falls in several signal classes: $H \to 4q$, $W \to qq'$ and $Z \to q\bar{q}$. The Higgs decay to four quarks loses the most, with 16% less rejection. The loss is spread over classes rather than concentrated in a class defined by flavor, and $H \to b\bar{b}$ shows no drop, but its rejection rests on so few passing background jets that its counting uncertainty is larger than the drops seen elsewhere (table 24). The drops appear only at the level that removes quark flavor; the 43-class model, whose labels still separate decays with b quarks, with c quarks and with neither, shows none of them. The fine-tuned 17-class network therefore likely starts from features that carry less of the charm and bottom content that W, Z and Higgs decays to quarks often have and that QCD jets rarely have.

On JetClass the random partitions recover part of the gap between the 17- and 188-class models, and how much depends on the number of prongs. With one run per partition, we read these values descriptively. The partitions' fine-tuned accuracy exceeds that of the 17-class model by +0.003 (absolute), which places them between the 17- and 43-class models, whereas on the pretraining classes they did better than the 43-class level. On the two-prong decays $H \to c\bar{c}$, $W \to qq'$ and $Z \to q\bar{q}$ they come close to the 188-class model; in the Higgs decay to four quarks and the two top-quark classes they stay near the 17-class model (table 22). The partitions were built to cut across the 17-class groups, and in doing so they put decays with different numbers of prongs into one group (table 25). They likely keep the flavor boundaries that help separate two-prong decays from the background and lose the prong-count boundaries that the three- and four-prong classes rely on.

4.3 Top tagging

Top tagging uses the landscape dataset of hadronic top-quark jets and QCD jets [20] with its standard training, validation and test sets, whose sizes are given with table 3. The dataset provides only the particle four-momenta, from which the kinematic inputs are computed. The charge, the five particle-type flags and the four

Table 3: Top tagging, fine-tuned on the full training set (1.2 million jets): accuracy, AUC and background rejection $1/\epsilon_B$ at 50% and 30% signal efficiency. Published rows from the tables cited; f.t. marks a model pretrained on JetClass and fine-tuned. Our pretrained models see the 10 inputs the top data lack set to zero; the model with kinematic inputs only, a diagnostic of that mismatch, has none missing (one run). Uncertainties: ours, the standard deviation over the pretraining runs, one fine-tuning each, and over three seeds from scratch; ParT, the spread over 9 fine-tunings of one pretrained model with different output-layer initializations, quoted for the run of median accuracy; L-GATr and OmniLearned, the standard deviation of five trainings. Bold: the best value of each column, and every value within one standard deviation of it.

	inputs set to zero	accuracy	AUC	R_{50}	R_{30}
ParticleNet [7]	–	0.940	0.9858	397 ± 7	$1{,}615 \pm 93$
ParT [8]	–	0.940	0.9858	413 ± 16	$1{,}602 \pm 81$
ParT-f.t. [8]	–	0.944	0.9877	$\mathbf{691 \pm 15}$	$2{,}766 \pm 130$
MIParT-L-f.t. [56]	–	0.944	0.9878	640 ± 10	$2{,}789 \pm 133$
L-GATr-f.t. [57]	–	**0.9446**	0.9879	651 ± 11	$2{,}894 \pm 84$
OmniLearned-l [31]	–	0.944	**0.9880**	$\mathbf{688 \pm 9}$	$\mathbf{3{,}486 \pm 157}$
this work (Sophon's network)					
from scratch	10	0.9387 ± 0.0002	0.9852 ± 0.0001	358 ± 11	$1{,}412 \pm 20$
188 classes	10	0.9400 ± 0.0002	0.9861 ± 0.0001	461 ± 4	$1{,}800 \pm 60$
162 classes	10	0.9401 ± 0.0002	0.9861 ± 0.0001	443 ± 17	$1{,}790 \pm 60$
43 classes	10	0.9402 ± 0.0004	0.9861 ± 0.0001	450 ± 13	$1{,}780 \pm 150$
17 classes	10	0.9398 ± 0.0001	0.9859 ± 0.0001	429 ± 10	$1{,}740 \pm 140$
188 classes, kinematic inputs only (diagnostic)	none	**[run]**	**[run]**	**[run]**	**[run]**

ParT's top-tagging training script, unchanged, on our cluster gives R_{30} of 1,711 (published $1{,}602 \pm 81$); with Sophon's output head, 1,577; with the head and Sophon's jet-p_T rescaling, 1,224 (one run each; our network from scratch, which uses the rescaling, $1{,}412 \pm 20$); table 16.

displacement variables, 10 of the 17 inputs of the network, are therefore set to zero for every model. Published results are quoted as their authors report them. The background rejection against signal efficiency of every label set, on top tagging and on quark/gluon discrimination, is drawn in figure 11.

Pretraining raises the background rejection on top tagging, and no label set changes it beyond what the test set resolves. Fine-tuned with the missing inputs set to zero, the 188-class model reaches 27% more rejection than the same network trained from scratch (R_{30}), and every label set improves on training from scratch in every run (table 3). The 17-class model matches the 188-class model (equal within 10%), as do the 162- and 43-class models, so unlike the 17-class labels on JetClass (section 4.2), no label set differs on top tagging by the criterion of section 3.4. The test set limits this comparison: R_{30} is set by the small number of QCD jets that pass at that working point, and their counting uncertainty is as large as the spread between runs (table 14). The agreement has a physical reason. A top-quark jet differs from a QCD jet by its three-prong substructure and its mass, which every label set rewards because even the 17-class labels separate decays by their number of prongs. The pairwise features of the network are computed from the four-momenta alone and do not depend on the missing inputs.

Our network trained from scratch rejects less background than ParT trained from scratch [8], and the difference comes from Sophon's preprocessing, which every pretrained model keeps. Run unchanged on our hardware, ParT's training script reproduces ParT's published accuracy and AUC, and replacing ParT's output head with Sophon's leaves the rejection within the published spread. Adding Sophon's

rescaling of every jet's constituents to one fixed jet transverse momentum [46] gives 28% less rejection than the unchanged script (one run each; table 16). We keep the rescaling in the network trained from scratch because the pretrained models use it, so the comparison of pretraining with training from scratch does not also change the preprocessing. The rescaling keeps the ratio of jet mass to transverse momentum but no longer the jet mass itself, so the spread of transverse momenta in the sample likely broadens the top-quark and W-boson mass peaks a top tagger relies on.

ParT fine-tuned after pretraining on JetClass [8] rejects more background than any of our models, and the 188-class model with kinematic inputs only measures how much of that gap the missing inputs cause. ParT pretrained a separate model on the kinematic features for top tagging, so nothing was missing for it. Our 188-class model, with 10 inputs set to zero, reaches 35% less rejection than ParT. The 188-class model with kinematic inputs only shares the data, the rescaling and the training settings of the 188-class model and lacks no input. Fine-tuned on top tagging, it closes part of the gap to ParT (kinematic-only model over the 188-class model: **[no run yet]**). It is one pretraining run, so it is read against the spread of the three 188-class runs rather than as a ratio paired by seed. The part of the gap it does not close comes from what else separates our models from ParT: the rescaling above, and pretraining on JetClass-II, with pileup, rather than on JetClass.

On top tagging, frozen features from pretraining give a linear probe no more than the randomly initialized network gives it. With the missing inputs set to zero, the rejection of a linear probe on the frozen 188-class features matches that of a linear probe on the randomly initialized network (at least 3.8% more rejection; the MLP probe gives the same result, equal within 40%), and both lie far below every fine-tuned model (figure 5(c)). Fine-tuning the same backbone on the same zero-substituted jets gives the gain of table 3, so the frozen features are limited by how they respond to the substituted inputs, not by what the backbone can represent.

The same zeros cost the pretrained features on JetClass-II itself, more than they cost the randomly initialized network. With the inputs the top data lack set to zero in the training and test jets of the linear probe, the 188-class model loses 89% of its accuracy on the 188 classes and the randomly initialized network 55–58%. Every pretrained label set loses more than the randomly initialized network (figure 5(a); table 11). Two explanations predict this pattern. The zeroed inputs are states the pretrained network never saw, so its features respond to them in ways no training jet shaped. Alternatively, the type flags and displacements carry information that the pretrained features use and the random features do not, so the pretrained network has more to lose. The other two substitutions separate these explanations.

The substitutions differ in how far they leave each particle from a real one, and that difference separates the two explanations. No JetClass-II particle has all of its type flags at zero, so the zeroed inputs put every particle of a top jet in a joint state the pretrained networks never saw. Each zeroed flag on its own still lies within the range of the training distribution after the input normalization (table 10); it is the combination that no real particle has. The training mean is no closer to a real particle, since it is the centroid of the training particles and carries fractional flags; only each random draw is a state a real particle takes. Figure 5(c) compares the linear probe on the 188-class model with random draws and with zeros (draws over zeros: **[no run yet]**), and the linear probe on the 188-class model with kinematic inputs only, which lacks nothing, with the zero-substituted 188-class model ([**no**

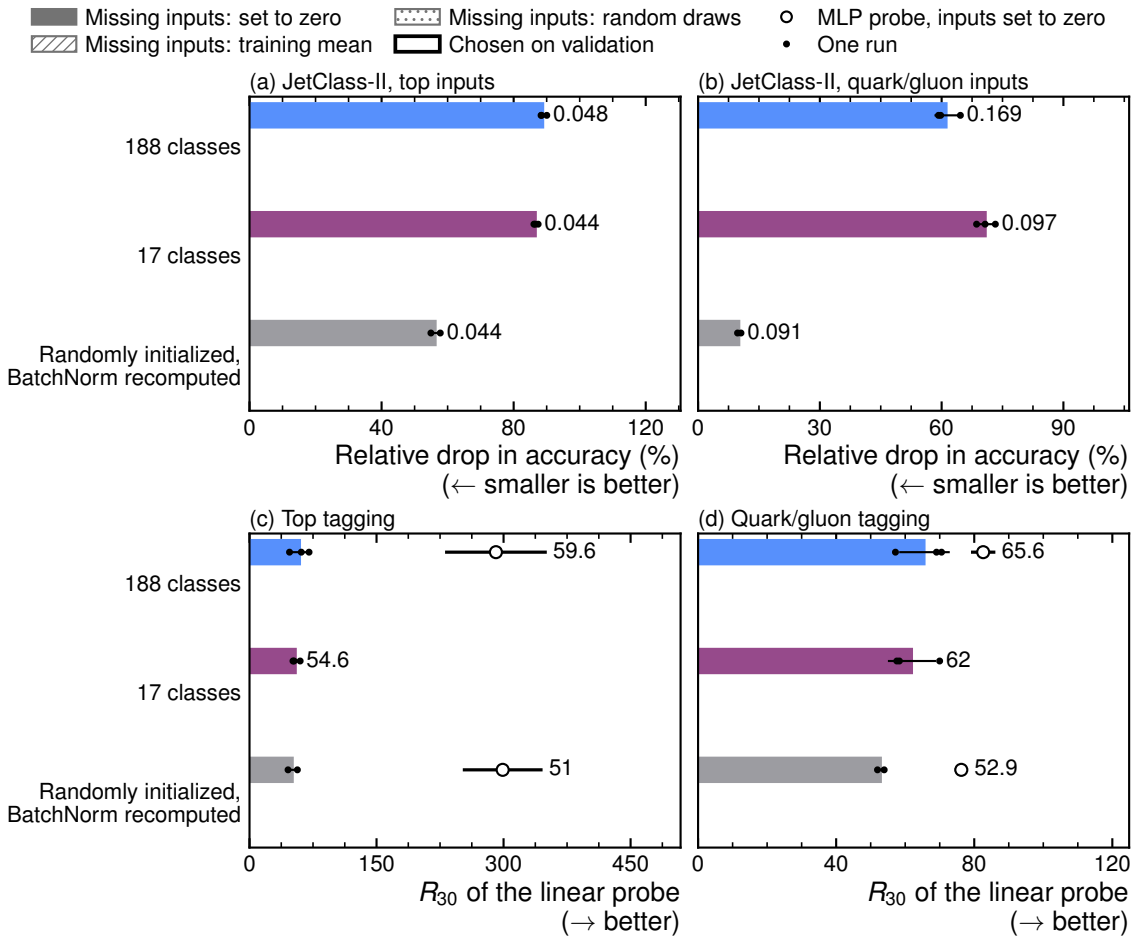


Figure 5: Linear probes on frozen features when the inputs a benchmark lacks are substituted. (a), (b) Relative drop in the accuracy of a linear probe on the 188 classes of the class-balanced JetClass-II test set when the 10 inputs the top-tagging data lack (a) or the 4 inputs the quark/gluon data lack (b) are substituted in the linear probe's training and test jets, against the same model with every input; the absolute accuracy is printed at each bar end. (c), (d) Background rejection at 30% signal efficiency of a linear probe on top tagging (c) and on quark/gluon discrimination with the Pythia sample (d), trained on the full benchmark training set. Each group of bars is one model: the 188-class model and the randomly initialized network with its BatchNorm statistics recomputed. Within a group the missing inputs are set to zero (solid), to the training mean (hatched) or to random draws of a real JetClass-II particle (dotted). Panel (c) adds the 188-class model with kinematic inputs only (dash-dot outline), which lacks no input; it is one run, not paired by seed with the other models. Open markers on the zero bars of (c) and (d): a multilayer perceptron trained on the same frozen features. After the input normalization a zeroed type flag lies at most **[a committed JSON of the input-normalization offsets of mtx-l188-s1 (A8)]** standard deviations from the training mean of that flag. Bars are the mean, error bars the standard deviation and points the individual runs, over the three pretraining runs of the 188-class model and the randomly initialized networks of the same seeds.

Table 4: Quark/gluon discrimination with the particle type as five flags and the charge (ParT's exp inputs), fine-tuned on the full training set (1.6 million Pythia jets) and scored on Pythia and on Herwig jets: accuracy, AUC and $1/\epsilon_B$ at 50% and 30% quark efficiency. Our models see the 4 displacement inputs set to zero. Published rows (exp inputs) from the tables cited; PFN-Ex's uncertainty is half the interquartile range of ten trainings, the others as in table 3, with nine trainings for OmniLearned. Bold as in table 3.

		Pythia				Herwig			
	inputs set to zero	accuracy	AUC	R_{50}	R_{30}	accuracy	AUC	R_{50}	R_{30}
PFN-Ex [21]	–	–	0.9005	34.7 ± 0.4	–	–	–	–	–
ParticleNet [7]	–	0.840	0.9116	39.8 ± 0.2	98.6 ± 1.3	–	–	–	–
ParT [8]	–	0.840	0.9121	41.3 ± 0.3	101.2 ± 1.1	–	–	–	–
ParticleNet-f.t. [8]	–	0.839	0.9115	40.1 ± 0.2	100.3 ± 1.0	–	–	–	–
ParT-f.t. [8]	–	0.843	0.9151	42.4 ± 0.2	107.9 ± 0.5	–	–	–	–
OmniLearned-m [31]	–	**0.845**	**0.9162**	$\mathbf{43.2 \pm 0.1}$	$\mathbf{111.2 \pm 1.5}$	–	–	–	–
this work (Sophon's network)									
from scratch	4	0.8397 ± 0.0007	0.9121 ± 0.0002	41.47 ± 0.29	104.0 ± 1.0	$\mathbf{0.7153 \pm 0.0024}$	0.8061 ± 0.0007	12.18 ± 0.11	35.5 ± 1.2
188 classes	4	0.8433 ± 0.0002	0.9148 ± 0.0001	42.56 ± 0.10	107.3 ± 2.2	0.712 ± 0.005	0.8078 ± 0.0023	12.3 ± 0.4	$\mathbf{35.2 \pm 2.1}$
162 classes	4	$0.8432 \pm <0.0001$	0.9148 ± 0.0001	42.39 ± 0.22	108.0 ± 1.4	0.7100 ± 0.0017	0.8074 ± 0.0010	12.33 ± 0.28	34.6 ± 2.0
43 classes	4	0.8430 ± 0.0005	0.9146 ± 0.0003	42.2 ± 0.8	108.3 ± 1.0	$\mathbf{0.713 \pm 0.006}$	$\mathbf{0.809 \pm 0.005}$	$\mathbf{12.5 \pm 0.6}$	$\mathbf{36.1 \pm 2.7}$
17 classes	4	0.8427 ± 0.0003	$0.9144 \pm <0.0001$	41.97 ± 0.25	106.5 ± 0.8	$\mathbf{0.7175 \pm 0.0033}$	$\mathbf{0.8103 \pm 0.0018}$	$\mathbf{12.79 \pm 0.20}$	$\mathbf{36.9 \pm 0.9}$

run yet]). A recovery with random draws places the loss in the zero value rather than in the missing information; a linear probe that neither the draws nor the mean recover, beside a kinematic-only linear probe far above it, places the loss in the missing information. Fine-tuning escapes both, because it re-learns the input layer on the substituted jets, and a constant substitution used in training and testing is consistent for a model trained with it [58, Theorem 3]. JetParticle-JEPA, which saw zeroed inputs during pretraining through input dropout, sets the missing top-tagging inputs to zero in the same way and reports a rejection below that of ParT trained from scratch [29]; its result does not distinguish the two explanations either.

4.4 Quark/gluon discrimination, Pythia and Herwig

Quark/gluon discrimination uses the Pythia sample of Komiske et al. [21, 22] with ParT's split into training, validation and test jets [8], and the companion sample generated with Herwig with the same selection [50, 23]; the sizes and cuts are given with table 4. The jets carry the particle type as five binary flags and the charge, ParT's "exp" inputs [8], and only the 4 displacement inputs are set to zero. Every model is fine-tuned on Pythia jets and tested on both samples. Published results are from the same input scenario.

Fine-tuned on the Pythia sample, the pretrained models reach the rejection of ParT fine-tuned after pretraining on JetClass [8], and the label sets do not differ. The 188-class model matches ParT (equal within 3%), and the 17-class model matches the 188-class model (equal within 3%; table 4), although only the 188- and 162-class labels separate resonance decays to gluons from decays to light quarks, the distinction quark/gluon discrimination asks for. Three differences between the tasks likely limit how much of that distinction transfers: the quark/gluon jets have a smaller radius than every pretraining jet; they are isolated jets from a single parton rather than prongs inside a resonance decay; and their displacement inputs are zero.

The change of generator costs the pretrained and the from-scratch networks the same fraction of their rejection, so pretraining gives no protection against it. Tested on the Herwig sample, the fine-tuned 188-class model keeps 0.33 times the rejection it has on Pythia jets, and the network trained from scratch 0.34 times the rejection

(table 4). Earlier generator comparisons found Herwig's quark and gluon jets less separable than Pythia's [59], which lowers the rejection of any tagger trained on Pythia jets. Pretraining does not offset this because JetClass-II is also showered and hadronized with Pythia [11, 41], so the pretrained network has seen no jet with Herwig's modeling.

On quark/gluon discrimination, frozen features from pretraining give a linear probe more than the randomly initialized network gives it, but not much more. With the displacement inputs substituted as chosen on validation (**[fill fits]**), a linear probe on the frozen 188-class features rejects more background than one on the randomly initialized network (**[no run yet]**; figure 5(d)). The frozen features likely do better here than on top tagging because the type flags and the charge are present. They gain little over the randomly initialized network because the missing displacements still leave every charged particle in a state no pretraining jet has, as the next paragraph explains.

Setting only the displacement inputs to zero already costs the pretrained features most of their accuracy on JetClass-II, and the randomly initialized network little. With the 4 inputs the quark/gluon data lack set to zero in the linear probe's training and test jets, the 188-class model loses 61% of its accuracy on the 188 classes and the randomly initialized network 9.7–11% (figure 5(b)). The mismatch differs from top tagging in one respect: a zeroed impact parameter lies at the training mean, but no charged JetClass-II track has zero measurement uncertainty on its impact parameters, so a charged particle with zeroed uncertainties is a joint state the input layer never saw. As on top tagging, the pretrained network may also lose information it uses, since the displacements mark the b- and c-hadron decays of many JetClass-II classes [6].

4.5 Fine-tuning with few training jets

[How the gain of the fine-tuned 188-class model over training from scratch on top tagging changes with the number of training jets, the data-efficiency equivalence of section 3.3, and the order of the label sets at the smaller training sets (table 19) are written once the fine-tuning runs on the smaller training sets are evaluated.]

Most of the gain from pretraining survives with the backbone frozen when no input is missing. On the smallest JetClass training set the linear probe on the 188-class model keeps **[no run yet]** of the gain of fine-tuning over training from scratch (section 3.3), and on class-balanced JetClass-II **[no run yet]**. On top tagging and quark/gluon discrimination the linear probe recovers little of the gain at any size (figure 6(c, d)), because fine-tuning re-learns the input layer on the substituted inputs, while the frozen features keep the mismatch described in section 4.3.

4.6 Jet mass: regression and sculpting

The jet-mass results use the JetClass-II test jets that have a generator-level match, so that the generator-level soft-drop mass m_{gen} is defined for resonant and QCD jets alike; a jet without a match has no m_{gen} and is left out. Every mass estimate is scored on the same jets through its response r and the resolution of equation (3), and the binned results use the reconstructed jet p_{T}. The reference is the reconstructed

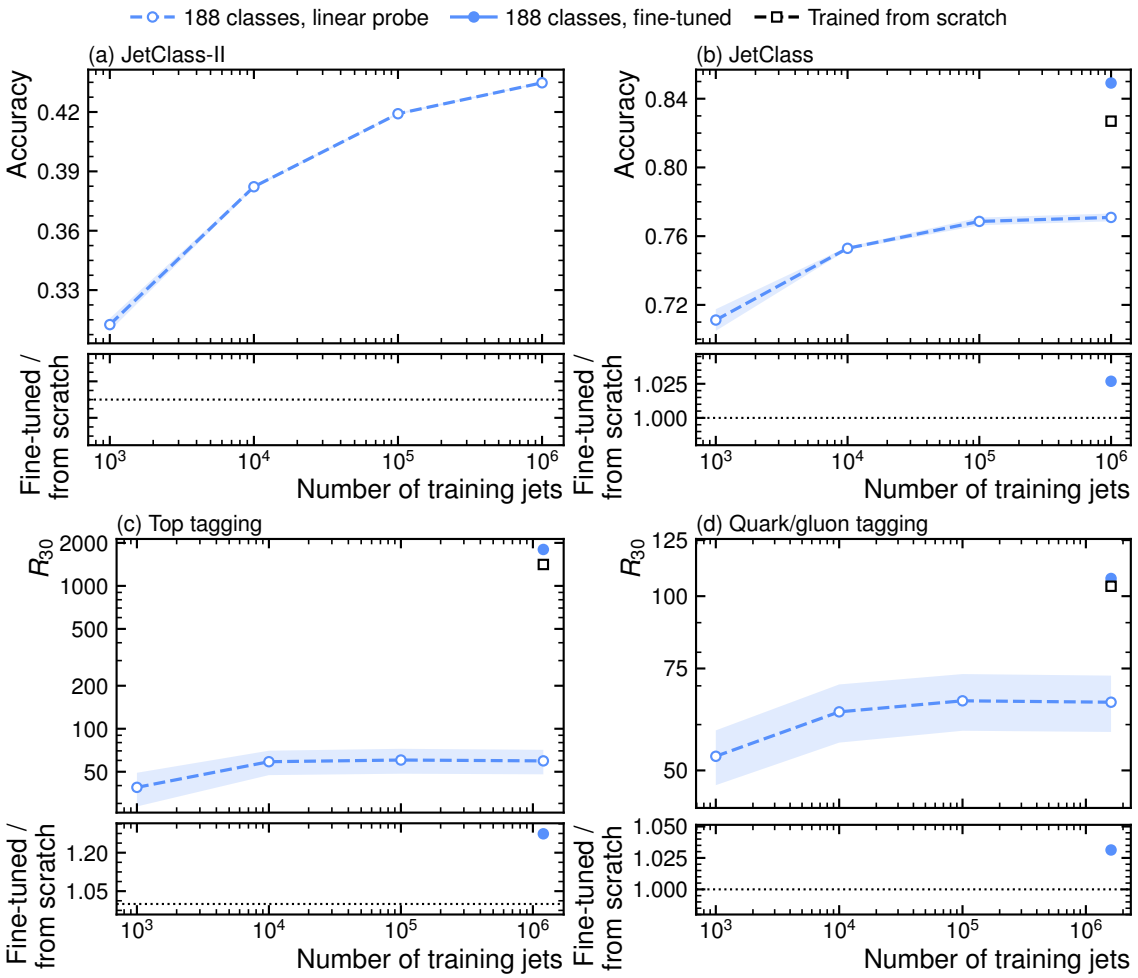


Figure 6: Fine-tuning against the number of training jets on (a) class-balanced JetClass-II with its 188 classes, (b) JetClass, (c) top tagging and (d) quark/gluon discrimination with the Pythia sample. Upper panels: accuracy (a, b) or background rejection at 30% signal efficiency on a logarithmic scale (c, d) for the fine-tuned 188-class model (solid line), a linear probe, a logistic regression on the frozen features of the same model (dashed line, same color), and the same network trained from scratch (black dashed line, open squares). Lower panels: fine-tuning over training from scratch at the same number of jets. The horizontal axes are logarithmic. Lines are the mean and bands the standard deviation over the three pretraining runs, one fine-tuning or linear probe each, and over the three training seeds from scratch. On top and quark/gluon discrimination the linear probe's missing inputs are substituted as chosen on validation ([**fill fits**] and [**fill fits**]), and fine-tuning sets them to zero. Every label set at every size is in table 19 (fine-tuning) and figure 10 (linear probes).

soft-drop mass m_{SD}, which no training changes. The networks give two further estimates: the regressed mass of the 162- and 17-class models with the auxiliary mass regression, and an MLP regression on the frozen features of every label set. Each model with the auxiliary mass regression has two pretraining runs; its tables give both values, and the text quotes the one less favorable to the claim. The sculpting results use the QCD jets of the JetClass-II test set and the models whose output layer scores every class: the four levels of the class hierarchy and the two models with the auxiliary mass regression.

The auxiliary mass regression gives a narrower mass response than the reconstructed soft-drop mass. Over all test jets, the regressed mass of the 162-class model with the auxiliary mass regression reaches at most 0.60 times the resolution of m_{SD}, that of the 17-class model is as narrow, and both stay narrower in every bin of jet p_{T} and of m_{gen} (figure 7; table 28). The regressed mass also reaches at most 0.67 times the resolution of an MLP regression on the frozen features of the 162-class model pretrained without it, although that regression adds back the median of each jet's true class and the auxiliary mass regression never sees the class. Because its target is $\ln(m_{\mathrm{gen}}/m_{\mathrm{SD}})$, the regression learns from the jet's constituents how detector response, pileup removal and grooming have moved the reconstructed mass away from the generator-level one, which is the spread that m_{SD} carries. That correction depends on the constituents rather than on which classes the labels separate, which is likely why the two label sets reach the same width.

The auxiliary mass regression also puts the jet mass into the frozen features themselves. An MLP regression on the frozen features of the 162-class model with the auxiliary mass regression reaches at most 0.73 times the resolution of the same regression on the 162-class model's features, and the 17-class pair gains as much (table 5). Without the auxiliary target the label sets differ far less (17- over 188-class model, 0.95 times the resolution). The reason lies in the training data: the reweighting flattens every reweighting category in p_{T} and m_{SD}, so a jet's mass says little about its class, and no classification label rewards features that keep the mass more precisely than the decay topology implies. The regression term does. The row of the randomly initialized network in table 5 shows how much of the mass a network that never trained already exposes, and every pretrained model's features carry more.

The auxiliary mass regression costs the classification of the pretraining classes little. A linear probe on the frozen features of the 162-class model with the auxiliary mass regression has up to 2.2% more errors on JetClass-II than one on the 162-class model (table 17). At 17 classes, where the mass term takes a larger share of the loss (section 3.2), the two runs move in opposite directions, so the larger share does not show a larger cost. A likely reason is that the frozen features have room for the mass beside the class information, so the regression adds what the features carry without displacing what the labels ask for. With two runs per model the data can only fail to show a cost that grows with the loss share; they cannot exclude a modest one.

Fine-tuning on top tagging gives no consistent change at 162 classes and a loss at 17 classes. With the auxiliary mass regression, the 162-class runs move R_{30} in opposite directions (equal within 20%), while both 17-class runs lose rejection (at least 8.1% less rejection; table 20). On the pretraining classes the larger loss share at 17 classes showed no larger cost, but after fine-tuning the two label sets fall in the order a cost growing with the share would give. That is a hypothesis resting on

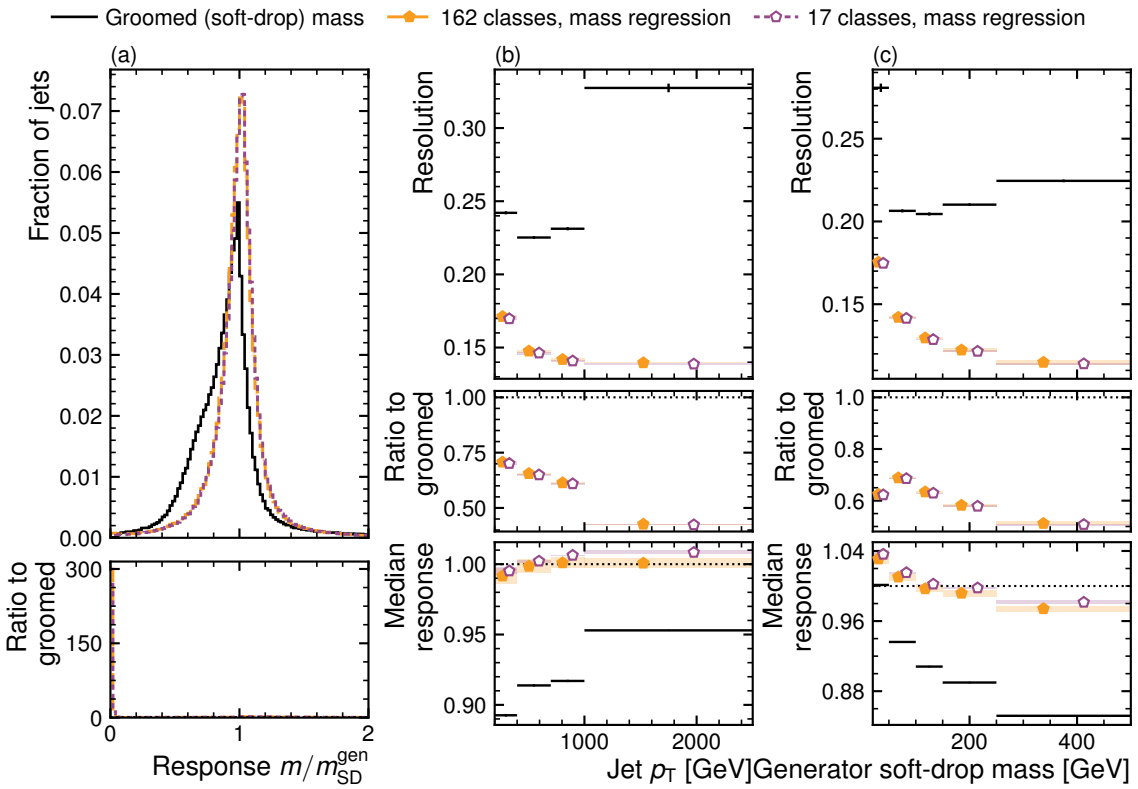


Figure 7: Jet-mass regression with the auxiliary mass regression, on the JetClass-II test jets with a generator-level match. (a) Distribution of the response $m_{\text{est}}/m_{\text{gen}}$, the estimated over the generator-level soft-drop mass, over all these jets: reconstructed soft-drop mass (black) and the regressed mass of the 162- and 17-class models pretrained with the auxiliary mass regression (colored, dashed); the lower panel is the ratio to the reconstructed soft-drop mass. (b) Resolution against the reconstructed jet p_{T} and (c) against the generator-level soft-drop mass, with the ratio to the reconstructed soft-drop mass below. Bands show the range of the two pretraining runs of each model; vertical bars on the reconstructed soft-drop mass are statistical (resampling of the jets), horizontal bars the bin width. The median response is 0.929 for the reconstructed soft-drop mass, and 0.995 and 1.003 and 1.005 and 1.005 for the two regressed masses (two runs each). The layout follows ATLAS [52, figures 7 and 8], except that the jet p_{T} is the reconstructed one and extends to the upper limit of our selection, the lower panels show the ratio rather than the root-square difference, and the panel against the generator-level mass is our addition. Resonant and QCD jets separately: table 28.

Table 5: Jet mass from frozen features: a multilayer perceptron regresses the mass from the frozen features of each configuration; the groomed (soft-drop) mass is the reference. Resolution: half the 15.9–84.1% inter-quantile range of $m_{\mathrm{pred}}/m_{\mathrm{gen}}$ divided by its median; and the median response, over all JetClass-II test jets with a generator match (resonant and QCD jets separately in table 28); lower resolution is better. The regression target is centred on the training median of each of the 188 classes, a truth-label input the groomed-mass row does not have. A multilayer perceptron is used because the mass is not linear in the features (ridge regression in table 28). The randomly initialized network shows what a network that never trained exposes. The mass regression in pretraining has the weight $\lambda = 5$ at both label sets, a share of the training loss of 13.78 and 13.87% (162 classes) and 28.65 and 28.67% (17 classes). Mean ± standard deviation over the three pretraining runs; the models with the mass regression have two runs, and both values are given.

	resolution	median response
groomed mass	0.245	0.929
MLP regression on frozen features		
randomly initialized network	0.2421 ± 0.0018	0.983 ± 0.005
188 classes	0.2200 ± 0.0014	0.991 ± 0.005
162 classes	0.2195 ± 0.0017	0.9876 ± 0.0014
43 classes	0.2158 ± 0.0029	0.991 ± 0.005
17 classes	0.2098 ± 0.0031	0.991 ± 0.004
162 classes, mass regression	0.1591, 0.1586	1.004, 1.006
17 classes, mass regression	0.1583, 0.1588	1.009, 1.006

two runs per model; a pair of models with the same loss share at both label sets would test it.

Coarser labels sculpt the QCD mass distribution more, but only for tight selections. The Jensen–Shannon distance between the soft-drop mass distributions of the selected and of all QCD jets is larger for the 17-class model than for the 188-class model at QCD efficiencies from 0.1% to 1.2011% (figure 8(d)). At looser cuts the label sets agree within their spread. At tighter cuts so few QCD jets pass that a random selection of the same number gives a distance almost as large: at the lowest efficiency in figure 8(d), 0.1%, only 410 jets pass, every configuration lies within 1.7 standard deviations of a random selection, and the label sets do not differ beyond their spread. The lower edge of the window is thus set by the size of the selected sample; the upper edge likely marks where a cut starts to reach the jets that the label sets rank differently.

Inside that window the 188-class model already sculpts: at the tighter of the two working points its selection lies at 1.4 times the distance of a random selection of the same size, and the 17-class model's at 1.3 times the distance of the 188-class model's (figure 8(a), (b); table 29). The difference does not come from the mass content of the labels: a class is defined by its decay products, so every class spans the whole mass range, and the reweighting flattens each category in p_{T} and m_{SD}. It lies instead in how each network rejects QCD jets. Without quark-flavor classes, the 17-class model likely relies more on prong structure, which for QCD jets changes with $m_{\mathrm{SD}}/p_{\mathrm{T}}$, and less on flavor and displaced vertices, which change little with the mass. The 43-class model keeps the quark-flavor boundaries that the 17-class level

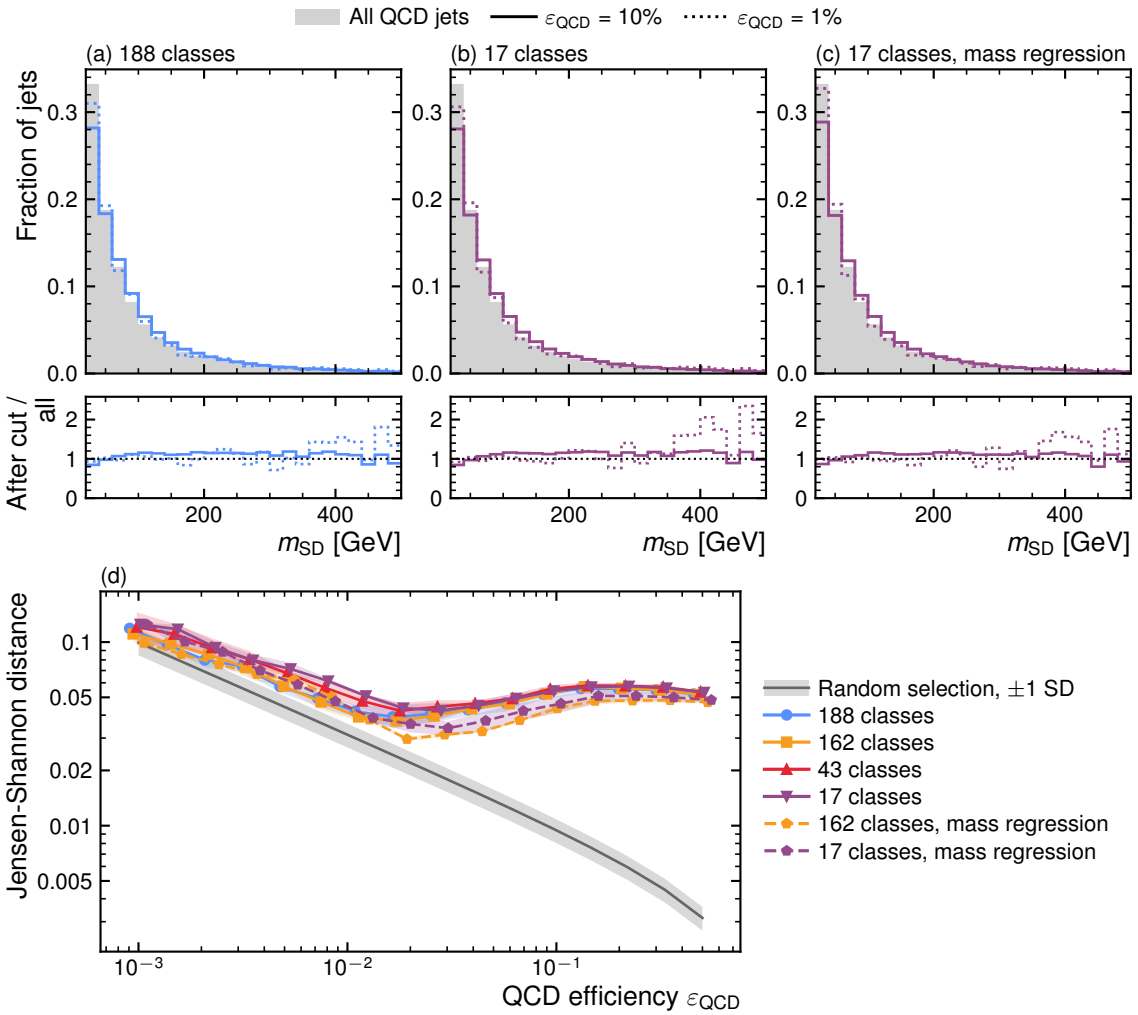


Figure 8: Mass sculpting by a cut on the summed score of the resonant classes, on the QCD jets (jets from light quarks and gluons) of the JetClass-II test set. (a)–(c) Soft-drop mass distribution of all QCD jets (grey, filled) and of the jets passing the cut at QCD efficiencies of 10% (solid) and 1% (dotted), which keep 41,028 and 4,103 jets, for the first pretraining run of the 188-class model (a), the 17-class model (b) and the 17-class model with the auxiliary mass regression (c); the lower panels give the ratio to all QCD jets. (d) Jensen–Shannon distance between the distributions of the passing and of all QCD jets against the QCD efficiency, for the four levels of the class hierarchy and the two models with the auxiliary mass regression: mean over the pretraining runs, with bands of one standard deviation over three runs or the range of two. The grey band is a random selection of the same number of jets (its expected distance and one standard deviation over random selections); at 0.1% every configuration lies within 1.7 standard deviations of it. The layout follows CMS [10, figures 8 and 9], with two working points instead of three, no split in jet p_T, a continuous efficiency axis and the random-selection band added.

removes while merging other distinctions, so it tests this explanation: if flavor is the cause, it sculpts like the 188-class model. Its distance lies between the two, with a spread comparable to their difference (table 29), so the test does not yet decide.

The auxiliary mass regression reduces the sculpting of the 17-class model at the tighter working point: with it, the selection reaches at most 0.82 times the distance of the 17-class model's (figure 8(c), (d)). A likely reason is that features which carry the jet mass let the classification judge prong structure against the mass itself, so the QCD jets it keeps follow their original mass distribution more closely. The 162-class pair tests this, since the same mechanism should act there: with the auxiliary mass regression its selection reaches at most 0.98 times the distance of the 162-class model's (table 29).

4.7 $X \to b\bar{b}$ and $X \to bs$ resonance tagging

The resonance benchmark repeats Sophon's [11, section III] at every label set, on samples of a resonance X decaying to $b\bar{b}$ or to bs, with the background, mass window and metrics of section 3.3; figure 9 gives the resonance mass, the mass window and the numbers of jets. The transfer heads are trained on QCD jets from files that no pretraining run read, drawn to match the signal in p_{T} and m_{SD}, so no jet seen in pretraining enters the benchmark, and every model is scored on the same signal and QCD jets. Figure 9 plots $\epsilon_S/\sqrt{\epsilon_B}$, the reach of a selection without event counts [54], where Sophon plots the significance itself [11, figure 2]; our curves rank the label sets on one background.

Transfer from frozen features tags $X \to bs$ better than a network trained from scratch for the task, at tight working points. Measured by R_{30}, a transfer head on the frozen features of the 188-class model gives **[X->bs fits and from-scratch runs]** than the network trained from scratch on the same jets (figure 9(b)), and the gain depends on the label set: the 17-class features give **[X->bs fits]** than the 188-class ones. The $X \to bq$ output node of the 188-class model, scored against QCD, tags $X \to bs$ worse than the transfer head, so the gain sits in the frozen features rather than in an output the network already had (table 31). An $X \to bs$ jet is told from QCD by two prongs of which exactly one carries a b quark. The 188-class labels separate two-prong decays by the flavor of each prong, while the 17-class labels merge them, so the coarser features likely keep less of the single-b information this signal needs.

Sophon's direct discriminant needs no new training but exists only where $X \to b\bar{b}$ is a class of its own, at 188 and 162 classes (table 1). On the 188-class model it reaches the rejection of a transfer head trained on the same frozen features (**[X->bb fits]**; figure 9(a)), so the output nodes already carry what the head learns for this signal. The transfer head compares every label set, and the 17-class features lose rejection against the 188-class ones (**[X->bb fits]**). The 17-class labels merge $X \to b\bar{b}$ with two-prong decays of other quark flavors, likely the reason, since that merge is also where a linear probe loses most in separating $X \to b\bar{b}$ from $X \to c\bar{c}$ (section 4.1).

[figure F9: not rendered yet]

Figure 9: Resonance tagging of $X \to b\bar{b}$ (a) and $X \to bs$ (b): $\epsilon_S/\sqrt{\epsilon_B}$, proportional to the discovery significance for a small signal, against the efficiency ϵ_B for QCD jets (jets from light quarks and gluons), for jets in the soft-drop mass window [samples not in]–[samples not in] GeV around a resonance mass of [samples not in] GeV. Solid: a transfer head on the frozen features of each label set. (a) Dotted, same color: Sophon's direct discriminant, which only the 188- and 162-class models can form. (b) Black dashed with open squares: the network trained from scratch on the $X \to bs$ training jets; grey dotted: the $X \to bq$ output node of the 188-class model scored against QCD, which tests whether the gain is already in a pretrained output. No network is trained from scratch for $X \to b\bar{b}$. Bands show one standard deviation over the three pretraining runs (from scratch: three training seeds). Every model is scored on the same [samples not in] $X \to b\bar{b}$ and [samples not in] $X \to bs$ jets and [samples not in] QCD jets; the heads are trained on [samples not in] ($X \to b\bar{b}$) and [samples not in] ($X \to bs$) jets per class, repeated [samples not in] and [samples not in] times to reach Sophon's training length. Compared with Sophon [11, figure 2], the vertical axis replaces the discovery significance at a fixed luminosity, the panels sit side by side, and the efficiency axis ends where too few QCD jets pass to draw a curve.

5 Discussion and conclusion

Frozen features keep what the pretraining labels separate, and the cost of a coarse label set sits where its merges remove flavor. On the pretraining classes a linear probe separates the 188 classes less well at every coarser level of the class hierarchy, and the step that removes quark flavor costs the most (figure 2). Merging the QCD classes costs only the QCD classes. Random partitions into 17 classes, built to cut across the physical groups, lose less than the physical 17-class label set, so the class count alone does not set the cost. Per class, decays merged with their lepton-flavor counterparts pay the most, while decays merged with their quark-flavor counterparts carry the largest summed cost (figure 3). A single flavor boundary placed at random among the four-quark decays keeps flavor readable even for pairs it does not cut (figure 4). Fine-tuning does not remove the effect where every input is present: on JetClass the 17-class model still rejects less background than the 188-class model in several signal classes (table 2). There the random partitions recover only part of that loss: they come close to the 188-class model on two-prong decays and stay near the 17-class model on the three- and four-prong classes, whose prong-count boundaries they mix (table 22). On top tagging and quark/gluon discrimination, which lack some inputs, the fine-tuned label sets lie much closer together (tables 3 and 4). Linear probes on their frozen features gain little over the randomly initialized network, and the substitutions for the missing inputs separate the part of that shortfall set by the zero value from the part set by the missing information (figure 5). The two effects act on one part of the input. Flavor reaches the network through the displacement and particle-type inputs, because b and c hadrons travel a few millimeters before they decay and leptons are identified by type. These are the inputs the benchmarks lack, so a label-set difference that lives in flavor likely has little to act on there.

The results translate into concrete choices for a pretraining label set. Keep ev-

ery distinction that a downstream discriminant sums over, because a merge removes the discriminant from the output layer whatever the frozen features keep: Sophon's $X \to b\bar{b}$ discriminant cannot be formed from the 43-class labels, and the D_{bc} discriminant of the boosted $|V_{cb}|$ measurement exists only at 188 and 162 classes (table 1). Keep the flavor boundaries, which the pretraining classes and JetClass both reward (figures 2–4 and table 2). Keep the QCD classes apart where a downstream task separates QCD jets by flavor: merging them leaves the resonant classes unchanged but costs the QCD pairs (table 26) and the one discriminant that sums QCD classes (table 1). Add an auxiliary mass regression: it puts the jet mass into the frozen features and lowers the sculpting of the QCD mass distribution at a QCD efficiency of 1% (table 5 and figures 7 and 8). Its cost to the classification of the pretraining classes is small; after fine-tuning on top tagging the 162-class models with it show no consistent change and the 17-class models lose some rejection, which two runs per model do not yet resolve (section 4.6). For benchmarks with missing inputs, compare label sets by fine-tuning with the missing input features set to zero, which gives every label set identical inputs; input dropout during pretraining, which replaces some inputs with zeros, is the published alternative [30, 29]. Report the variation between pretraining runs, because the label-set effects here are measured against it (section 3.4) and a single pretrained model cannot tell the two apart.

The study has four limitations. First, all pretraining and every evaluation but one use a single simulation family; the Herwig quark/gluon sample changes the generator for one task only. Second, one architecture and one model size are studied. Third, each level of the class hierarchy has three pretraining runs and every other configuration fewer. Fourth, the networks are initialized with a different scheme from Sophon's code, so their absolute performance need not match Sophon's published model; every comparison is between models trained here.

Which distinctions a pretraining label set keeps, more than how many classes it has, shapes which discriminants an analysis can still form and how well frozen features separate decays by flavor. A label set built for transfer should therefore keep the classes that analyses sum into discriminants and its flavor boundaries, and an auxiliary target such as the jet mass can add to the frozen features a quantity that no classification label rewards, for little loss in classification.

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We used Claude (Anthropic), through its Claude Code command-line tool, to write and test analysis, training and plotting code, to check statements about cited papers against their full texts, and to draft and edit the manuscript text under the authors' direction. The output was verified in three ways: every number in the paper is written by a script from the result files and none is typed by hand, the code is covered by automated tests that are run before every analysis, and every entry of the reference list was checked against its INSPIRE, arXiv or publisher record. The authors read and revised all text and take full responsibility for it.

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Data availability

The pretraining dataset JetClass-II [13] is public, as are the downstream datasets: JetClass [60], the top-quark tagging reference dataset [61] of the landscape study [20], and the Pythia and Herwig quark/gluon samples [22, 23]. The $X \to b\bar{b}$ and $X \to bs$ samples are held by [holder of the X to bb and X to bs samples and how to obtain them]. The pretrained models, the per-jet scores of every evaluation and the result files behind every number are archived at [DOI of the models and results archive]. The code, including the label maps of every label set, is archived at [DOI of the code archive of the released version]; the network and its training follow Sophon's released code [46]. Every number in the paper is written from those result files by a script that records the hash of each input file beside the number. Pretraining every model reported here took 3,435 GPU-hours, training and validation passes included, on NVIDIA L4, RTX 3090 and V100 GPUs; the GPU types and hours of each run are listed in table 9.

A Label sets

To give what each JetClass-II class becomes at every label set, table 6 lists the levels of the class hierarchy and the groups of the 17-class label set with the number of classes each merges, and table 7 gives the group of every class at every level of the hierarchy, in each random partition and in each random partition with controlled flavor boundaries. The label maps behind both tables, one row per class and one column per label set, are part of the code release.

Table 6: The levels of the class hierarchy, finest first: the number of classes, what each level labels, and whether models were pretrained on it. Below, the groups of the 17-class label set, by the group number used in Table 7, with the number of JetClass-II classes each merges. ℓ is an electron or a muon; τ_ℓ a leptonic τ decay. A four-prong group with a neutrino has three visible prongs.

classes	labels	pretrained
188	the JetClass-II classes	yes
162	one QCD class; every resonant class kept	yes
64	each decay topology by its numbers of b and c quarks	no
43	each decay topology by whether it holds a b quark, else a c, else neither	yes
30	each decay topology by whether it holds a b or c quark	no
17	the number of prongs and whether each is a quark or gluon (without distinguishing them), an electron or muon, a τ lepton or a neutrino, without flavor	yes
4	two, three or four visible prongs (a neutrino is not counted), or QCD	no
2	resonance decay or QCD	no
the 17 classes of the 17-class label set		
0	two-prong: two quarks or gluons	10 classes
3	two-prong: $\ell\ell$	2 classes
2	two-prong: τ_h, τ_ℓ	2 classes
1	two-prong: τ_h, τ_h	1 class
9	four-prong: four quarks or gluons	27 classes
11	four-prong: two quarks or gluons, $\ell\ell$	10 classes
12	four-prong: two quarks or gluons, τ_h, τ_ℓ	10 classes
10	four-prong: two quarks or gluons, τ_h, τ_h	5 classes
4	three-prong: three quarks or gluons	29 classes
6	three-prong: two quarks or gluons, ℓ	10 classes
7	three-prong: one quark or gluon, $\ell\ell$	10 classes
8	three-prong: one quark or gluon, τ_h, τ_ℓ	10 classes
5	three-prong: one quark or gluon, τ_h, τ_h	5 classes
13	four-prong: two quarks or gluons, ℓ, ν	12 classes
14	four-prong: two quarks or gluons, τ_ℓ, ν	12 classes
15	four-prong: two quarks or gluons, τ_h, ν	6 classes
16	QCD	27 classes

Table 7: Every JetClass-II class and its group at each coarser level of the class hierarchy (columns headed by the number of classes at that level: 64, 43, 30, 17), in each random partition, numbered as in Table 26, and in each random partition with controlled flavor boundaries: no flavor boundary, a b-quark boundary and a random flavor boundary (Section 3.2). Group numbers are those of the released label maps; classes that share a number in a column share a class in that label set. At the 162-class level every resonant class keeps its own label and the QCD classes share one; at 4 classes the groups are decays with two, three or four visible prongs (a neutrino is not counted) and QCD. Rows are ordered by the 17-class group (Table 6), with a gap between groups. q is a u or d quark; τ_h a hadronic and τ_e, τ_μ leptonic τ decays.

	level				random partition					flavor boundary		
JetClass-II class	64	43	30	17	1	2	3	4	5	none	b quark	random
$X \to bb$	0	0	0	0	0	9	0	9	0	0	0	0
$X \to bc$	1	0	0	0	0	0	0	9	0	0	0	0
$X \to bq$	2	0	0	0	0	0	9	0	0	0	0	0
$X \to cc$	3	1	0	0	0	0	0	0	0	0	0	0
$X \to cs$	4	1	0	0	0	9	9	0	0	0	0	0
$X \to cq$	4	1	0	0	9	0	9	0	0	0	0	0
$X \to ss$	5	2	1	0	9	9	9	9	0	0	0	0
$X \to qq$	5	2	1	0	0	0	9	0	0	0	0	0
$X \to sq$	5	2	1	0	9	0	9	0	0	0	0	0
$X \to gg$	5	2	1	0	3	3	9	13	9	15	15	15
$X \to ee$	8	5	4	3	11	9	13	11	0	0	0	0
$X \to \mu\mu$	8	5	4	3	11	9	9	11	9	11	9	9
$X \to \tau_h\tau_e$	7	4	3	2	15	3	12	12	1	4	4	4
$X \to \tau_h\tau_\mu$	7	4	3	2	4	4	10	10	12	0	0	0
$X \to \tau_h\tau_h$	6	3	2	1	4	10	12	3	4	7	7	7
$X \to YY \to bbbb$	31	21	15	9	0	9	11	0	11	9	11	9
$X \to YY \to bbcc$	32	21	15	9	9	0	11	0	9	9	11	9
$X \to YY \to bbss$	33	21	15	9	9	0	0	0	9	9	11	11
$X \to YY \to bbqq$	33	21	15	9	9	0	11	0	9	9	11	11
$X \to YY \to bbgg$	33	21	15	9	15	15	3	3	13	13	13	13
$X \to YY \to qqqb$	34	21	15	9	0	0	0	0	9	9	11	11
$X \to YY \to qqbs$	34	21	15	9	0	9	11	9	11	9	11	9
$X \to YY \to bbcq$	35	21	15	9	9	0	11	9	11	9	11	11
$X \to YY \to ccbs$	36	21	15	9	0	9	0	0	9	9	11	9
$X \to YY \to ccbq$	36	21	15	9	0	9	11	0	11	9	11	11
$X \to YY \to qqbc$	37	21	15	9	0	9	0	9	9	9	11	11
$X \to YY \to bcsq$	37	21	15	9	0	9	0	0	11	9	11	9
$X \to YY \to cccc$	38	22	15	9	9	9	0	9	9	9	9	9
$X \to YY \to ccss$	39	22	15	9	0	0	0	0	11	9	9	11
$X \to YY \to ccqq$	39	22	15	9	9	9	11	9	11	9	9	11
$X \to YY \to ccgg$	39	22	15	9	10	12	4	13	4	13	13	13
$X \to YY \to ccsq$	39	22	15	9	9	0	0	9	9	9	9	9
$X \to YY \to qqqc$	40	22	15	9	0	9	0	0	11	9	9	11
$X \to YY \to sscq$	40	22	15	9	9	0	0	9	9	9	9	11
$X \to YY \to qqcs$	40	22	15	9	0	9	11	9	11	9	9	9
$X \to YY \to ssss$	41	23	16	9	9	0	11	9	11	9	9	11
$X \to YY \to ssqq$	41	23	16	9	9	0	0	9	9	9	9	9
$X \to YY \to ssgg$	41	23	16	9	3	15	3	7	13	13	13	13
$X \to YY \to qqqq$	41	23	16	9	0	9	11	9	9	9	9	9
$X \to YY \to qqgg$	41	23	16	9	12	10	13	12	12	13	13	13
$X \to YY \to gggg$	41	23	16	9	4	14	12	12	9	4	4	4

JetClass-II class	level				random partition					flavor boundary		
	64	43	30	17	1	2	3	4	5	none	b quark	random
$X \to YY \to qqqs$	41	23	16	9	9	0	11	0	11	9	9	9
$X \to YY \to bbee$	45	27	19	11	9	12	0	4	11	3	3	3
$X \to YY \to bb\mu\mu$	45	27	19	11	11	11	0	11	9	11	9	9
$X \to YY \to ccee$	46	28	19	11	11	11	0	11	9	3	3	3
$X \to YY \to cc\mu\mu$	46	28	19	11	11	11	11	4	9	11	9	9
$X \to YY \to ssee$	47	29	20	11	9	13	0	9	9	3	3	3
$X \to YY \to ss\mu\mu$	47	29	20	11	11	13	0	11	13	11	9	9
$X \to YY \to qqee$	47	29	20	11	11	11	11	11	11	3	3	3
$X \to YY \to qq\mu\mu$	47	29	20	11	11	11	0	9	9	11	9	9
$X \to YY \to ggee$	47	29	20	11	9	11	9	9	4	4	4	4
$X \to YY \to gg\mu\mu$	47	29	20	11	10	11	3	11	3	9	9	9
$X \to YY \to bb\tau_h\tau_e$	48	30	21	12	14	13	3	10	12	13	13	13
$X \to YY \to bb\tau_h\tau_\mu$	48	30	21	12	4	9	3	15	7	4	4	4
$X \to YY \to cc\tau_h\tau_e$	49	31	21	12	15	2	10	1	9	13	13	13
$X \to YY \to cc\tau_h\tau_\mu$	49	31	21	12	12	3	4	10	0	4	4	4
$X \to YY \to ss\tau_h\tau_e$	50	32	22	12	15	10	12	6	12	13	13	13
$X \to YY \to ss\tau_h\tau_\mu$	50	32	22	12	12	3	12	6	14	4	4	4
$X \to YY \to qq\tau_h\tau_e$	50	32	22	12	2	4	4	12	1	13	13	13
$X \to YY \to qq\tau_h\tau_\mu$	50	32	22	12	4	0	12	12	4	4	4	4
$X \to YY \to gg\tau_h\tau_e$	50	32	22	12	12	4	10	1	14	1	1	1
$X \to YY \to gg\tau_h\tau_\mu$	50	32	22	12	2	6	2	3	12	14	14	14
$X \to YY \to bb\tau_h\tau_h$	42	24	17	10	4	7	4	14	15	11	9	9
$X \to YY \to cc\tau_h\tau_h$	43	25	17	10	12	13	4	3	15	11	9	9
$X \to YY \to ss\tau_h\tau_h$	44	26	18	10	13	4	15	2	15	11	9	9
$X \to YY \to qq\tau_h\tau_h$	44	26	18	10	13	11	4	9	10	11	9	9
$X \to YY \to gg\tau_h\tau_h$	44	26	18	10	1	11	7	13	10	1	1	1
$X \to YY \to bbb$	9	6	5	4	7	1	3	15	10	10	10	10
$X \to YY \to bbc$	10	6	5	4	4	1	13	10	1	10	10	10
$X \to YY \to bbs$	11	6	5	4	3	4	10	15	6	10	10	10
$X \to YY \to bbq$	11	6	5	4	4	15	10	6	1	10	10	10
$X \to YY \to bbg$	11	6	5	4	6	0	1	14	13	14	14	14
$X \to YY \to ccb$	12	6	5	4	6	12	14	3	2	10	10	10
$X \to YY \to ssb$	13	6	5	4	10	4	13	10	6	10	10	10
$X \to YY \to qqb$	13	6	5	4	9	12	12	10	3	10	10	10
$X \to YY \to ggb$	13	6	5	4	12	7	7	1	13	6	6	6
$X \to YY \to bsq$	13	6	5	4	3	10	12	15	7	10	10	10
$X \to YY \to bcs$	14	6	5	4	1	13	15	3	10	10	10	10
$X \to YY \to bcq$	14	6	5	4	9	12	3	3	10	10	10	10
$X \to YY \to ccc$	15	7	5	4	3	1	10	3	6	10	10	10
$X \to YY \to ccs$	16	7	5	4	7	9	13	12	7	10	10	10
$X \to YY \to ccq$	16	7	5	4	7	4	10	10	7	10	10	10
$X \to YY \to ccg$	16	7	5	4	12	6	14	12	15	14	14	14
$X \to YY \to ssc$	17	7	5	4	10	13	4	12	10	10	10	10
$X \to YY \to qqc$	17	7	5	4	3	1	4	13	15	10	10	10
$X \to YY \to ggc$	17	7	5	4	6	2	1	14	0	6	6	6
$X \to YY \to csq$	17	7	5	4	7	1	15	15	6	10	10	10
$X \to YY \to sss$	18	8	6	4	9	12	9	10	3	10	10	10
$X \to YY \to ssq$	18	8	6	4	3	1	6	10	15	10	10	10
$X \to YY \to ssg$	18	8	6	4	6	10	14	12	14	14	14	14
$X \to YY \to qqs$	18	8	6	4	4	15	10	10	6	10	10	10
$X \to YY \to qqq$	18	8	6	4	4	4	13	3	7	10	10	10
$X \to YY \to qqg$	18	8	6	4	12	7	7	13	0	14	14	14
$X \to YY \to ggs$	18	8	6	4	12	6	14	1	13	6	6	6
$X \to YY \to ggq$	18	8	6	4	1	2	2	14	4	6	6	6
$X \to YY \to ggg$	18	8	6	4	14	14	6	9	2	2	2	2

JetClass-II class	level				random partition					flavor boundary		
	64	43	30	17	1	2	3	4	5	none	b quark	random
$X \to YY \to bbe$	22	12	9	6	7	12	10	10	10	14	14	14
$X \to YY \to bb\mu$	22	12	9	6	12	12	12	10	4	0	0	0
$X \to YY \to cce$	23	13	9	6	14	4	1	3	13	14	14	14
$X \to YY \to cc\mu$	23	13	9	6	10	0	14	10	13	0	0	0
$X \to YY \to sse$	24	14	10	6	12	4	14	15	10	14	14	14
$X \to YY \to ss\mu$	24	14	10	6	4	10	6	12	0	0	0	0
$X \to YY \to qqe$	24	14	10	6	7	6	1	15	15	14	14	14
$X \to YY \to qq\mu$	24	14	10	6	6	6	2	10	14	0	0	0
$X \to YY \to gge$	24	14	10	6	3	7	7	12	15	2	2	2
$X \to YY \to gg\mu$	24	14	10	6	6	2	14	1	13	2	2	2
$X \to YY \to bee$	25	15	11	7	14	10	1	10	5	2	2	2
$X \to YY \to b\mu\mu$	25	15	11	7	4	4	6	10	2	4	4	4
$X \to YY \to cee$	26	16	11	7	10	10	6	10	4	2	2	2
$X \to YY \to c\mu\mu$	26	16	11	7	5	10	1	10	15	4	4	4
$X \to YY \to see$	27	17	12	7	14	14	6	9	15	2	2	2
$X \to YY \to qee$	27	17	12	7	10	10	4	10	2	2	2	2
$X \to YY \to gee$	27	17	12	7	10	14	5	9	2	5	5	5
$X \to YY \to s\mu\mu$	27	17	12	7	10	14	5	9	15	4	4	4
$X \to YY \to q\mu\mu$	27	17	12	7	5	10	6	4	15	4	4	4
$X \to YY \to g\mu\mu$	27	17	12	7	14	5	1	9	15	5	5	5
$X \to YY \to b\tau_h\tau_e$	28	18	13	8	8	5	1	5	2	8	8	8
$X \to YY \to b\tau_h\tau_\mu$	28	18	13	8	10	5	8	8	5	8	8	8
$X \to YY \to c\tau_h\tau_e$	29	19	13	8	10	5	1	5	2	8	8	8
$X \to YY \to c\tau_h\tau_\mu$	29	19	13	8	14	14	1	5	2	8	8	8
$X \to YY \to s\tau_h\tau_e$	30	20	14	8	10	14	1	5	5	8	8	8
$X \to YY \to q\tau_h\tau_e$	30	20	14	8	10	8	1	5	5	8	8	8
$X \to YY \to g\tau_h\tau_e$	30	20	14	8	14	8	1	8	8	8	8	8
$X \to YY \to s\tau_h\tau_\mu$	30	20	14	8	14	14	1	5	5	8	8	8
$X \to YY \to q\tau_h\tau_\mu$	30	20	14	8	14	5	1	5	8	8	8	8
$X \to YY \to g\tau_h\tau_\mu$	30	20	14	8	8	14	8	5	2	8	8	8
$X \to YY \to b\tau_h\tau_h$	19	9	7	5	8	14	8	8	8	4	4	4
$X \to YY \to c\tau_h\tau_h$	20	10	7	5	14	8	8	5	8	4	4	4
$X \to YY \to s\tau_h\tau_h$	21	11	8	5	8	8	8	8	8	4	4	4
$X \to YY \to q\tau_h\tau_h$	21	11	8	5	10	5	1	8	5	4	4	4
$X \to YY \to g\tau_h\tau_h$	21	11	8	5	8	8	1	5	2	4	4	4
$X \to YY \to bce\nu$	51	33	23	13	9	0	10	11	3	11	11	11
$X \to YY \to bc\mu\nu$	51	33	23	13	9	13	0	4	13	9	9	9
$X \to YY \to bqe\nu$	52	33	23	13	0	4	13	11	12	11	11	11
$X \to YY \to bq\mu\nu$	52	33	23	13	15	13	10	11	3	9	9	9
$X \to YY \to cse\nu$	53	34	23	13	13	15	0	0	3	11	11	11
$X \to YY \to cqe\nu$	53	34	23	13	15	3	15	15	10	11	11	11
$X \to YY \to cs\mu\nu$	53	34	23	13	9	9	15	3	3	9	9	9
$X \to YY \to cq\mu\nu$	53	34	23	13	15	4	10	3	13	9	9	9
$X \to YY \to sqe\nu$	54	35	24	13	13	0	15	15	3	11	11	11
$X \to YY \to qqe\nu$	54	35	24	13	13	4	15	4	3	11	11	11
$X \to YY \to sq\mu\nu$	54	35	24	13	13	13	0	3	3	9	9	9
$X \to YY \to qq\mu\nu$	54	35	24	13	9	11	3	4	13	9	9	9
$X \to YY \to bc\tau_e\nu$	55	36	25	14	7	0	7	14	14	6	6	6
$X \to YY \to bc\tau_\mu\nu$	55	36	25	14	6	12	12	3	10	9	9	9
$X \to YY \to bq\tau_e\nu$	56	36	25	14	14	10	6	1	4	6	6	6
$X \to YY \to bq\tau_\mu\nu$	56	36	25	14	6	7	2	10	4	9	9	9
$X \to YY \to cs\tau_e\nu$	57	37	25	14	4	6	14	12	14	6	6	6
$X \to YY \to cq\tau_e\nu$	57	37	25	14	3	4	14	15	15	6	6	6
$X \to YY \to cs\tau_\mu\nu$	57	37	25	14	7	12	2	13	0	9	9	9
$X \to YY \to cq\tau_\mu\nu$	57	37	25	14	6	6	1	10	15	9	9	9

	level				random partition					flavor boundary		
JetClass-II class	64	43	30	17	1	2	3	4	5	none	b quark	random
$X \to YY \to sq\tau_e\nu$	58	38	26	14	1	4	14	12	10	6	6	6
$X \to YY \to qq\tau_e\nu$	58	38	26	14	12	2	14	10	15	6	6	6
$X \to YY \to sq\tau_\mu\nu$	58	38	26	14	10	10	10	10	13	9	9	9
$X \to YY \to qq\tau_\mu\nu$	58	38	26	14	12	0	1	15	13	9	9	9
$X \to YY \to bc\tau_h\nu$	59	39	27	15	0	11	3	9	9	12	12	12
$X \to YY \to bq\tau_h\nu$	60	39	27	15	13	11	15	15	9	12	12	12
$X \to YY \to cs\tau_h\nu$	61	40	27	15	13	11	13	9	3	12	12	12
$X \to YY \to cq\tau_h\nu$	61	40	27	15	13	3	0	0	9	12	12	12
$X \to YY \to sq\tau_h\nu$	62	41	28	15	13	11	9	0	10	12	12	12
$X \to YY \to qq\tau_h\nu$	62	41	28	15	0	15	0	15	12	12	12	12
QCD, *bbccss*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *bbccs*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *bbcc*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *bbcss*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *bbcs*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *bbc*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *bbss*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *bbs*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *bb*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *bccss*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *bccs*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *bcc*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *bcss*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *bcs*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *bc*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *bss*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *bs*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *b*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *ccss*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *ccs*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *cc*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *css*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *cs*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *c*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *ss*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *s*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, light	63	42	29	16	16	16	16	16	16	16	16	16

Three rules fixed before any pretraining select the random partitions. A draw assigns the resonant classes at random to as many groups as the 17-class level has resonant groups, plus one group for all QCD jets, and matches each group to a resonant group of the 17-class level whose share of the reweighted training jets it reproduces within 5%, so that every group carries the weight in the loss of the group it replaces. Each of 32 draws is then refined by a search that lowers the number of class pairs it merges that the 17-class level also merges, and the draw with the fewest such pairs becomes a candidate. Of 6,600 candidates built from consecutive seeds, 27 keep every realized share within the tolerance after the refinement. The five partitions are a random sample of these that meets a balance rule: each of seven class pairs, among them the pairs of the linear probes on class pairs, is merged in two or three partitions and kept apart in the others. The balance rule makes every one of these pairs available both merged and kept apart, which the comparison of merged against separated classes needs. Because the refinement pushes the partitions away

from the grouping by decay topology, a group can mix decays with different numbers of prongs and the partitions keep many flavor boundaries; table 25 lists the prong-count boundaries each one keeps. Each partition is pretrained once, the first and third with the initial weights and training examples of seed one and the other three with those of seed two, and is compared with the 188-class model of its seed.

The three random partitions with controlled flavor boundaries share one grouping and differ only in where the four-quark decays $X \to YY \to QQQQ$ go. In the partition with no flavor boundary, all decays that differ only in quark flavor share a group. The partition with a b-quark boundary starts from it and moves the 11 of the 22 four-quark decays that contain a b quark into another group, while the 9 decays of the forms $X \to YY \to QQ\mu\mu$, $X \to YY \to QQ\tau_h\tau_h$ and $X \to \mu\mu$ move in the opposite direction, so that every group keeps its share exactly. Its only flavor boundary lies between the four-quark decays with and without a b quark, and it keeps 8 of the 11 pairs of four-quark decays that differ by exchanging b and c quarks apart. The partition with a random flavor boundary makes the same compensating move but moves 11 four-quark decays drawn at random, 6 of them with a b quark, under rules fixed before the draw: about as many of the moved decays contain a b quark as do not; no pair that differs by exchanging b and c quarks is kept apart; the pairs of the balance rule are merged or kept apart as in the partition with no flavor boundary; and every group's share of the training examples stays within 1% of that in the partition with a b-quark boundary. It therefore keeps none of the b–c exchange pairs apart, and the two partitions differ in whether the moved decays are chosen by flavor, not in how many move. Each of the three is pretrained with two seeds, one and two.

B Training settings

The settings follow the order of Sophon's appendix B [11]: architecture, inputs, reweighting, optimization and training length, then the checkpoint rule. Each subsection after the first gives only what differs from pretraining, so that every model can be rebuilt from the data and the code release.

B.1 Pretraining at every label set

Every model is the ParT network [8] in the configuration of Sophon's released code [46], as implemented in weaver-core 0.4.17, the training framework of ParT and Sophon. It has 8 particle-attention blocks with a pairwise interaction bias, then 2 class-attention blocks, with 128-dimensional particle representations and 8 attention heads. The head has one hidden layer of 512 units with a ReLU activation and dropout of 0.1 after it, followed by an output layer as wide as the label set. Where Sophon's paper and its released code differ, in the number of particle-attention blocks (section 3.1) and in the paper's GELU activation throughout [11, App. B.1], we follow the code.

The initial weights follow PyTorch's default initialization, except for the class token, which is drawn from a narrow truncated normal distribution. The weaver-core version used by Sophon's code instead draws the linear layers from a Xavier-uniform distribution with zero biases and rescales the residual projections by depth; we did not reproduce that initialization (section 3.1).

Each particle carries 17 input features: 7 computed from its four-momentum and that of the jet, its electric charge, five binary flags for its type (charged hadron, neutral hadron, photon, electron and muon) and four impact-parameter features, the transverse and longitudinal impact parameters with their uncertainties. The pairwise features of the attention bias are computed from the four-momenta, which are rescaled so that every jet has the same transverse momentum, as in Sophon [11, Sec. II]. The features first pass a BatchNorm layer whose running mean and variance are accumulated in pretraining; the substitutions of appendix C are defined with respect to this normalization.

The training jets are drawn with Sophon's reweighting in $(p_{\mathrm{T}}, m_{\mathrm{SD}})$ within each of its reweighting categories of the 188 classes [11, Sec. II], with the same weights for every label set.

Training proceeds in cycles of five epochs. In each cycle the rows of every training file are divided at random into five disjoint parts, and each epoch reads one part, or 20% of the rows, of nearly every file; a row is therefore read in one epoch per cycle, and within that epoch the reweighting can draw it several times. The order of the training examples and the seeds of the dropout masks are functions of the seed and the epoch number alone, which is what makes models with the same seed see the same training examples in the same order whatever their labels.

Every model is trained for 80 epochs of 10,240,000 jets, 8.192×10^8 training examples in all, in batches of 512 with mixed precision. Sophon's paper gives 10 million jets per epoch [11, App. B.1]; the exact number is that of its code [46]. The optimizer is Ranger: Lookahead [47] with $k = 6$ and $\alpha = 0.5$ over RAdam [48] with $\beta_1 = 0.95$, $\beta_2 = 0.999$ and ϵ of 10^{-5}, the values of Sophon [11, App. B.1]. There is no weight decay, as in Sophon's code; Sophon applies weight decay only to its ParT reference [11, App. B.3]. The learning rate is 5×10^{-4} for the first 70% of the epochs. From epoch 56, counting from zero, it falls by a constant factor at every epoch, reaching 5×10^{-6}, 1% of its initial value, in the last epoch.

The validation accuracy is computed after every epoch on every selected jet of 25 validation files that no run trains on. The checkpoint evaluated throughout this paper is the first maximum of the validation accuracy among epochs 70–79, where the learning rate is lowest, so that every model is read at the same stage of its schedule; Sophon keeps the checkpoint with the highest validation accuracy [11, App. B.1]. For the check of appendix D.2, a second copy of each selected checkpoint has its BatchNorm statistics recomputed on 200,000 training jets, and the saved initial weights of the randomly initialized network are treated the same way.

B.2 Auxiliary mass regression and kinematic inputs only

The models with the auxiliary mass regression differ from the others by one output, which regresses the target y of equation (1) with a log-cosh loss, and one loss term. A jet without a generator-level match has a stored generator mass of exactly zero and is left out of the regression term, though not out of the classification. The total loss is the classification loss plus $\lambda = 5$ times the regression loss at both label sets. Every other setting is that of appendix B.1, and each of these models shares its initial weights, except the output layer, and its training examples with the model of the same label set and seed trained without the regression.

The 188-class model with kinematic inputs only keeps the 7 kinematic features

of each particle and the pairwise features, and drops the charge, the type flags and the impact-parameter features. It is trained once, on the training examples of seed one in their order and with the settings of appendix B.1. Its first layer is narrower than that of the other models, so its initial weights are its own.

B.3 Linear and MLP probes

Each linear probe is fitted as a multinomial logistic regression on the standardized frozen features by full-batch L-BFGS in double precision. A fit counts as converged when the largest component of its gradient falls below 1×10^{-5}, and its inverse regularization strength is chosen on validation jets from a logarithmic grid between 1×10^{-3} and 10^4. The frozen features are stored in half precision. Each MLP probe has one hidden layer of 256 units and is trained on the same frozen features at the largest training set of each task; its result is the average over three seeds.

On JetClass-II the linear probes are scored on a class-balanced test set of 903,922 jets from the test files: 5,000 jets of each class, or every selected jet of the 11 classes that have fewer, the rarest with 224. The training sets hold equal numbers of jets per class from files that no pretraining run trained or validated on, and they are nested from 10^3 to 10^6 jets, each contained in the next. At 10^6 jets, 11 classes have fewer held-out jets than their share and contribute all they have. The regularization is chosen on up to 109 validation jets per class. The AUC averaged over classes takes each of the 188 classes against the rest on every test jet, and table 17 also gives it over the classes with a full test sample and over the resonant and the QCD classes.

The linear probes on class pairs are binary. They are trained on up to 10^5 jets per side and tested on up to 2×10^5, each side at its natural composition, and choose their regularization on 19,759 to 20,195 jets held out from the training jets.

On JetClass the linear probes are trained on up to 1,000,000 jets, choose their regularization on 200,000 validation jets and are scored on 2,000,000 test jets, taken from the first test files of each class with equal numbers per class; the AUC averaged over pairs of classes uses 2,000,000 of them. On top tagging the largest training set holds 1,200,000 jets and the test set 404,000; on quark/gluon discrimination they hold 1,600,000 Pythia jets and 200,000. The Herwig jets are only scored, by the linear probe trained on Pythia jets. On both benchmarks the missing input features are substituted as appendix C describes.

The full-set linear probe on top is not limited by the number of training jets: for the 188-class model, its rejection at 30% signal efficiency with 10^5 training jets and with the full set agree within the spread over pretraining runs (table 18).

B.4 Fine-tuning and training from scratch

Each pretrained model is fine-tuned once per task and training-set size, from its selected checkpoint, whose hash every fine-tuning run records. The whole head, hidden layer included, is initialized afresh, with dropout 0.1 after the hidden layer. Every label set therefore starts fine-tuning from its class token with the same head, whereas a pretrained hidden layer was trained to serve its own label set's outputs. The optimizer is Ranger, as in pretraining, with mixed precision, and the missing input features of top and quark/gluon are set to zero. All runs of one task and size

Table 8: Learning-rate choice at 10^4 training jets: the best validation accuracy of each rate on top and quark/gluon and their sum. *: the rate chosen, the largest sum; the test jets play no part.

network	learning rate	top	quark/gluon	sum
pretrained, backbone	10^{-4}	0.9204	0.8204	1.741
pretrained, backbone	3×10^{-4}	0.9192	0.8202	1.739
pretrained, backbone	10^{-3}	0.9212	0.8211	1.742*
from scratch	3×10^{-4}	0.9152	0.8052	1.720
from scratch	10^{-3}	0.9188	0.8029	1.722
from scratch	3×10^{-3}	0.9177	0.8059	1.724*

use the same training jets, so the three networks trained from scratch differ only in their training seed.

On the full training sets of top and quark/gluon, and on 10^6 JetClass jets, fine-tuning uses ParT's settings [8, Sec. 5.2]. The pretrained weights train at a constant learning rate of 10^{-4} and the head at 50 times that rate, with weight decay 0.01 and batches of 512 jets. Training lasts 20 epochs on top and quark/gluon and 10 on JetClass, where the head's learning rate falls to 1% of its value over the last 30% of the epochs. The epoch with the highest validation accuracy is kept, the earliest in a tie. The networks trained from scratch on these sets follow ParT's settings for training from scratch [8, Secs. 5.1 and 5.2]: a learning rate of 10^{-3} for every weight, constant and then lowered to 1% of its value over the last 30% of the epochs; weight decay 0.01 on top and quark/gluon and none on JetClass; and the epochs and batches of fine-tuning. The kinematic-only model is fine-tuned on top with the same settings and has no missing input.

Below the full training sets, and on JetClass-II at every size, pretrained networks and networks trained from scratch follow one rule. They train in batches of 512 jets for at most 20,000 steps and are evaluated on a fixed validation set every 50 steps below 10^5 training jets and every 500 steps at and above that size. The validation set holds 200,000 jets, or on JetClass-II the 20,480 jets of the class-balanced validation set. Training stops after 10 evaluations without improvement, and the checkpoint with the best validation metric is kept. The pretrained weights train at a constant learning rate of 10^{-3} and the head at 50 times that rate, at most 5×10^{-3}. The networks trained from scratch train every weight at 3×10^{-3} with the training framework's default schedule, which is constant and lowered only near the end of the maximum training length. The weight decay is 0.01 throughout. Top and quark/gluon are fine-tuned on 10^3, 10^4 and 10^5 jets below their full sets; JetClass on 10^4 and 10^5; and class-balanced JetClass-II on 10^4, 10^5 and 10^6. On JetClass-II the fine-tuned models are the levels of the class hierarchy and the models with the auxiliary mass regression.

The two learning rates of these smaller sets were chosen on validation (table 8). The 188-class model of seed one was fine-tuned with its pretrained weights at 10^{-4}, 3×10^{-4} and 10^{-3}, and the network trained from scratch at 3×10^{-4}, 10^{-3} and 3×10^{-3}, each on 10,000 jets of top and of quark/gluon. The rate with the largest best validation accuracy summed over the two tasks was kept; the test jets play no part. In both grids the chosen rate is the largest one swept, so the sweep does not bracket a maximum, and the summed accuracies of the three rates lie close together.

Fine-tuning keeps the missing inputs at zero for every model. Replacing zero by the training mean in the fine-tuning of the 188-class model of seed one on the full top training set moves its rejection at 30% signal efficiency by less than the spread over pretraining runs, so the benchmark comparison does not depend on that choice.

B.5 Transfer to $X \to b\bar{b}$ and $X \to bs$

For $X \to b\bar{b}$ and $X \to bs$ the transfer head follows Sophon's [11, App. B.5]: a hidden layer of 512 units with a ReLU activation and two outputs on the frozen class token, with the hidden layer initialized from that of the pretrained head. It is trained for one pass over 2.5 million samples in batches of 1024 at a constant learning rate of $5 \times 10{-}4$. Sophon states neither an optimizer nor a regularization; we use RAdam with the settings of pretraining, dropout after the hidden layer and no weight decay, and train several seeds per pretraining run, which the tables average.

The signal and QCD training jets are drawn in matched bins of $(p_{\mathrm{T}}, m_{\mathrm{SD}})$, **[samples not in]** per class for $X \to b\bar{b}$ and **[samples not in]** for $X \to bs$. They are kept in a fixed shuffled order, repeated **[samples not in]** and **[samples not in]** times to reach Sophon's number of samples. The training QCD jets come from JetClass-II files that no pretraining run read and the evaluation QCD jets from its test files, so no pretraining jet enters either. Whether the signal mass is fixed or variable is recorded for every fit; Sophon trains its $X \to bs$ transfer head on a variable mass and evaluates it at 200 GeV [11, Sec. III].

Sophon's direct discriminant can be formed only by the 188- and 162-class models, in which $X \to b\bar{b}$ is a class of its own. The closest-output-node control scores the pretrained $X \to bq$ output against QCD; where a label set merges $X \to bq$ with other decays, it scores the output of that group. The network trained from scratch for $X \to bs$ is the Sophon network on the same inputs and jets, trained for Sophon's number of epochs for its reference, 50, with weight decay [11, App. B.3], its epoch chosen on our validation jets, with three seeds. No network is trained from scratch for $X \to b\bar{b}$.

B.6 Compute

To report what each result cost and on which hardware, table 9 gives every pretraining run's GPU types with the hours spent on each, and its total GPU-hours. The hours sum the training and validation passes of the run's epochs as each epoch recorded its device. A run could resume on another GPU type after an interruption, so several runs list more than one. The pretraining runs used 3,435 GPU-hours on L4, RTX 3090 and V100 GPUs; the V100 carries the most hours, between 56% and 71% of each label set's. The fine-tuning records hold the device but not the time, so the table gives the GPU types of each run's fine-tuning without hours.

C Missing input features

To show what each substitution puts in each missing slot, table 10 lists the 17 per-particle inputs in the layout of ParT's table of input features [8, Table 2] and marks those that the top and the quark/gluon data provide. It also gives where a zero lies after the input normalization of the 188-class model of seed one, in standard

Table 9: GPU types and GPU-hours of every pretraining run, summed over the training and validation passes of its epochs as each epoch recorded its device, and the GPU types its fine-tuning ran on.

run	GPU types (hours)	GPU-hours	fine-tuning GPU types
188 classes, seed 1	V100 (135)	135	V100
188 classes, seed 2	V100 (145)	145	V100
188 classes, seed 3	RTX 3090 (116)	116	V100
162 classes, seed 1	V100 (75), RTX 3090 (54)	129	V100
162 classes, seed 2	V100 (86), RTX 3090 (40)	126	V100
162 classes, seed 3	V100 (123), L4 (39)	162	V100
43 classes, seed 1	RTX 3090 (113)	113	V100
43 classes, seed 2	V100 (101), RTX 3090 (24)	125	V100
43 classes, seed 3	V100 (107), RTX 3090 (20)	127	V100
17 classes, seed 1	RTX 3090 (108), L4 (3)	111	V100
17 classes, seed 2	V100 (133)	133	V100
17 classes, seed 3	V100 (101), L4 (72)	174	V100
162 classes, mass regression, seed 1	V100 (129), RTX 3090 (26)	155	V100
162 classes, mass regression, seed 2	V100 (130), RTX 3090 (15)	145	V100
17 classes, mass regression, seed 1	V100 (126), RTX 3090 (16)	142	V100
17 classes, mass regression, seed 2	RTX 3090 (101)	101	V100
random partition 1, seed 1	RTX 3090 (78), V100 (38)	116	V100
random partition 2, seed 2	RTX 3090 (110)	110	V100
random partition 3, seed 1	RTX 3090 (109)	109	V100
random partition 4, seed 2	RTX 3090 (107)	107	V100
random partition 5, seed 2	RTX 3090 (98)	98	V100
random partition, no flavor boundary, seed 1	RTX 3090 (99)	99	V100
random partition, no flavor boundary, seed 2	RTX 3090 (130)	130	V100
random partition, b-quark boundary, seed 1	V100 (98), RTX 3090 (24)	122	V100
random partition, b-quark boundary, seed 2	RTX 3090 (137)	137	V100
random partition, random flavor boundary, seed 1	RTX 3090 (125)	125	V100
random partition, random flavor boundary, seed 2	V100 (135), RTX 3090 (7)	142	V100
188 classes, kinematic inputs only	**[training]**		–

deviations of the training particles. The top data lack 10 inputs: the charge, the five type flags and the four impact-parameter features. The quark/gluon data provide the particle type and the charge, ParT's exp inputs [8], and lack the 4 impact-parameter features.

The three substitutions differ in what they put in a missing slot. Zero is also the value every model sees in fine-tuning. The training mean is the running mean of the model's own input normalization, which the normalization maps to its bias: it is the centroid of the training particles, with fractional flags that no particle has, and it differs from model to model. A random draw replaces the missing features of a particle by those of a real JetClass-II particle. For top, the draw takes a species with the JetClass-II species frequencies, a charge given the species and the impact-parameter features of a random particle of that species; for quark/gluon, the flags and the charge are kept and only the impact-parameter features are drawn, from particles of the same species. Each particle receives four draws whose features are averaged. The draws come from the class-balanced JetClass-II training set and depend only on the batch counter and the seed, so every model sees the same draws.

Each single draw is a state a real particle takes, but the average of four is not, and neither are zero and the mean: in JetClass-II every particle has exactly one type flag set, and only neutral particles have zero impact parameters. A zeroed flag alone lies at most **[a committed JSON of the input-normalization offsets of mtx-l188-s1 (A8)]** standard deviations from the training particles after the normalization (table 10); the state that no pretraining particle has is the joint one, with no flag set.

Table 10: The per-particle inputs: which the top and the quark/gluon data provide (•), where a zero lies after the network's input normalization (in standard deviations of the training particles), and what the mean and random-draw substitutions put in a missing slot. A random draw is averaged over four draws.

input	top	quark/gluon	zero after normalization	mean	random draws
$\log p_T$, rescaled	•	•	–	–	–
$\log E$, rescaled	•	•	–	–	–
$\log(p_T/p_T^{jet})$	•	•	–	–	–
$\log(E/E^{jet})$	•	•	–	–	–
ΔR	•	•	–	–	–
electric charge	—	•	[offsets]	running mean of the input normalization	drawn given the species (top)
charged hadron	—	•	[offsets]	running mean of the input normalization	a species drawn from JetClass-II's frequencies (top)
neutral hadron	—	•	[offsets]	running mean of the input normalization	a species drawn from JetClass-II's frequencies (top)
photon	—	•	[offsets]	running mean of the input normalization	a species drawn from JetClass-II's frequencies (top)
electron	—	•	[offsets]	running mean of the input normalization	a species drawn from JetClass-II's frequencies (top)
muon	—	•	[offsets]	running mean of the input normalization	a species drawn from JetClass-II's frequencies (top)
d_0	—	—	[offsets]	running mean of the input normalization	a JetClass-II particle of the same species
σ_{d_0}	—	—	[offsets]	running mean of the input normalization	a JetClass-II particle of the same species
d_z	—	—	[offsets]	running mean of the input normalization	a JetClass-II particle of the same species
σ_{d_z}	—	—	[offsets]	running mean of the input normalization	a JetClass-II particle of the same species
$\Delta\eta$	•	•	–	–	–
$\Delta\phi$	•	•	–	–	–

The substitution for each task is chosen on validation jets and applied to every model, the randomly initialized network included: **[fill fits]** for top and **[fill fits]** for quark/gluon. Table 11 gives all three substitutions for every configuration and so completes figure 5, which shows the 188-class model and the randomly initialized network.

D Method checks

D.1 MLP probes and the hidden layer of the head

MLP probes answer whether a linear null means that the information is absent or only that a linear classifier cannot read it. Wherever the main text reports that a linear probe finds no difference between label sets, table 12 gives the MLP probe on the same frozen features at the largest training set of JetClass-II, JetClass and the class pairs, and of top and quark/gluon with the missing inputs set to zero.

The last rows of the same table address a second question: whether the hidden layer of the head, where Sophon's transfer head starts, holds information that a linear probe cannot read from the class token. For each 188-class model the pretrained first layer of the head and its activation, $\mathrm{ReLU}(W_1 x + b_1)$, are applied to the saved class token x. A linear probe on these 512 outputs is trained on JetClass-II with 10^6 training jets and on the full quark/gluon training set, and the table sets it beside the linear probe on the class token itself.

D.2 BatchNorm statistics

Recomputing the BatchNorm statistics tests whether the label-set comparison depends on the statistics each model accumulated during pretraining. Every linear probe on natural-composition JetClass-II was repeated on the copy of the selected checkpoint whose BatchNorm statistics were recomputed on 200,000 training jets (table 13); the class-balanced linear probes of the main text use the checkpoints as trained, which is why this check runs on the natural-composition test set. The ratio of $1-\mathrm{AUC}$ of a label set to the 188-class model of its seed changes by at most 0.2%,

Table 11: The three substitutions of the missing inputs (zero, the training mean, random draws of a real particle) for every configuration, at the largest training set. Upper part: relative drop in JetClass-II accuracy (per cent) when the slots the top (10) or the quark/gluon (4) data lack are substituted, against the same model with every input. Lower part: R_{30} of the linear probe on top and quark/gluon jets under each substitution; the model with kinematic inputs only has nothing missing. The substitution chosen on validation is **[fill fits]** for top and **[fill fits]** for quark/gluon. Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run.

	JetClass-II, top slots			JetClass-II, quark/gluon slots		
pretraining	zero	mean	draws	zero	mean	draws
188 classes	88.9 ± 1.0%	[no run yet]	[no run yet]	61.2 ± 3.0%	[no run yet]	[no run yet]
162 classes	89.0 ± 0.9%	[no run yet]	[no run yet]	64 ± 5%	[no run yet]	[no run yet]
43 classes	87.81 ± 0.28%	[no run yet]	[no run yet]	68 ± 5%	[no run yet]	[no run yet]
17 classes	86.7 ± 0.7%	[no run yet]	[no run yet]	70.9 ± 2.3%	[no run yet]	[no run yet]
162 classes, mass regression	89.20, 88.29%	[no run yet]	[no run yet]	61.8, 67.9%	[no run yet]	[no run yet]
17 classes, mass regression	84.8, 87.4%	[no run yet]	[no run yet]	66.7, 65.4%	[no run yet]	[no run yet]
random partition 1, seed 1	90.2%	[no run yet]	[no run yet]	66.5%	[no run yet]	[no run yet]
random partition 2, seed 2	90.4%	[no run yet]	[no run yet]	69.5%	[no run yet]	[no run yet]
random partition 3, seed 1	90.8%	[no run yet]	[no run yet]	63.7%	[no run yet]	[no run yet]
random partition 4, seed 2	89.9%	[no run yet]	[no run yet]	72.9%	[no run yet]	[no run yet]
random partition 5, seed 2	89.5%	[no run yet]	[no run yet]	69.5%	[no run yet]	[no run yet]
random partition, no flavor boundary	88.88, 88.46%	[no run yet]	[no run yet]	62.72, 62.24%	[no run yet]	[no run yet]
random partition, *b*-quark boundary	87.80, 87.65%	[no run yet]	[no run yet]	66.2, 62.8%	[no run yet]	[no run yet]
random partition, random flavor boundary	89.58, 90.28%	[no run yet]	[no run yet]	67.2, 69.0%	[no run yet]	[no run yet]
randomly initialized network, BatchNorm recomputed	57.8, 54.9% **[2 of 3]**	[no run yet]	[no run yet]	10.57, 9.73% **[2 of 3]**	[no run yet]	[no run yet]

	top, R_{30}			quark/gluon, R_{30}		
pretraining	zero	mean	draws	zero	mean	draws
188 classes	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
162 classes	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
43 classes	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
17 classes	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
162 classes, mass regression	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
17 classes, mass regression	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition 1, seed 1	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition 2, seed 2	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition 3, seed 1	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition 4, seed 2	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition 5, seed 2	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition, no flavor boundary	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition, *b*-quark boundary	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition, random flavor boundary	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
randomly initialized network, BatchNorm recomputed	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
188 classes, kinematic inputs only	[run]	nothing missing		–	–	–

less than the smallest difference between label sets that the main text reports, so no result rests on the statistics accumulated in pretraining.

The randomly initialized network enters every comparison in its recomputed form, because its saved statistics are PyTorch's defaults, zero mean and unit variance, which do not normalize the inputs; a comparison with it would otherwise set a normalized network against an unnormalized one. Table 13 gives it both ways.

D.3 Test-sample uncertainty

A bootstrap of the test jets shows whether the finite test set or the variation between pretraining runs is the larger uncertainty. Every model is scored on the same test jets, so the test sample moves all of them together, and much of its effect cancels in a ratio between models. On JetClass-II the test jets are resampled with replacement within each class 200 times, every model is scored on each resample, and the standard deviation of the logarithm of a label-set ratio of $1 - \mathrm{AUC}$ over the resamples is its relative uncertainty from the test sample. On top and quark/gluon, the uncertainty

of a ratio of R_{30} follows from the number of background jets that pass the cut. Table 14 sets each beside the standard deviation over pretraining runs. The test sample contributes at most [bootstrap]% to a ratio from the linear probes on JetClass-II and [bootstrap]% after fine-tuning there, and at most 3.6% and 13% to a ratio of R_{30} on top and quark/gluon from the linear probe and from fine-tuning; in each case this is below the standard deviation over pretraining runs of the same ratio.

D.4 Class-balanced and natural-composition test sets

Comparing two test sets tests whether class balancing changes the order of the label sets on JetClass-II. Table 15 gives the linear probes with 10^6 training jets on the natural-composition test set, whose class shares follow the selection, beside those on the class-balanced set of the main text. The label sets fall in the same order on both.

D.5 Reproduction of ParT on top tagging

To locate the gap between our network trained from scratch and ParT on top tagging, ParT's top-tagging training script [8] was run on our cluster unchanged and then with Sophon's changes added one at a time (table 16). Unchanged, it reaches a rejection at 30% signal efficiency of 1,711, within twice the published standard deviation of ParT's 1602, so ParT reproduces here. Adding Sophon's output head gives 7.8% less rejection than the unchanged script, which stays within ParT's published spread. Adding Sophon's rescaling of every jet to a common transverse momentum on top of the head gives 28% less rejection than the unchanged script. Each variant is a single run. The network trained from scratch in table 3 keeps the rescaling because the pretrained models use it, so the comparison of pretrained and from-scratch networks shares every preprocessing step.

E Complete results

The tables and figures of this appendix complete the results of the main text, each for the item named with it. Their uncertainties follow the convention of the main text, and a configuration with two runs shows both values.

E.1 Linear probes and fine-tuning for every configuration

Table 17 gives the linear probes at the largest training set of every task for every configuration, each random partition included; on JetClass-II it gives the accuracy beside the averages of the AUC defined in appendix B.3. Table 18 gives the linear probes against the number of training jets in each task's metric. Figure 10 draws the four label sets against the number of training jets, to show whether their order holds at every size. Fine-tuning is in table 19 against the number of training jets and in table 20 on the full top and quark/gluon training sets, for every configuration. Table 21 gives each pretraining run's own values.

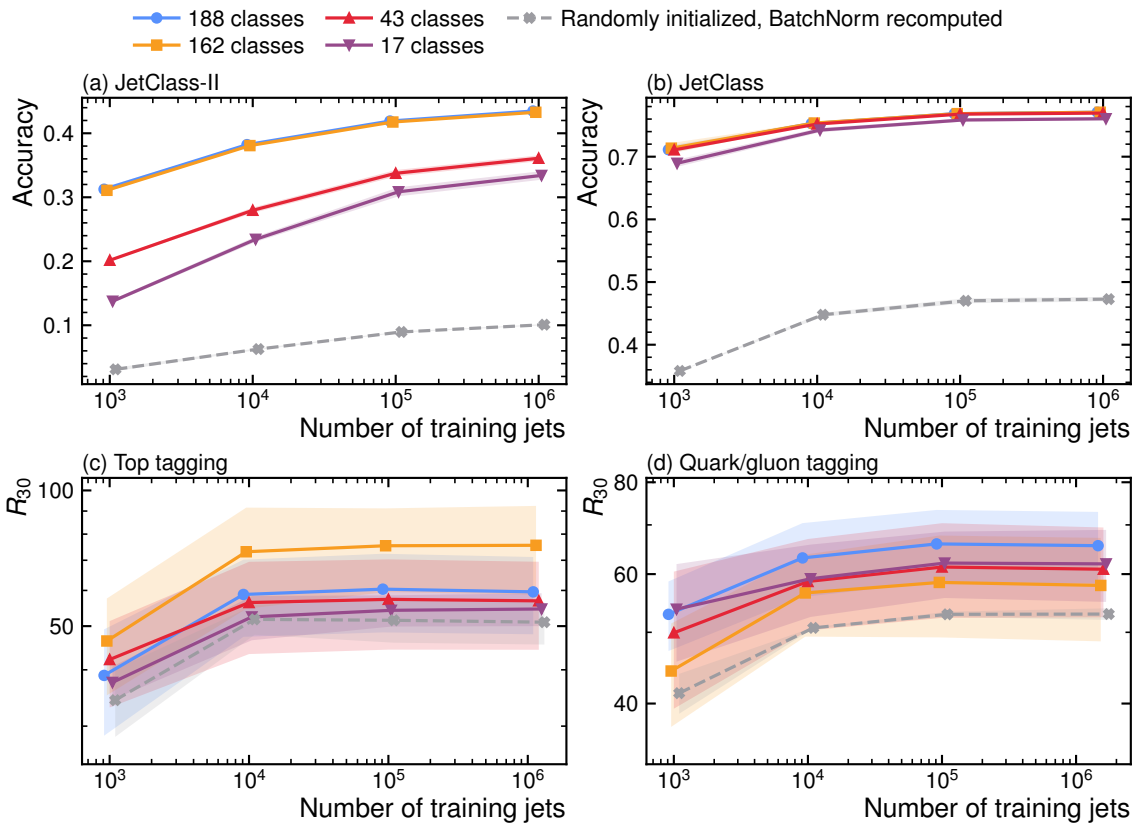


Figure 10: Linear probes on frozen features against the number of training jets, for the 188-, 162-, 43- and 17-class models and the randomly initialized network with BatchNorm statistics recomputed. Panels: (a) accuracy on class-balanced JetClass-II; (b) accuracy on JetClass; background rejection at 30% signal efficiency on (c) top tagging and (d) quark/gluon discrimination, with the missing inputs substituted as chosen on validation. Colors and axes as in figure 6. Lines join the means; bands show the standard deviation over the three pretraining runs of each label set.

Table 21: Every pretraining run's values at the largest training set: linear probe and fine-tuning. Linear probes on top and quark/gluon with the missing inputs substituted as chosen on validation ([fill fits] for top, [fill fits] for quark/gluon); fine-tuning with them set to zero.

run	linear probe				fine-tun	
	JetClass-II AUC	JetClass AUC	top R_{30}	q/g R_{30}	JetClass-II AUC	JetClass accuracy
188 classes, seed 1	0.9801	0.9683	47.3	57.2	–	0.8489
188 classes, seed 2	0.9801	0.9687	61.1	70.5	–	0.8497
188 classes, seed 3	0.9801	0.9675	70.3	69.1	–	0.8487
162 classes, seed 1	0.9795	0.9682	74.9	60.5	–	0.8492
162 classes, seed 2	0.9797	0.9683	59.0	47.6	–	0.8496
162 classes, seed 3	0.9796	0.9685	92.7	65.6	–	0.8496
43 classes, seed 1	0.9758	0.9678	59.7	61.8	–	0.8475
43 classes, seed 2	0.9755	0.9678	43.1	52.0	–	0.8479
43 classes, seed 3	0.9755	0.9685	67.8	69.0	–	0.8486
17 classes, seed 1	0.9695	0.9658	52.6	57.6	–	0.8434
17 classes, seed 2	0.9689	0.9643	51.3	70.0	–	0.8435
17 classes, seed 3	0.9688	0.9651	59.8	58.3	–	0.8433
162 classes, mass regression, seed 1	0.9791	0.9684	76.6	65.4	–	0.8478
162 classes, mass regression, seed 2	0.9794	0.9697	94.8	61.1	–	0.8490
17 classes, mass regression, seed 1	0.9679	0.9655	103	66.0	–	0.8420
17 classes, mass regression, seed 2	0.9696	0.9651	68.5	65.5	–	0.8419
random partition 1, seed 1	0.9776	0.9636	44.2	66.0	–	0.8450
random partition 2, seed 2	0.9776	0.9641	60.5	59.0	–	0.8458
random partition 3, seed 1	0.9779	0.9638	50.0	65.2	–	0.8475
random partition 4, seed 2	0.9778	0.9636	30.2	51.0	–	0.8464
random partition 5, seed 2	0.9775	0.9639	73.9	54.9	–	0.8459
random partition, no flavor boundary, seed 1	0.9717	0.9633	69.7	63.1	–	0.8446
random partition, no flavor boundary, seed 2	0.9710	0.9622	69.3	66.4	–	0.8442
random partition, b-quark boundary, seed 1	0.9758	0.9661	73.7	54.6	–	0.8451
random partition, b-quark boundary, seed 2	0.9759	0.9665	80.7	59.4	–	0.8469
random partition, random flavor boundary, seed 1	0.9760	0.9654	64.5	65.6	–	0.8456
random partition, random flavor boundary, seed 2	0.9760	0.9645	40.9	54.3	–	0.8454
188 classes, kinematic inputs only	–	–	–	–	–	–
from scratch, seed 1	–	–	–	–	–	0.8273
from scratch, seed 2	–	–	–	–	–	0.8276
from scratch, seed 3	–	–	–	–	–	0.8258

E.2 JetClass

Table 22 extends table 2 to every configuration, and table 23 gives the linear probes in the same columns. The rejection of every signal class rests on the QCD jets that pass its working point, and table 24 gives these counts, from which the daggers of both tables follow.

To show which boundaries the random partitions lose on JetClass, table 25 lists the prong-count boundaries each partition keeps beside its fine-tuned rejection of each signal class, relative to the 188-class model of its seed.

E.3 JetClass-II class pairs and classes

Table 26 completes figure 4 with all six class pairs. Beside the three pairs of the main text it gives QCD $b\bar{b}$ against QCD $c\bar{c}$, QCD jets with a b quark against QCD jets without b, c or s quarks, and QCD jets with a c but no b quark against the same light QCD jets. Each pair has its AUC and, in a separate column, the ratio of $1 - \text{AUC}$ to the 188-class model. Table 27 gives the numbers behind figure 3, one row per JetClass-II class.

Table 27: Linear probes per class of JetClass-II (class-balanced test set, largest training set): the AUC of the class against the rest for the 188-class model (mean ± standard deviation over its runs) and the ratio of $1 - \text{AUC}$ to it of every configuration, run by run and averaged; p1–p5 are the random partitions, then the partitions with no, a b-quark and a random flavor boundary. Classes grouped by their 17-class group.

class	AUC, 188	162	43	17	p1	p2	p3	p4	p5	no flavor	b quark	ra
$X \to bb$	0.9893 ± 0.0002	1.011 ± 0.029	1.066 ± 0.018	1.445 ± 0.007	1.14	1.11	1.15	1.13	1.18	1.49, 1.64	1.189, 1.136	
$X \to cc$	0.9778 ± 0.0001	0.998 ± 0.011	1.048 ± 0.012	1.563 ± 0.020	1.16	1.17	1.15	1.15	1.15	1.56, 1.72	1.252, 1.221	
$X \to ss$	0.9760 ± 0.0001	1.002 ± 0.005	1.078 ± 0.004	1.401 ± 0.030	1.08	1.08	1.07	1.04	1.10	1.389, 1.390	1.211, 1.222	
$X \to qq$	0.9751 ± 0.0002	1.009 ± 0.005	1.062 ± 0.012	1.28 ± 0.06	1.06	1.09	1.06	1.08	1.08	1.234, 1.221	1.168, 1.182	
$X \to bc$	0.9810 ± 0.0001	1.010 ± 0.013	1.084 ± 0.017	1.458 ± 0.026	1.19	1.18	1.14	1.19	1.20	1.470, 1.499	1.254, 1.239	
$X \to cs$	0.9690 ± 0.0001	1.002 ± 0.009	1.069 ± 0.005	1.343 ± 0.015	1.11	1.13	1.09	1.10	1.13	1.356, 1.377	1.238, 1.200	
$X \to bq$	0.9812 ± 0.0001	1.025 ± 0.015	1.141 ± 0.019	1.607 ± 0.027	1.22	1.22	1.17	1.18	1.26	1.611, 1.569	1.366, 1.326	
$X \to cq$	0.9700 ± 0.0003	1.005 ± 0.020	1.102 ± 0.017	1.358 ± 0.014	1.11	1.12	1.10	1.10	1.15	1.375, 1.418	1.240, 1.204	
$X \to sq$	0.9745 ± 0.0001	1.002 ± 0.005	1.044 ± 0.004	1.299 ± 0.027	1.08	1.08	1.08	1.05	1.10	1.277, 1.251	1.178, 1.171	
$X \to gg$	0.9614 ± 0.0001	1.008 ± 0.006	1.037 ± 0.017	1.125 ± 0.003	1.04	1.04	1.05	1.06	1.05	1.089, 1.084	1.050, 1.044	
$X \to \tau_h\tau_h$	$0.9994 \pm <0.0001$	1.04 ± 0.04	1.028 ± 0.029	1.32 ± 0.22	1.29	1.11	1.38	1.09	1.26	1.19, 1.06	1.112, 1.068	
$X \to \tau_h\tau_e$	$0.9994 \pm <0.0001$	0.99 ± 0.06	4.0 ± 0.6	3.4 ± 0.6	1.09	1.07	1.21	1.04	1.26	1.080, 1.075	1.16, 1.04	
$X \to \tau_h\tau_\mu$	$0.9993 \pm <0.0001$	1.05 ± 0.10	3.80 ± 0.29	3.32 ± 0.30	1.20	1.02	1.20	1.03	1.13	1.160, 1.245	1.064, 1.096	
$X \to ee$	$0.9996 \pm <0.0001$	1.01 ± 0.05	3.64 ± 0.16	3.6 ± 0.8	1.00	1.03	1.00	1.09	1.06	1.050, 1.088	1.04, 1.15	
$X \to \mu\mu$	$0.9995 \pm <0.0001$	1.00 ± 0.05	3.45 ± 0.32	3.5 ± 0.5	0.995	1.04	1.07	1.16	1.06	1.34, 1.09	1.057, 1.104	
$X \to YY \to bbb$	0.9912 ± 0.0001	1.004 ± 0.004	1.105 ± 0.031	1.516 ± 0.022	1.14	1.15	1.15	1.15	1.15	1.59, 1.85	1.220, 1.267	
$X \to YY \to bbc$	0.9774 ± 0.0001	1.001 ± 0.009	1.102 ± 0.003	1.462 ± 0.018	1.12	1.12	1.12	1.11	1.09	1.530, 1.627	1.273, 1.199	
$X \to YY \to bbs$	0.9756 ± 0.0001	0.9978 ± 0.0028	1.123 ± 0.010	1.533 ± 0.028	1.13	1.11	1.11	1.10	1.12	1.526, 1.618	1.240, 1.213	
$X \to YY \to bbq$	0.9740 ± 0.0001	1.010 ± 0.005	1.109 ± 0.003	1.484 ± 0.014	1.13	1.13	1.10	1.11	1.13	1.55, 1.66	1.219, 1.200	
$X \to YY \to ccb$	0.9705 ± 0.0004	0.996 ± 0.013	1.079 ± 0.020	1.575 ± 0.029	1.14	1.19	1.11	1.17	1.14	1.580, 1.617	1.227, 1.292	
$X \to YY \to ccc$	0.9756 ± 0.0001	0.996 ± 0.008	1.039 ± 0.016	1.75 ± 0.06	1.15	1.18	1.11	1.15	1.19	1.78, 2.00	1.230, 1.247	
$X \to YY \to ccs$	0.9597 ± 0.0001	1.003 ± 0.009	1.058 ± 0.009	1.488 ± 0.024	1.11	1.10	1.09	1.08	1.09	1.535, 1.630	1.199, 1.175	
$X \to YY \to ccq$	0.9569 ± 0.0001	0.9974 ± 0.0032	1.058 ± 0.005	1.49 ± 0.04	1.09	1.11	1.09	1.07	1.10	1.50, 1.67	1.149, 1.179	
$X \to YY \to ssb$	0.9715 ± 0.0002	0.997 ± 0.008	1.149 ± 0.002	1.92 ± 0.09	1.13	1.13	1.10	1.16	1.18	1.780, 1.770	1.414, 1.367	
$X \to YY \to ssc$	0.9561 ± 0.0005	0.998 ± 0.014	1.080 ± 0.009	1.434 ± 0.035	1.05	1.06	1.05	1.07	1.07	1.429, 1.464	1.198, 1.181	
$X \to YY \to sss$	0.9727 ± 0.0001	1.006 ± 0.005	1.093 ± 0.007	1.54 ± 0.05	1.07	1.07	1.05	1.05	1.06	1.456, 1.524	1.272, 1.220	
$X \to YY \to ssq$	0.9631 ± 0.0001	1.000 ± 0.003	1.051 ± 0.009	1.32 ± 0.05	1.05	1.06	1.03	1.06	1.05	1.267, 1.292	1.142, 1.151	
$X \to YY \to qqb$	0.9666 ± 0.0003	1.000 ± 0.007	1.091 ± 0.020	1.60 ± 0.04	1.11	1.11	1.08	1.13	1.11	1.571, 1.515	1.299, 1.280	
$X \to YY \to qqc$	0.9519 ± 0.0001	1.000 ± 0.006	1.083 ± 0.006	1.337 ± 0.007	1.08	1.08	1.10	1.06	1.07	1.333, 1.410	1.186, 1.184	
$X \to YY \to qqs$	0.9626 ± 0.0002	1.006 ± 0.006	1.034 ± 0.005	1.26 ± 0.04	1.04	1.05	1.04	1.05	1.04	1.201, 1.218	1.129, 1.125	
$X \to YY \to qqq$	0.9691 ± 0.0001	1.010 ± 0.007	1.085 ± 0.006	1.40 ± 0.08	1.05	1.07	1.03	1.07	1.05	1.306, 1.279	1.223, 1.228	
$X \to YY \to bcs$	0.9583 ± 0.0002	1.001 ± 0.005	1.112 ± 0.011	1.429 ± 0.015	1.10	1.13	1.10	1.13	1.11	1.399, 1.420	1.278, 1.248	
$X \to YY \to bcq$	0.9582 ± 0.0003	1.010 ± 0.010	1.119 ± 0.010	1.388 ± 0.012	1.15	1.15	1.13	1.13	1.12	1.417, 1.419	1.214, 1.203	
$X \to YY \to bsq$	0.9648 ± 0.0003	0.998 ± 0.015	1.081 ± 0.007	1.491 ± 0.023	1.13	1.10	1.11	1.14	1.15	1.479, 1.461	1.261, 1.198	
$X \to YY \to csq$	0.9477 ± 0.0004	1.001 ± 0.005	1.053 ± 0.019	1.257 ± 0.015	1.07	1.07	1.07	1.07	1.06	1.252, 1.308	1.111, 1.110	
$X \to YY \to bbg$	0.9770 ± 0.0002	1.015 ± 0.019	1.095 ± 0.025	1.345 ± 0.029	1.10	1.11	1.10	1.12	1.13	1.31, 1.43	1.100, 1.097	
$X \to YY \to ccg$	0.9635 ± 0.0001	1.005 ± 0.003	1.063 ± 0.002	1.53 ± 0.05	1.11	1.11	1.10	1.10	1.10	1.43, 1.58	1.158, 1.156	
$X \to YY \to ssg$	$0.9608 \pm <0.0001$	1.003 ± 0.001	1.104 ± 0.008	1.386 ± 0.016	1.05	1.07	1.06	1.07	1.06	1.311, 1.302	1.167, 1.142	
$X \to YY \to qqg$	0.9599 ± 0.0001	0.9977 ± 0.0023	1.089 ± 0.007	1.28 ± 0.04	1.04	1.03	1.04	1.06	1.05	1.185, 1.169	1.149, 1.134	
$X \to YY \to ggb$	0.9677 ± 0.0001	1.000 ± 0.008	1.100 ± 0.024	1.384 ± 0.007	1.08	1.08	1.06	1.10	1.10	1.223, 1.321	1.097, 1.113	
$X \to YY \to ggc$	0.9555 ± 0.0003	0.999 ± 0.005	1.104 ± 0.012	1.337 ± 0.012	1.10	1.08	1.07	1.07	1.11	1.230, 1.273	1.121, 1.106	
$X \to YY \to ggs$	0.9547 ± 0.0001	0.9989 ± 0.0010	1.069 ± 0.010	1.233 ± 0.020	1.04	1.04	1.04	1.04	1.06	1.142, 1.154	1.096, 1.093	
$X \to YY \to ggq$	0.9563 ± 0.0001	1.004 ± 0.010	1.059 ± 0.007	1.19 ± 0.04	1.05	1.04	1.05	1.06	1.04	1.124, 1.114	1.090, 1.088	
$X \to YY \to ggg$	0.9681 ± 0.0001	$1.002 \pm <0.001$	1.036 ± 0.020	1.120 ± 0.021	1.02	1.04	1.04	1.05	1.05	1.060, 1.055	1.031, 1.034	
$X \to YY \to b\tau_h\tau_h$	0.9965 ± 0.0001	0.97 ± 0.05	0.987 ± 0.028	1.366 ± 0.024	1.07	1.08	1.13	1.07	1.11	1.50, 1.60	1.228, 1.194	
$X \to YY \to c\tau_h\tau_h$	0.9929 ± 0.0001	1.002 ± 0.004	1.018 ± 0.031	1.35 ± 0.04	1.06	1.09	1.07	1.13	1.11	1.345, 1.420	1.203, 1.234	
$X \to YY \to s\tau_h\tau_h$	$0.9920 \pm <0.0001$	1.004 ± 0.006	1.068 ± 0.015	1.303 ± 0.029	1.06	1.05	1.05	1.05	1.07	1.286, 1.249	1.189, 1.225	
$X \to YY \to q\tau_h\tau_h$	0.9919 ± 0.0001	0.997 ± 0.018	1.047 ± 0.024	1.212 ± 0.011	1.06	1.05	1.05	1.03	1.04	1.167, 1.150	1.149, 1.152	
$X \to YY \to g\tau_h\tau_h$	0.9908 ± 0.0001	1.007 ± 0.008	1.030 ± 0.013	1.147 ± 0.016	1.05	1.07	1.04	1.08	1.07	1.068, 1.075	1.007, 1.040	
$X \to YY \to bbe$	0.9977 ± 0.0001	1.045 ± 0.029	1.96 ± 0.26	2.10 ± 0.30	1.20	1.15	1.18	1.17	1.28	1.55, 1.88	1.266, 1.244	
$X \to YY \to cce$	0.9939 ± 0.0001	1.009 ± 0.015	1.70 ± 0.14	2.02 ± 0.19	1.19	1.13	1.16	1.13	1.20	1.63, 1.74	1.278, 1.302	
$X \to YY \to sse$	0.9922 ± 0.0001	1.021 ± 0.011	1.55 ± 0.08	1.75 ± 0.13	1.09	1.10	1.10	1.07	1.10	1.464, 1.467	1.283, 1.282	
$X \to YY \to qqe$	0.9911 ± 0.0001	1.007 ± 0.010	1.47 ± 0.09	1.54 ± 0.14	1.06	1.08	1.07	1.05	1.05	1.243, 1.240	1.178, 1.217	
$X \to YY \to bb\mu$	0.9976 ± 0.0001	1.019 ± 0.025	2.04 ± 0.08	1.99 ± 0.19	1.14	1.12	1.23	1.15	1.19	1.68, 1.88	1.257, 1.168	
$X \to YY \to cc\mu$	0.9942 ± 0.0001	0.996 ± 0.014	1.87 ± 0.12	2.27 ± 0.25	1.09	1.17	1.12	1.24	1.24	1.61, 1.85	1.324, 1.371	
$X \to YY \to ss\mu$	0.9923 ± 0.0001	1.02 ± 0.04	1.636 ± 0.018	1.80 ± 0.09	1.07	1.10	1.10	1.07	1.10	1.495, 1.464	1.297, 1.303	
$X \to YY \to qq\mu$	0.9903 ± 0.0001	1.001 ± 0.003	1.55 ± 0.08	1.56 ± 0.15	1.04	1.05	1.04	1.06	1.06	1.236, 1.243	1.173, 1.213	
$X \to YY \to gge$	0.9935 ± 0.0002	0.995 ± 0.029	1.55 ± 0.07	1.49 ± 0.13	1.08	1.09	1.12	1.03	1.13	1.151, 1.121	1.108, 1.044	
$X \to YY \to gg\mu$	0.9936 ± 0.0001	1.003 ± 0.016	1.673 ± 0.026	1.45 ± 0.17	1.07	1.02	1.10	1.06	1.09	1.081, 1.082	1.043, 1.031	
$X \to YY \to bee$	$0.9981 \pm <0.0001$	0.992 ± 0.020	3.07 ± 0.11	3.48 ± 0.18	1.14	1.12	1.10	1.09	1.10	1.479, 1.501	1.250, 1.233	

class	AUC, 188	162	43	17	p1	p2	p3	p4	p5	no flavor	b quark	ra
$X \to YY \to cee$	$0.9954 \pm <0.0001$	0.999 ± 0.015	2.43 ± 0.11	3.03 ± 0.13	1.09	1.09	1.06	1.07	1.09	1.453, 1.456	1.213, 1.245	
$X \to YY \to see$	$0.9943 \pm <0.0001$	0.998 ± 0.009	2.29 ± 0.08	2.55 ± 0.09	1.04	1.02	1.02	1.05	1.04	1.213, 1.224	1.149, 1.165	
$X \to YY \to qee$	$0.9942 \pm <0.0001$	0.996 ± 0.004	2.30 ± 0.09	2.41 ± 0.21	1.03	1.01	1.04	1.03	1.04	1.140, 1.130	1.130, 1.145	
$X \to YY \to b\mu\mu$	$0.9983 \pm <0.0001$	0.997 ± 0.011	3.13 ± 0.10	3.7 ± 0.4	1.10	1.07	1.07	1.10	1.10	1.548, 1.562	1.267, 1.224	
$X \to YY \to c\mu\mu$	$0.9952 \pm <0.0001$	1.004 ± 0.015	2.33 ± 0.05	2.87 ± 0.20	1.05	1.07	1.04	1.07	1.09	1.409, 1.367	1.186, 1.220	
$X \to YY \to s\mu\mu$	$0.9943 \pm <0.0001$	1.003 ± 0.004	2.28 ± 0.10	2.55 ± 0.13	1.04	1.03	1.03	1.05	1.04	1.236, 1.229	1.199, 1.214	
$X \to YY \to q\mu\mu$	$0.9940 \pm <0.0001$	0.995 ± 0.005	2.25 ± 0.07	2.34 ± 0.09	1.04	1.01	1.02	1.04	1.03	1.139, 1.154	1.145, 1.133	
$X \to YY \to gee$	$0.9942 \pm <0.0001$	1.005 ± 0.006	2.28 ± 0.10	2.29 ± 0.13	1.03	1.01	1.01	1.05	1.02	1.049, 1.062	1.018, 1.028	
$X \to YY \to g\mu\mu$	0.9941 ± 0.0001	1.011 ± 0.029	2.23 ± 0.11	2.27 ± 0.12	1.03	1.04	1.02	1.02	1.03	1.059, 1.050	1.044, 1.014	
$X \to YY \to b\tau_h\tau_e$	0.9960 ± 0.0001	0.988 ± 0.016	2.13 ± 0.07	2.46 ± 0.14	1.11	1.18	1.15	1.16	1.18	1.561, 1.639	1.195, 1.237	
$X \to YY \to c\tau_h\tau_e$	$0.9916 \pm <0.0001$	1.000 ± 0.020	1.94 ± 0.05	2.21 ± 0.12	1.12	1.13	1.09	1.11	1.12	1.349, 1.361	1.172, 1.219	
$X \to YY \to s\tau_h\tau_e$	0.9916 ± 0.0001	1.002 ± 0.014	2.04 ± 0.08	2.25 ± 0.18	1.09	1.09	1.08	1.08	1.09	1.355, 1.338	1.229, 1.274	
$X \to YY \to q\tau_h\tau_e$	0.9907 ± 0.0001	1.024 ± 0.017	2.06 ± 0.06	2.04 ± 0.20	1.07	1.09	1.06	1.07	1.11	1.184, 1.216	1.171, 1.196	
$X \to YY \to b\tau_h\tau_\mu$	0.9965 ± 0.0001	1.017 ± 0.034	2.286 ± 0.032	2.60 ± 0.09	1.15	1.18	1.20	1.13	1.12	1.499, 1.571	1.194, 1.188	
$X \to YY \to c\tau_h\tau_\mu$	0.9921 ± 0.0001	1.010 ± 0.002	1.99 ± 0.04	2.29 ± 0.18	1.07	1.12	1.06	1.10	1.11	1.325, 1.395	1.180, 1.231	
$X \to YY \to s\tau_h\tau_\mu$	0.9918 ± 0.0001	0.995 ± 0.021	2.09 ± 0.12	2.23 ± 0.18	1.08	1.09	1.06	1.08	1.09	1.248, 1.289	1.184, 1.226	
$X \to YY \to q\tau_h\tau_\mu$	0.9915 ± 0.0001	1.001 ± 0.011	2.05 ± 0.08	2.05 ± 0.23	1.06	1.08	1.06	1.06	1.06	1.176, 1.172	1.133, 1.171	
$X \to YY \to g\tau_h\tau_e$	0.9905 ± 0.0001	1.007 ± 0.005	1.94 ± 0.07	1.88 ± 0.13	1.07	1.08	1.07	1.06	1.11	1.054, 1.104	1.038, 1.042	
$X \to YY \to g\tau_h\tau_\mu$	0.9907 ± 0.0001	1.017 ± 0.022	1.94 ± 0.06	1.88 ± 0.14	1.06	1.09	1.06	1.06	1.11	1.066, 1.089	1.010, 1.036	
$X \to YY \to bbbb$	0.9964 ± 0.0002	1.02 ± 0.08	1.210 ± 0.022	2.12 ± 0.15	1.44	1.35	1.34	1.31	1.25	2.444, 2.421	1.543, 1.592	
$X \to YY \to bbcc$	$0.9846 \pm <0.0001$	0.997 ± 0.017	1.154 ± 0.001	1.77 ± 0.04	1.20	1.20	1.14	1.20	1.18	1.78, 1.89	1.310, 1.304	
$X \to YY \to bbss$	$0.9815 \pm <0.0001$	1.008 ± 0.011	1.256 ± 0.011	1.98 ± 0.06	1.18	1.19	1.18	1.16	1.18	1.93, 2.10	1.425, 1.373	
$X \to YY \to bbqq$	0.9770 ± 0.0002	0.997 ± 0.017	1.157 ± 0.024	1.713 ± 0.020	1.17	1.14	1.14	1.13	1.12	1.790, 1.838	1.340, 1.305	
$X \to YY \to cccc$	0.9851 ± 0.0002	1.006 ± 0.032	1.098 ± 0.008	2.32 ± 0.15	1.30	1.29	1.22	1.20	1.26	2.32, 2.50	1.386, 1.354	
$X \to YY \to ccss$	0.9657 ± 0.0002	0.999 ± 0.007	1.119 ± 0.011	1.62 ± 0.06	1.11	1.11	1.12	1.12	1.12	1.684, 1.733	1.255, 1.243	
$X \to YY \to ccqq$	0.9613 ± 0.0001	0.997 ± 0.005	1.087 ± 0.008	1.541 ± 0.024	1.10	1.12	1.08	1.09	1.10	1.56, 1.72	1.213, 1.203	
$X \to YY \to ssss$	0.9801 ± 0.0001	0.998 ± 0.007	1.140 ± 0.020	1.72 ± 0.04	1.09	1.10	1.05	1.06	1.07	1.598, 1.660	1.338, 1.298	
$X \to YY \to ssqq$	0.9682 ± 0.0001	1.002 ± 0.009	1.051 ± 0.006	1.373 ± 0.030	1.06	1.07	1.04	1.04	1.06	1.320, 1.322	1.153, 1.156	
$X \to YY \to qqqq$	0.9755 ± 0.0002	1.000 ± 0.014	1.096 ± 0.011	1.47 ± 0.04	1.08	1.06	1.05	1.05	1.04	1.372, 1.313	1.256, 1.248	
$X \to YY \to qqqb$	0.9764 ± 0.0003	1.008 ± 0.003	1.143 ± 0.011	1.75 ± 0.06	1.18	1.17	1.14	1.16	1.21	1.714, 1.709	1.381, 1.369	
$X \to YY \to qqqc$	0.9610 ± 0.0003	1.005 ± 0.018	1.130 ± 0.005	1.458 ± 0.007	1.12	1.11	1.10	1.09	1.11	1.456, 1.496	1.239, 1.232	
$X \to YY \to qqqs$	0.9703 ± 0.0002	1.001 ± 0.012	1.046 ± 0.010	1.362 ± 0.003	1.07	1.06	1.05	1.06	1.05	1.310, 1.282	1.174, 1.149	
$X \to YY \to bbcq$	0.9778 ± 0.0003	0.999 ± 0.009	1.176 ± 0.014	1.65 ± 0.05	1.18	1.22	1.17	1.12	1.16	1.70, 1.87	1.339, 1.281	
$X \to YY \to ccbs$	0.9696 ± 0.0002	0.995 ± 0.016	1.157 ± 0.013	1.636 ± 0.019	1.16	1.17	1.12	1.15	1.17	1.601, 1.635	1.328, 1.325	
$X \to YY \to ccbq$	0.9691 ± 0.0004	1.015 ± 0.031	1.151 ± 0.018	1.628 ± 0.009	1.22	1.15	1.12	1.13	1.15	1.618, 1.616	1.309, 1.251	
$X \to YY \to ccsq$	0.9597 ± 0.0002	1.004 ± 0.010	1.079 ± 0.015	1.514 ± 0.013	1.14	1.12	1.12	1.13	1.12	1.512, 1.611	1.176, 1.155	
$X \to YY \to sscq$	0.9589 ± 0.0002	1.005 ± 0.008	1.117 ± 0.004	1.44 ± 0.04	1.12	1.12	1.10	1.11	1.11	1.476, 1.488	1.199, 1.198	
$X \to YY \to qqbc$	0.9642 ± 0.0002	1.006 ± 0.010	1.154 ± 0.002	1.518 ± 0.004	1.17	1.18	1.13	1.15	1.16	1.505, 1.486	1.254, 1.219	
$X \to YY \to qqbs$	0.9725 ± 0.0003	1.016 ± 0.019	1.120 ± 0.011	1.69 ± 0.05	1.21	1.18	1.17	1.16	1.20	1.654, 1.638	1.314, 1.277	
$X \to YY \to qqcs$	0.9542 ± 0.0001	1.005 ± 0.013	1.058 ± 0.012	1.309 ± 0.012	1.10	1.11	1.08	1.08	1.09	1.300, 1.347	1.117, 1.103	
$X \to YY \to bcsq$	0.9632 ± 0.0003	1.008 ± 0.016	1.132 ± 0.013	1.462 ± 0.007	1.19	1.16	1.15	1.14	1.17	1.443, 1.455	1.239, 1.230	
$X \to YY \to bbgg$	0.9782 ± 0.0002	1.002 ± 0.008	1.125 ± 0.016	1.51 ± 0.09	1.08	1.12	1.09	1.08	1.11	1.36, 1.53	1.097, 1.112	
$X \to YY \to ccgg$	0.9641 ± 0.0003	1.005 ± 0.019	1.091 ± 0.009	1.55 ± 0.04	1.08	1.08	1.07	1.06	1.07	1.421, 1.498	1.167, 1.134	
$X \to YY \to ssgg$	0.9638 ± 0.0001	1.001 ± 0.006	1.088 ± 0.007	1.337 ± 0.004	1.05	1.06	1.03	1.03	1.05	1.254, 1.273	1.131, 1.126	
$X \to YY \to qqgg$	0.9623 ± 0.0002	0.9981 ± 0.0028	1.069 ± 0.002	1.290 ± 0.034	1.04	1.05	1.02	1.04	1.03	1.154, 1.153	1.114, 1.126	
$X \to YY \to gggg$	0.9737 ± 0.0001	0.998 ± 0.013	1.058 ± 0.011	1.188 ± 0.004	1.05	1.04	1.03	1.03	1.05	1.081, 1.096	1.031, 1.020	
$X \to YY \to bb\tau_h\tau_h$	0.9971 ± 0.0002	1.01 ± 0.08	1.04 ± 0.10	1.86 ± 0.13	1.36	1.15	1.46	1.09	1.24	2.12, 2.25	1.360, 1.339	
$X \to YY \to cc\tau_h\tau_h$	0.9934 ± 0.0001	0.989 ± 0.012	1.022 ± 0.024	1.67 ± 0.05	1.11	1.17	1.12	1.14	1.17	1.64, 1.75	1.302, 1.364	
$X \to YY \to ss\tau_h\tau_h$	$0.9914 \pm <0.0001$	1.018 ± 0.012	1.099 ± 0.015	1.504 ± 0.028	1.08	1.08	1.10	1.02	1.09	1.433, 1.433	1.304, 1.300	
$X \to YY \to qq\tau_h\tau_h$	0.9913 ± 0.0001	1.008 ± 0.003	1.077 ± 0.017	1.34 ± 0.04	1.07	1.08	1.06	1.08	1.10	1.284, 1.310	1.216, 1.262	
$X \to YY \to gg\tau_h\tau_h$	0.9896 ± 0.0002	0.991 ± 0.031	1.029 ± 0.011	1.203 ± 0.020	1.05	1.05	1.08	1.03	1.04	1.097, 1.094	1.050, 1.056	
$X \to YY \to bbee$	$0.9991 \pm <0.0001$	0.972 ± 0.018	4.01 ± 0.33	5.2 ± 0.4	1.37	1.14	1.34	1.18	1.71	2.22, 2.39	1.32, 1.58	
$X \to YY \to ccee$	$0.9976 \pm <0.0001$	0.992 ± 0.019	2.84 ± 0.06	4.16 ± 0.30	1.12	1.12	1.08	1.14	1.14	1.77, 1.92	1.393, 1.384	
$X \to YY \to ssee$	$0.9961 \pm <0.0001$	0.994 ± 0.009	2.58 ± 0.07	3.11 ± 0.20	1.04	1.03	1.03	1.05	1.04	1.462, 1.505	1.336, 1.316	
$X \to YY \to qqee$	$0.9960 \pm <0.0001$	1.003 ± 0.008	2.52 ± 0.10	2.86 ± 0.10	1.07	1.04	1.03	1.08	1.04	1.235, 1.296	1.21, 1.33	
$X \to YY \to bb\mu\mu$	$0.9991 \pm <0.0001$	0.980 ± 0.011	4.06 ± 0.14	4.8 ± 0.4	1.15	1.15	1.19	1.07	1.26	2.00, 2.33	1.361, 1.369	
$X \to YY \to cc\mu\mu$	0.9972 ± 0.0001	1.00 ± 0.04	2.75 ± 0.06	4.0 ± 0.4	1.08	1.13	1.07	1.12	1.13	1.804, 1.806	1.312, 1.350	
$X \to YY \to ss\mu\mu$	$0.9956 \pm <0.0001$	1.005 ± 0.018	2.64 ± 0.08	3.05 ± 0.23	1.05	1.05	1.04	1.04	1.06	1.440, 1.456	1.291, 1.294	
$X \to YY \to qq\mu\mu$	$0.9956 \pm <0.0001$	1.000 ± 0.001	2.61 ± 0.04	2.74 ± 0.20	1.07	1.05	1.05	1.05	1.06	1.285, 1.286	1.254, 1.306	
$X \to YY \to ggee$	$0.9961 \pm <0.0001$	1.003 ± 0.010	2.47 ± 0.08	2.47 ± 0.23	1.03	1.03	1.02	1.03	1.03	1.053, 1.067	1.020, 1.051	
$X \to YY \to gg\mu\mu$	$0.9955 \pm <0.0001$	1.006 ± 0.004	2.61 ± 0.10	2.57 ± 0.27	1.06	1.07	1.05	1.05	1.04	1.083, 1.070	1.076, 1.057	
$X \to YY \to bb\tau_h\tau_e$	$0.9973 \pm <0.0001$	1.00 ± 0.04	2.39 ± 0.16	3.13 ± 0.10	1.22	1.22	1.29	1.16	1.27	2.01, 2.33	1.308, 1.378	
$X \to YY \to cc\tau_h\tau_e$	0.9932 ± 0.0001	1.004 ± 0.007	1.97 ± 0.07	2.90 ± 0.05	1.20	1.17	1.13	1.11	1.21	1.732, 1.831	1.370, 1.452	
$X \to YY \to ss\tau_h\tau_e$	0.9909 ± 0.0001	1.005 ± 0.011	2.09 ± 0.09	2.59 ± 0.12	1.07	1.08	1.11	1.08	1.08	1.498, 1.460	1.305, 1.361	
$X \to YY \to qq\tau_h\tau_e$	0.9899 ± 0.0001	0.991 ± 0.012	2.03 ± 0.05	2.27 ± 0.17	1.08	1.10	1.10	1.07	1.06	1.331, 1.339	1.284, 1.309	
$X \to YY \to bb\tau_h\tau_\mu$	0.9974 ± 0.0001	0.99 ± 0.05	2.439 ± 0.013	3.17 ± 0.16	1.14	1.25	1.16	1.17	1.15	1.76, 2.29	1.23, 1.37	
$X \to YY \to cc\tau_h\tau_\mu$	0.9943 ± 0.0001	1.012 ± 0.027	2.16 ± 0.08	3.19 ± 0.22	1.17	1.18	1.10	1.15	1.23	1.73, 1.90	1.405, 1.442	
$X \to YY \to ss\tau_h\tau_\mu$	$0.9919 \pm <0.0001$	1.010 ± 0.006	2.29 ± 0.06	2.69 ± 0.18	1.11	1.10	1.07	1.09	1.06	1.463, 1.436	1.324, 1.300	
$X \to YY \to qq\tau_h\tau_\mu$	0.9911 ± 0.0001	1.005 ± 0.016	2.16 ± 0.07	2.31 ± 0.19	1.08	1.10	1.11	1.09	1.09	1.306, 1.273	1.226, 1.260	

class	AUC, 188	162	43	17	p1	p2	p3	p4	p5	no flavor	b quark	ra
$X \to YY \to gg\tau_h\tau_e$	0.9891 ± 0.0001	0.996 ± 0.010	1.93 ± 0.08	1.95 ± 0.15	1.05	1.12	1.04	1.07	1.09	1.076, 1.122	1.033, 1.058	
$X \to YY \to gg\tau_h\tau_\mu$	$0.9908 \pm <0.0001$	1.002 ± 0.013	2.14 ± 0.08	2.12 ± 0.20	1.03	1.12	1.09	1.09	1.08	1.122, 1.150	1.081, 1.067	
$X \to YY \to bce\nu$	0.9946 ± 0.0001	1.008 ± 0.014	1.77 ± 0.12	1.84 ± 0.16	1.11	1.11	1.06	1.06	1.09	1.618, 1.618	1.318, 1.278	
$X \to YY \to cse\nu$	0.9897 ± 0.0001	1.003 ± 0.013	1.61 ± 0.08	1.69 ± 0.13	1.08	1.06	1.06	1.05	1.05	1.438, 1.456	1.359, 1.292	
$X \to YY \to bqe\nu$	0.9951 ± 0.0001	0.998 ± 0.027	1.87 ± 0.13	2.15 ± 0.18	1.14	1.21	1.08	1.10	1.16	1.79, 1.89	1.404, 1.450	
$X \to YY \to cqe\nu$	0.9899 ± 0.0001	1.010 ± 0.013	1.64 ± 0.09	1.81 ± 0.13	1.09	1.09	1.06	1.08	1.11	1.555, 1.622	1.376, 1.371	
$X \to YY \to sqe\nu$	0.9898 ± 0.0001	1.001 ± 0.016	1.42 ± 0.07	1.55 ± 0.09	1.03	1.04	1.04	1.04	1.06	1.344, 1.312	1.234, 1.267	
$X \to YY \to qqe\nu$	$0.9901 \pm <0.0001$	1.010 ± 0.013	1.44 ± 0.08	1.47 ± 0.11	1.08	1.04	1.04	1.03	1.07	1.237, 1.210	1.180, 1.200	
$X \to YY \to bc\mu\nu$	0.9949 ± 0.0001	1.008 ± 0.019	2.01 ± 0.07	2.05 ± 0.06	1.14	1.12	1.12	1.08	1.13	1.728, 1.703	1.363, 1.375	
$X \to YY \to cs\mu\nu$	0.9889 ± 0.0001	0.9957 ± 0.0023	1.78 ± 0.08	1.77 ± 0.15	1.05	1.07	1.06	1.09	1.12	1.433, 1.489	1.326, 1.329	
$X \to YY \to bq\mu\nu$	$0.9946 \pm <0.0001$	0.991 ± 0.015	1.93 ± 0.10	2.18 ± 0.15	1.08	1.08	1.11	1.08	1.10	1.798, 1.787	1.371, 1.411	
$X \to YY \to cq\mu\nu$	0.9891 ± 0.0001	1.003 ± 0.010	1.77 ± 0.10	1.84 ± 0.11	1.09	1.08	1.07	1.08	1.09	1.552, 1.583	1.383, 1.390	
$X \to YY \to sq\mu\nu$	0.9896 ± 0.0001	1.006 ± 0.009	1.50 ± 0.05	1.56 ± 0.10	1.04	1.08	1.03	1.06	1.06	1.310, 1.354	1.259, 1.267	
$X \to YY \to qq\mu\nu$	0.9896 ± 0.0001	1.015 ± 0.008	1.56 ± 0.04	1.54 ± 0.07	1.06	1.06	1.04	1.06	1.07	1.250, 1.257	1.223, 1.289	
$X \to YY \to bc\tau_e\nu$	0.9908 ± 0.0001	1.007 ± 0.019	1.84 ± 0.08	2.24 ± 0.09	1.18	1.18	1.12	1.11	1.18	1.622, 1.601	1.274, 1.280	
$X \to YY \to cs\tau_e\nu$	0.9848 ± 0.0002	1.016 ± 0.021	1.87 ± 0.08	2.09 ± 0.11	1.07	1.08	1.08	1.08	1.13	1.407, 1.463	1.262, 1.264	
$X \to YY \to bq\tau_e\nu$	$0.9907 \pm <0.0001$	1.023 ± 0.009	1.89 ± 0.04	2.30 ± 0.05	1.12	1.22	1.11	1.16	1.19	1.61, 1.78	1.371, 1.354	
$X \to YY \to cq\tau_e\nu$	0.9835 ± 0.0001	1.018 ± 0.020	1.86 ± 0.05	2.11 ± 0.08	1.12	1.10	1.11	1.12	1.15	1.450, 1.530	1.290, 1.285	
$X \to YY \to sq\tau_e\nu$	0.9869 ± 0.0001	0.998 ± 0.006	1.79 ± 0.06	2.04 ± 0.11	1.08	1.09	1.09	1.07	1.11	1.345, 1.351	1.256, 1.224	
$X \to YY \to qq\tau_e\nu$	$0.9868 \pm <0.0001$	1.007 ± 0.005	1.83 ± 0.07	1.97 ± 0.13	1.07	1.07	1.05	1.08	1.08	1.259, 1.307	1.202, 1.221	
$X \to YY \to bc\tau_\mu\nu$	0.9917 ± 0.0001	1.003 ± 0.027	1.86 ± 0.06	2.302 ± 0.010	1.20	1.18	1.15	1.14	1.15	1.53, 1.64	1.286, 1.252	
$X \to YY \to cs\tau_\mu\nu$	0.9865 ± 0.0001	1.006 ± 0.010	1.95 ± 0.04	2.19 ± 0.15	1.09	1.09	1.11	1.14	1.13	1.459, 1.499	1.287, 1.284	
$X \to YY \to bq\tau_\mu\nu$	0.9918 ± 0.0002	1.014 ± 0.025	1.99 ± 0.07	2.65 ± 0.11	1.11	1.22	1.11	1.18	1.23	1.794, 1.879	1.401, 1.477	
$X \to YY \to cq\tau_\mu\nu$	0.9863 ± 0.0001	1.005 ± 0.012	1.96 ± 0.05	2.28 ± 0.12	1.10	1.12	1.08	1.11	1.12	1.512, 1.574	1.353, 1.343	
$X \to YY \to sq\tau_\mu\nu$	0.9887 ± 0.0001	1.004 ± 0.010	1.90 ± 0.06	2.10 ± 0.16	1.08	1.11	1.08	1.09	1.10	1.359, 1.369	1.269, 1.286	
$X \to YY \to qq\tau_\mu\nu$	0.9885 ± 0.0001	1.003 ± 0.011	1.925 ± 0.035	2.10 ± 0.12	1.10	1.09	1.11	1.10	1.10	1.344, 1.298	1.289, 1.266	
$X \to YY \to bc\tau_h\nu$	0.9902 ± 0.0001	0.994 ± 0.020	1.05 ± 0.04	1.50 ± 0.08	1.26	1.25	1.17	1.21	1.24	1.51, 1.61	1.313, 1.363	
$X \to YY \to cs\tau_h\nu$	0.9836 ± 0.0001	1.005 ± 0.007	1.058 ± 0.016	1.391 ± 0.013	1.13	1.10	1.10	1.12	1.14	1.382, 1.405	1.267, 1.247	
$X \to YY \to bq\tau_h\nu$	0.9899 ± 0.0001	0.995 ± 0.024	1.072 ± 0.023	1.53 ± 0.04	1.16	1.16	1.08	1.10	1.22	1.51, 1.62	1.291, 1.271	
$X \to YY \to cq\tau_h\nu$	0.9830 ± 0.0001	1.006 ± 0.011	1.063 ± 0.011	1.358 ± 0.001	1.12	1.09	1.09	1.13	1.13	1.369, 1.416	1.232, 1.262	
$X \to YY \to sq\tau_h\nu$	0.9876 ± 0.0001	1.015 ± 0.003	1.056 ± 0.006	1.409 ± 0.028	1.11	1.11	1.11	1.07	1.09	1.406, 1.352	1.273, 1.346	
$X \to YY \to qq\tau_h\nu$	$0.9871 \pm <0.0001$	1.002 ± 0.014	1.072 ± 0.006	1.35 ± 0.05	1.08	1.09	1.08	1.08	1.10	1.327, 1.262	1.229, 1.275	
QCD, b	0.9699 ± 0.0003	1.084 ± 0.019	1.107 ± 0.015	1.446 ± 0.025	1.20	1.20	1.18	1.17	1.20	1.437, 1.469	1.241, 1.226	
QCD, bb	0.9722 ± 0.0001	1.086 ± 0.006	1.132 ± 0.021	1.512 ± 0.015	1.21	1.21	1.18	1.16	1.21	1.589, 1.648	1.209, 1.222	
QCD, bbc	0.9694 ± 0.0004	1.091 ± 0.008	1.144 ± 0.019	1.578 ± 0.030	1.21	1.23	1.19	1.21	1.26	1.68, 1.79	1.261, 1.246	
QCD, $bbcc$	$0.9761 \pm <0.0001$	1.077 ± 0.007	1.128 ± 0.022	1.66 ± 0.06	1.24	1.27	1.24	1.22	1.25	1.729, 1.741	1.291, 1.265	
QCD, $bbccs$	0.9803 ± 0.0004	1.088 ± 0.005	1.198 ± 0.030	1.63 ± 0.12	1.29	1.24	1.30	1.23	1.27	1.717, 1.692	1.288, 1.225	
QCD, $bbccss$	0.9842 ± 0.0002	1.115 ± 0.031	1.188 ± 0.014	1.63 ± 0.09	1.28	1.27	1.32	1.23	1.33	1.788, 1.770	1.257, 1.268	
QCD, $bbcs$	0.9661 ± 0.0006	1.053 ± 0.016	1.129 ± 0.005	1.58 ± 0.05	1.20	1.25	1.23	1.16	1.13	1.621, 1.628	1.260, 1.218	
QCD, $bbcss$	0.9716 ± 0.0008	1.086 ± 0.016	1.16 ± 0.05	1.53 ± 0.07	1.14	1.23	1.23	1.23	1.25	1.613, 1.634	1.212, 1.224	
QCD, bbs	0.9694 ± 0.0001	1.082 ± 0.002	1.134 ± 0.006	1.49 ± 0.04	1.19	1.21	1.18	1.17	1.19	1.567, 1.591	1.196, 1.203	
QCD, $bbss$	0.9737 ± 0.0001	1.129 ± 0.016	1.191 ± 0.001	1.684 ± 0.029	1.25	1.25	1.27	1.22	1.28	1.754, 1.760	1.277, 1.291	
QCD, bc	0.9520 ± 0.0003	1.062 ± 0.004	1.096 ± 0.003	1.337 ± 0.029	1.15	1.14	1.10	1.14	1.17	1.337, 1.396	1.180, 1.160	
QCD, bcc	0.9534 ± 0.0006	1.067 ± 0.006	1.162 ± 0.009	1.598 ± 0.020	1.25	1.22	1.22	1.20	1.20	1.606, 1.605	1.356, 1.283	
QCD, $bccs$	0.9478 ± 0.0008	1.088 ± 0.022	1.16 ± 0.06	1.52 ± 0.07	1.23	1.26	1.20	1.22	1.26	1.396, 1.491	1.252, 1.311	
QCD, $bccss$	0.9571 ± 0.0008	1.09 ± 0.04	1.19 ± 0.04	1.503 ± 0.004	1.25	1.27	1.20	1.26	1.29	1.40, 1.55	1.318, 1.323	
QCD, bcs	0.9456 ± 0.0020	1.10 ± 0.06	1.14 ± 0.05	1.38 ± 0.15	1.37	1.10	1.22	1.11	1.25	1.440, 1.392	1.222, 1.244	
QCD, $bcss$	0.9400 ± 0.0006	1.122 ± 0.032	1.22 ± 0.04	1.52 ± 0.05	1.24	1.27	1.30	1.32	1.30	1.443, 1.486	1.326, 1.329	
QCD, bs	0.9610 ± 0.0002	1.078 ± 0.012	1.105 ± 0.009	1.408 ± 0.015	1.18	1.19	1.14	1.16	1.18	1.380, 1.436	1.234, 1.222	
QCD, bss	$0.9615 \pm <0.0001$	1.113 ± 0.006	1.167 ± 0.005	1.615 ± 0.025	1.26	1.24	1.22	1.24	1.27	1.579, 1.621	1.337, 1.333	
QCD, c	0.9593 ± 0.0002	1.059 ± 0.005	1.074 ± 0.008	1.288 ± 0.006	1.12	1.12	1.11	1.12	1.13	1.262, 1.292	1.153, 1.137	
QCD, cc	0.9558 ± 0.0002	1.070 ± 0.010	1.075 ± 0.007	1.475 ± 0.013	1.18	1.17	1.14	1.16	1.18	1.470, 1.541	1.224, 1.189	
QCD, ccs	0.9543 ± 0.0002	1.113 ± 0.019	1.105 ± 0.002	1.563 ± 0.011	1.21	1.24	1.22	1.21	1.22	1.560, 1.554	1.291, 1.273	
QCD, $ccss$	0.9595 ± 0.0003	1.129 ± 0.010	1.137 ± 0.005	1.540 ± 0.007	1.22	1.25	1.23	1.20	1.22	1.526, 1.553	1.290, 1.274	
QCD, cs	0.9517 ± 0.0002	1.083 ± 0.009	1.094 ± 0.004	1.291 ± 0.022	1.14	1.16	1.14	1.15	1.18	1.271, 1.289	1.169, 1.162	
QCD, css	0.9522 ± 0.0001	1.101 ± 0.010	1.126 ± 0.005	1.376 ± 0.009	1.16	1.19	1.17	1.16	1.20	1.383, 1.389	1.215, 1.223	
QCD, light	0.9791 ± 0.0001	1.047 ± 0.004	1.050 ± 0.006	1.148 ± 0.011	1.10	1.08	1.10	1.07	1.07	1.159, 1.114	1.103, 1.100	
QCD, s	$0.9727 \pm <0.0001$	$1.048 \pm <0.001$	1.044 ± 0.007	1.171 ± 0.009	1.09	1.08	1.09	1.07	1.09	1.172, 1.177	1.103, 1.113	
QCD, ss	0.9695 ± 0.0002	1.055 ± 0.007	1.071 ± 0.014	1.237 ± 0.019	1.09	1.10	1.09	1.09	1.10	1.191, 1.204	1.120, 1.129	

E.4 Background rejection against signal efficiency on the benchmarks

Tables 3 and 4 give the benchmark results at two working points. Figure 11 draws the whole receiver operating characteristic curve after fine-tuning, so that a difference

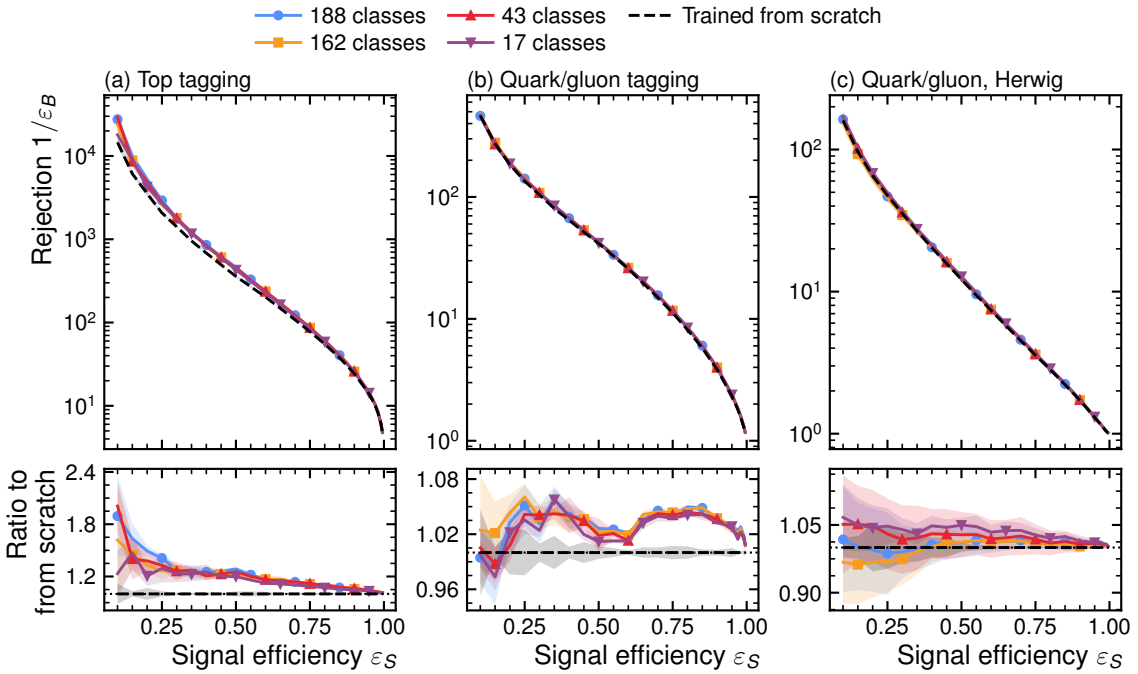


Figure 11: Background rejection $1/\epsilon_B$ against signal efficiency ϵ_S after fine-tuning on the full training sets: (a) top tagging, (b) quark/gluon discrimination on Pythia jets, with quark jets as the signal, and (c) the same networks scored on Herwig jets. Solid lines: the 188-, 162-, 43- and 17-class models fine-tuned with the missing inputs set to zero; dashed: the network trained from scratch; dash-dotted, on top only: the 188-class model with kinematic inputs only (one run). Lower panels: ratio to the network trained from scratch. Bands: standard deviation over the three pretraining runs, one fine-tuning each, and for the network trained from scratch over three training seeds.

between label sets at another signal efficiency would show.

E.5 Jet mass

Table 28 completes table 5 and figure 7. It gives the resolution and the median response of the regressed mass and of ridge and MLP regressions on frozen features, over all test jets and over resonant and QCD jets separately, and against jet p_{T} and generator soft-drop mass. The median response of the regressed mass lies close to one because its target, equation (1), is the logarithm of the mass ratio itself. Figure 12 shows the distributions of the response for resonant and QCD jets.

Table 29 gives the values behind figure 8 at QCD efficiencies of 10% and 1%, beside the distance that a random selection of as many QCD test jets gives.

E.6 Resonance tagging of $X \to b\bar{b}$ and $X \to bs$

Table 30 gives the signal efficiencies at fixed QCD efficiency behind figure 9, and table 31 the rejection of QCD jets at 50% and 30% signal efficiency for every configuration, with the transfer head, the direct discriminant and the closest-output-node control.

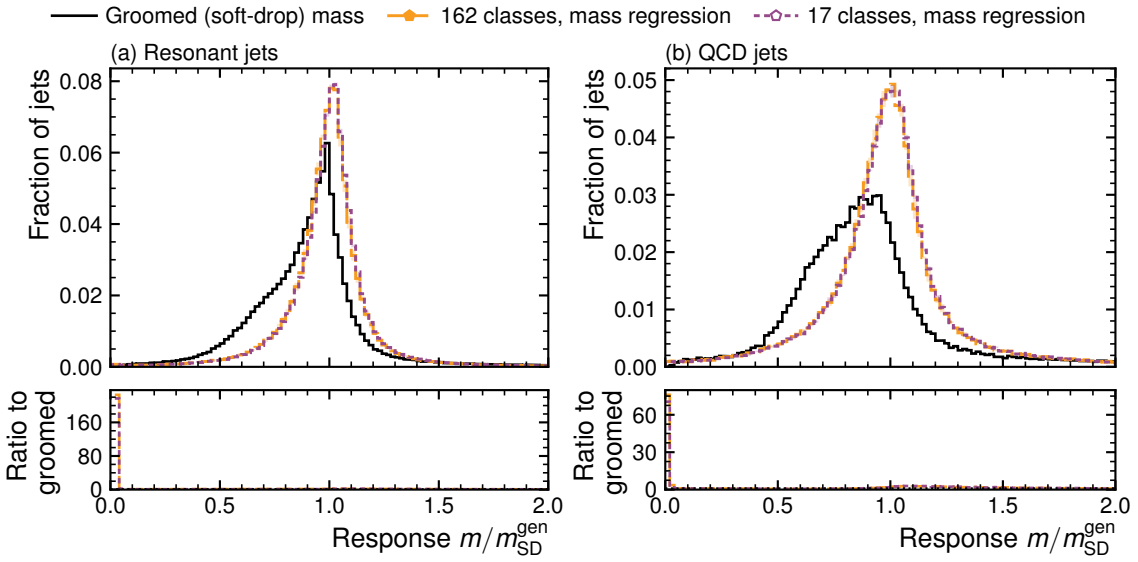


Figure 12: Distribution of the response $m_{\mathrm{est}}/m_{\mathrm{gen}}$, the estimated over the generator-level soft-drop mass, for (a) resonant jets and (b) QCD jets of the JetClass-II test set with a generator-level match: the reconstructed soft-drop mass, and the regressed mass of the 162- and 17-class models with the auxiliary mass regression (dashed). Bands span the two pretraining runs of each model with the auxiliary mass regression. The resolution and the median response in each group, and against jet transverse momentum and generator mass, are in table 28.

Table 12: Nonlinear probes (a multilayer perceptron with one hidden layer of 256 units, three seeds averaged) at the largest training set: JetClass-II AUC (all, resonant and QCD classes) with the ratio of $1-\text{AUC}$ to the 188-class run, JetClass AUC, and AUC and R_{30} on top, quark/gluon and Herwig jets (missing inputs set to zero); then the class pairs of table 26; last, the linear probe on the 512-unit hidden layer of the 188-class models' pretraining head beside the one on the class token. Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run.

	JetClass-II				JetClass	top		quark/gluon		Herwig	
pretraining	AUC	resonant	QCD	$\frac{1-\text{AUC}}{(1-\text{AUC})_{188}}$	AUC	AUC	R_{30}	AUC	R_{30}	AUC	R_{30}
188 classes	0.9806 ± <0.0001	0.9834 ± <0.0001	0.9643 ± 0.0001	1.00	0.9786 ± 0.0004	0.9713 ± 0.0034	290 ± 60	0.894 ± 0.004	82.6 ± 3.5	0.8077 ± 0.0006	36.64 ± 0.25
162 classes	0.9802 ± 0.0001	0.9833 ± 0.0001	0.9618 ± 0.0003	1.019 ± 0.007	0.9786 ± 0.0002	0.971 ± 0.004	280 ± 90	0.889 ± 0.005	77.0 ± 2.2	0.8060 ± 0.0014	35.6 ± 0.6
43 classes	0.9773 ± 0.0001	0.9802 ± 0.0002	0.9605 ± 0.0001	1.170 ± 0.006	0.9782 ± 0.0003	0.9688 ± 0.0029	250 ± 60	0.8904 ± 0.0031	81 ± 4	0.8071 ± 0.0007	35.67 ± 0.28
17 classes	0.9731 ± 0.0004	0.9766 ± 0.0004	0.9524 ± 0.0004	1.388 ± 0.021	0.9776 ± 0.0005	0.9688 ± 0.0018	224 ± 24	0.887 ± 0.005	78.9 ± 1.7	0.8062 ± 0.0024	36.2 ± 0.5
162 classes, mass regression	[no run yet]	[no run yet]	[no run yet]	[no run yet]	0.9794, 0.9799	0.9731, 0.9754	317, 376	0.8932, 0.8914	81.8, 78.4	0.8094, 0.8071	37.64, 36.83
17 classes, mass regression	[no run yet]	[no run yet]	[no run yet]	[no run yet]	0.9787, 0.9785	0.9747, 0.9704	356, 261	0.8933, 0.8927	80.91, 81.90	0.8079, 0.8084	35.95, 36.47
random partition 1, seed 1	[no run yet]	[no run yet]	[no run yet]	[no run yet]	0.974	0.964	161	0.893	80.9	0.809	36.6
random partition 2, seed 2	[no run yet]	[no run yet]	[no run yet]	[no run yet]	0.975	0.968	210	0.888	79.0	0.806	35.2
random partition 3, seed 1	[no run yet]	[no run yet]	[no run yet]	[no run yet]	0.974	0.967	190	0.894	80.5	0.806	35.2
random partition 4, seed 2	[no run yet]	[no run yet]	[no run yet]	[no run yet]	0.974	0.959	125	0.881	72.0	0.806	35.5
random partition 5, seed 2	[no run yet]	[no run yet]	[no run yet]	[no run yet]	0.975	0.970	251	0.886	75.5	0.805	35.4
random partition, no flavor boundary	0.9749, 0.9746	0.9786, 0.9783	0.9528, 0.9522	1.294, 1.310	0.9764, 0.9763	0.9668, 0.9682	210.1, 218.3	0.8912, 0.8921	77.5, 83.1	0.8073, 0.8071	35.27, 34.98
random partition, b-quark boundary	0.9774, 0.9775	0.9806, 0.9807	0.9582, 0.9584	1.168, 1.159	0.9767, 0.9776	0.9718, 0.9724	304.1, 313.5	0.8887, 0.8911	80.8, 78.6	0.8069, 0.8067	36.08, 35.57
random partition, random flavor boundary	0.9776, 0.9777	0.9810, 0.9811	0.9571, 0.9576	1.158, 1.150	0.9767, 0.9763	0.9670, 0.9626	215, 160	0.8906, 0.8870	80.6, 77.9	0.8074, 0.8052	35.31, 34.53
randomly initialized network	0.8648 ± 0.0014	0.8694 ± 0.0017	0.8372 ± 0.0015	–	0.8859 ± 0.0021	0.9667 ± 0.0006	210 ± 4	0.8844 ± 0.0001	76.6 ± 0.8	0.8074 ± 0.0007	38.2 ± 0.4
randomly initialized network, BatchNorm recomputed	0.8974 ± 0.0018	0.9035 ± 0.0020	0.8610 ± 0.0008	–	0.9146 ± 0.0012	0.9712 ± 0.0024	300 ± 50	0.8831 ± 0.0003	76.3 ± 1.9	0.8087 ± 0.0009	37.8 ± 1.5

pretraining	$X \to bc$ vs bq, cs		$X \to b\bar{b}$ vs $c\bar{c}$		$X \to YY \to bbqq$ vs $ccqq$	
	AUC	$\frac{1-\text{AUC}}{(1-\text{AUC})_{188}}$	AUC	$\frac{1-\text{AUC}}{(1-\text{AUC})_{188}}$	AUC	$\frac{1-\text{AUC}}{(1-\text{AUC})_{188}}$
188 classes	0.9434 ± 0.0001	1.00	0.9978 ± <0.0001	1.00	0.9941 ± 0.0001	1.00
162 classes	0.9432 ± 0.0002	1.002 ± 0.006	0.9977 ± <0.0001	1.023 ± 0.021	0.9941 ± 0.0001	1.011 ± 0.030
43 classes	0.9375 ± 0.0008	1.104 ± 0.017	0.9975 ± <0.0001	1.125 ± 0.011	0.9935 ± <0.0001	1.112 ± 0.006
17 classes	0.9080 ± 0.0013	1.624 ± 0.022	0.9932 ± 0.0002	3.04 ± 0.09	0.9828 ± 0.0011	2.93 ± 0.17
162 classes, mass regression	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
17 classes, mass regression	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition 1, seed 1	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition 2, seed 2	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition 3, seed 1	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition 4, seed 2	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition 5, seed 2	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition, no flavor boundary	0.9077, 0.9046	1.633, 1.679	0.9926, 0.9902	3.3, 4.3	0.9813, 0.9776	3.20, 3.78
random partition, b-quark boundary	0.9216, 0.9227	1.385, 1.360	0.9961, 0.9963	1.74, 1.64	0.9916, 0.9919	1.442, 1.372
random partition, random flavor boundary	0.9231, 0.9242	1.359, 1.335	0.9951, 0.9946	2.21, 2.37	0.9889, 0.9891	1.890, 1.843
randomly initialized network	0.630 ± 0.004	–	0.759 ± 0.009	–	0.747 ± 0.006	–
randomly initialized network, BatchNorm recomputed	0.6865 ± 0.0030	–	0.866 ± 0.008	–	0.837 ± 0.006	–

pretraining	QCD bb vs cc		QCD with b vs light		QCD with c, no b, vs light	
	AUC	$\frac{1-\text{AUC}}{(1-\text{AUC})_{188}}$	AUC	$\frac{1-\text{AUC}}{(1-\text{AUC})_{188}}$	AUC	$\frac{1-\text{AUC}}{(1-\text{AUC})_{188}}$
188 classes	0.9871 ± 0.0001	1.00	0.9882 ± <0.0001	1.00	0.9329 ± 0.0001	1.00
162 classes	0.9860 ± 0.0003	1.085 ± 0.030	0.9876 ± 0.0001	1.050 ± 0.004	0.9303 ± 0.0002	1.039 ± 0.005
43 classes	0.9855 ± 0.0001	1.121 ± 0.017	0.9875 ± 0.0001	1.055 ± 0.005	0.9292 ± 0.0003	1.055 ± 0.006
17 classes	0.9701 ± 0.0008	2.31 ± 0.04	0.9842 ± 0.0002	1.337 ± 0.021	0.9145 ± 0.0004	1.274 ± 0.008
162 classes, mass regression	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
17 classes, mass regression	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition 1, seed 1	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition 2, seed 2	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition 3, seed 1	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition 4, seed 2	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition 5, seed 2	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition, no flavor boundary	0.9680, 0.9647	2.50, 2.72	0.9845, 0.9847	1.315, 1.288	0.9174, 0.9186	1.234, 1.211
random partition, b-quark boundary	0.9817, 0.9821	1.432, 1.372	0.9863, 0.9865	1.158, 1.140	0.9224, 0.9229	1.159, 1.147
random partition, random flavor boundary	0.9776, 0.9775	1.745, 1.725	0.9858, 0.9859	1.201, 1.187	0.9236, 0.9247	1.141, 1.120
randomly initialized network	0.7228 ± 0.0031	–	0.8185 ± 0.0019	–	0.682 ± 0.005	–
randomly initialized network, BatchNorm recomputed	0.822 ± 0.005	–	0.8879 ± 0.0033	–	0.7143 ± 0.0013	–

188 classes, linear probe on	JetClass-II AUC	quark/gluon AUC	quark/gluon R_{30}
class token	0.9801 ± <0.0001	0.878 ± 0.006	66 ± 7
512-unit hidden layer	0.9802 ± <0.0001	0.880 ± 0.007	70 ± 6

Table 13: Linear probes on JetClass-II (natural composition, largest training set) with the BatchNorm statistics as accumulated in pretraining and recomputed on training jets: the AUC averaged over the classes and the ratio of $1 - \mathrm{AUC}$ to the 188-class run. The class-balanced probes use the checkpoints as trained, so this check uses the natural-composition test set. Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run.

	AUC		$\frac{1-\mathrm{AUC}}{(1-\mathrm{AUC})_{188}}$	
pretraining	as trained	recomputed	as trained	recomputed
188 classes	$0.9760 \pm <0.0001$	$0.9760 \pm <0.0001$	1.00	1.00
162 classes	0.9748 ± 0.0003	0.9748 ± 0.0003	1.049 ± 0.014	1.050 ± 0.014
43 classes	0.9719 ± 0.0006	0.9719 ± 0.0006	1.171 ± 0.024	1.172 ± 0.025
17 classes	0.9631 ± 0.0006	0.9631 ± 0.0006	1.538 ± 0.025	1.537 ± 0.026
162 classes, mass regression	0.9735, 0.9740	0.9735, 0.9740	1.106, 1.082	1.105, 1.083
17 classes, mass regression	0.9623, 0.9632	0.9623, 0.9632	1.571, 1.530	1.572, 1.530
random partition 1, seed 1	0.972	0.972	1.17	1.17
random partition 2, seed 2	0.972	0.972	1.17	1.17
random partition 3, seed 1	0.972	0.972	1.15	1.15
random partition 4, seed 2	0.972	0.972	1.15	1.15
random partition 5, seed 2	0.972	0.972	1.16	1.16
random partition, no flavor boundary	0.9657, 0.9645	0.9657, 0.9645	1.432, 1.477	1.432, 1.477
random partition, b-quark boundary	0.9698, 0.9701	0.9698, 0.9701	1.260, 1.245	1.260, 1.245
random partition, random flavor boundary	0.9698, 0.9698	0.9698, 0.9698	1.259, 1.254	1.260, 1.254
randomly initialized network	0.8058 ± 0.0018	0.8525 ± 0.0019	–	–

Table 14: Uncertainty of the label-set comparisons at the largest training set, in per cent: the standard deviation of the logarithm of the ratio over the pretraining runs, then that from the finite test sample. JetClass-II: the ratio of $1 - \mathrm{AUC}$, the test-sample part from a bootstrap of the class-balanced test jets (200 resamples within each class). Top and quark/gluon: the ratio of R_{30}, the test-sample part from the number of background jets passing the cut.

	linear probe			fine-tuning		
ratio	JetClass-II, $1 - \mathrm{AUC}$	top, R_{30}	q/g, R_{30}	JetClass-II, $1 - \mathrm{AUC}$	top, R_{30}	q/g, R_{30}
162 classes / 188 classes	– / –	25 / 2.6	24 / 3.5	– / –	3.2 / 13	2.6 / 4.6
43 classes / 188 classes	– / –	29 / 2.4	20 / 3.6	– / –	7.3 / 13	2.8 / 4.6
17 classes / 188 classes	– / –	16 / 2.4	9.8 / 3.6	– / –	9.9 / 13	2.6 / 4.6

Table 15: JetClass-II linear probes with 10^6 training jets on the natural-composition test set and on the class-balanced one: AUC averaged over the classes and the ratio of $1 - \text{AUC}$ to the 188-class run. Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run.

	natural composition		class-balanced	
pretraining	AUC	$\frac{1-\text{AUC}}{(1-\text{AUC})_{188}}$	AUC	$\frac{1-\text{AUC}}{(1-\text{AUC})_{188}}$
188 classes	$0.9760 \pm <0.0001$	1.00	$0.9801 \pm <0.0001$	1.00
162 classes	0.9748 ± 0.0003	1.049 ± 0.014	0.9796 ± 0.0001	1.026 ± 0.005
43 classes	0.9719 ± 0.0006	1.171 ± 0.024	0.9756 ± 0.0002	1.227 ± 0.009
17 classes	0.9631 ± 0.0006	1.538 ± 0.025	0.9691 ± 0.0004	1.557 ± 0.019
162 classes, mass regression	0.9735, 0.9740	1.106, 1.082	0.9791, 0.9794	1.053, 1.037
17 classes, mass regression	0.9623, 0.9632	1.571, 1.530	0.9679, 0.9696	1.617, 1.528
random partition 1, seed 1	0.972	1.17	0.978	1.13
random partition 2, seed 2	0.972	1.17	0.978	1.13
random partition 3, seed 1	0.972	1.15	0.978	1.11
random partition 4, seed 2	0.972	1.15	0.978	1.12
random partition 5, seed 2	0.972	1.16	0.977	1.13
random partition, no flavor boundary	0.9657, 0.9645	1.432, 1.477	0.9717, 0.9710	1.424, 1.462
random partition, b-quark boundary	0.9698, 0.9701	1.260, 1.245	0.9758, 0.9759	1.218, 1.212
random partition, random flavor boundary	0.9698, 0.9698	1.259, 1.254	0.9760, 0.9760	1.206, 1.208
randomly initialized network, BatchNorm recomputed	0.8525 ± 0.0019	–	0.8660, 0.8630 **[2 of 3]**	–

Table 16: ParT's top-tagging training script on our cluster, then with Sophon's output head, then also with Sophon's rescaling of every jet's four-momenta to $p_\text{T} = 500$ GeV (cumulative, one run each), beside ParT's published row and our network trained from scratch, which uses the rescaling.

	accuracy	AUC	R_{50}	R_{30}	background jets passing at R_{30}
ParT's script, unchanged	0.9398	0.9858	424	1,711	118
+ Sophon's output head	0.9403	0.9859	416	1,577	128
+ Sophon's jet-p_T rescaling	0.9385	0.9850	357	1,224	165
ParT, published [8]	0.940	0.9858	413 ± 16	1602 ± 81	–
our network from scratch	0.9387 ± 0.0002	0.9852 ± 0.0001	358 ± 11	$1{,}412 \pm 20$	–

Table 17: Linear probes on the frozen features at the largest training set of each task. Upper part: JetClass-II (class-balanced test set) accuracy and AUC averaged over its 188 classes each against the rest, over the classes with the full 5,000 test jets, over the resonant and over the QCD classes; JetClass accuracy and AUC. Lower part: top tagging and quark/gluon discrimination (also on Herwig jets) with the missing inputs substituted as chosen on validation for every model ([**fill fits**] for top, [**fill fits**] for quark/gluon; column headings): accuracy, AUC and the background rejection $1/\epsilon_B$ at 50% and 30% signal efficiency. The ratio column is $(1 - \text{AUC})$ over that of the 188-class run of the same seed (a random partition: of its seed). Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run. AUC: area under the curve of signal efficiency against background efficiency. *: a probe that did not converge.

	JetClass-II						JetClass	
pretraining	accuracy	AUC	AUC, full count	AUC, resonant	AUC, QCD	$\frac{1-\text{AUC}}{(1-\text{AUC})_{188}}$	accuracy	AUC
188 classes	0.4348 ± 0.0003	$0.9801 \pm <0.0001$	$0.9813 \pm <0.0001$	$0.9830 \pm <0.0001$	$0.9632 \pm <0.0001$	1.00	0.7709 ± 0.0023	0.9682 ± 0.0006
162 classes	0.4328 ± 0.0008	0.9796 ± 0.0001	0.9810 ± 0.0001	0.9829 ± 0.0001	0.9600 ± 0.0002	1.026 ± 0.005	0.7709 ± 0.0005	0.9683 ± 0.0002
43 classes	0.361 ± 0.005	0.9756 ± 0.0002	0.9769 ± 0.0002	0.9785 ± 0.0002	0.9583 ± 0.0003	1.227 ± 0.009	0.7695 ± 0.0014	0.9680 ± 0.0004
17 classes	0.334 ± 0.007	0.9691 ± 0.0004	0.9708 ± 0.0005	0.9729 ± 0.0005	0.9460 ± 0.0006	1.557 ± 0.019	0.7605 ± 0.0027	0.9650 ± 0.0008
162 classes, mass regression	0.4271, 0.4305	0.9791, 0.9794	0.9805, 0.9808	0.9824, 0.9827	0.9592, 0.9598	1.053, 1.037	0.7718, 0.7760	0.9684, 0.9697
17 classes, mass regression	0.312, 0.350	0.9679, 0.9696	0.9695, 0.9714	0.9716, 0.9737	0.9458, 0.9457	1.617, 1.528	0.7624, 0.7612	0.9655, 0.9651
random partition 1, seed 1	0.420	0.978	0.979	0.981	0.956	1.13	0.753	0.964
random partition 2, seed 2	0.421	0.978	0.979	0.981	0.956	1.13	0.755	0.964
random partition 3, seed 1	0.423	0.978	0.979	0.982	0.956	1.11	0.755	0.964
random partition 4, seed 2	0.423	0.978	0.979	0.981	0.956	1.12	0.754	0.964
random partition 5, seed 2	0.420	0.977	0.979	0.981	0.955	1.13	0.754	0.964
random partition, no flavor boundary	0.3769, 0.3723	0.9717, 0.9710	0.9736, 0.9729	0.9760, 0.9753	0.9460, 0.9449	1.424, 1.462	0.7535, 0.7486	0.9633, 0.9622
random partition, b-quark boundary	0.3945, 0.3963	0.9758, 0.9759	0.9774, 0.9775	0.9794, 0.9795	0.9543, 0.9545	1.218, 1.212	0.7615, 0.7644	0.9661, 0.9665
random partition, random flavor boundary	0.4071, 0.4089	0.9760, 0.9760	0.9777, 0.9777	0.9798, 0.9798	0.9536, 0.9535	1.206, 1.208	0.7593, 0.7564	0.9654, 0.9645
randomly initialized network	0.07183, 0.07119 [2 of 3]	0.8110, 0.8111 [2 of 3]	0.8113, 0.8108 [2 of 3]	0.8148, 0.8131 [2 of 3]	0.788, 0.799 [2 of 3]	–	0.4062, 0.4087 [2 of 3]	0.8268, 0.8282 [2 of 3]
randomly initialized network, BatchNorm recomputed	0.1019, 0.0999 [2 of 3]	0.8660, 0.8630 [2 of 3]	0.8685, 0.8650 [2 of 3]	0.8733, 0.8694 [2 of 3]	0.8223, 0.8246 [2 of 3]	–	0.4764, 0.4691 [2 of 3]	0.8649, 0.8618 [2 of 3]

	top missing inputs: zero [choice on validation]				quark/gluon missing inputs: zero [choice on validation]				quark/gluon, Herwig missing inputs: zero [choice on validation]			
pretraining	accuracy	AUC	R_{50}	R_{30}	accuracy	AUC	R_{50}	R_{30}	accuracy	AUC	R_{50}	R_{30}
188 classes	0.883 ± 0.008	0.941 ± 0.008	28 ± 6	60 ± 12	0.803 ± 0.006	0.878 ± 0.006	23.5 ± 2.4	66 ± 7	0.7192 ± 0.0020	0.8038 ± 0.0017	11.18 ± 0.14	32.8 ± 1.0
162 classes	0.889 ± 0.009	0.948 ± 0.006	33 ± 7	76 ± 17	0.796 ± 0.008	0.869 ± 0.009	21.0 ± 2.6	58 ± 9	0.7195 ± 0.0020	0.800 ± 0.006	10.5 ± 0.7	30.4 ± 3.2
43 classes	0.875 ± 0.010	0.938 ± 0.009	27 ± 6	57 ± 13	0.796 ± 0.004	0.871 ± 0.006	21.8 ± 1.9	61 ± 9	0.7189 ± 0.0029	0.8004 ± 0.0027	10.60 ± 0.22	30.8 ± 1.9
17 classes	0.886 ± 0.006	0.942 ± 0.004	26.9 ± 2.8	55 ± 5	0.793 ± 0.010	0.866 ± 0.011	21.6 ± 2.6	62 ± 7	0.718 ± 0.004	0.799 ± 0.005	11.0 ± 0.5	33.4 ± 1.4
162 classes, mass regression	0.8918, 0.8944	0.9491, 0.9529	34.4, 39.9	77, 95	0.8010, 0.7998	0.8764, 0.8748	23.9, 22.7	65.4, 61.1	0.7233, 0.7210	0.8055, 0.8008	11.63, 10.97	34.6, 32.1
17 classes, mass regression	0.9026, 0.8944	0.9582, 0.9482	45, 32	103, 69	0.7978, 0.7959	0.8735, 0.8719	22.85, 22.64	66.01, 65.53	0.7171, 0.7194	0.8004, 0.8008	10.98, 11.17	32.7, 34.3
random partition 1, seed 1	0.868	0.939	21.0	44.2	0.798	0.875	22.6	66.0	0.718	0.802	11.0	32.8
random partition 2, seed 2	0.882	0.943	28.1	60.5	0.794	0.869	21.2	59.0	0.716	0.799	10.5	30.3
random partition 3, seed 1	0.879	0.939	24.4	50.0	0.800	0.877	22.8	65.2	0.717	0.799	10.4	29.8
random partition 4, seed 2	0.862	0.918	16.3	30.2	0.788	0.860	19.2	51.0	0.717	0.796	10.2	28.4
random partition 5, seed 2	0.887	0.947	32.2	73.9	0.787	0.862	19.8	54.9	0.718	0.794	10.1	27.8
random partition, no flavor boundary	0.8806, 0.8862	0.9420, 0.9453	28.73, 29.14	69.67, 69.31	0.8015, 0.8066	0.8762, 0.8748	22.2, 23.3	63.1, 66.4	0.7210, 0.7154	0.8041, 0.8004	11.22, 11.03	32.9, 31.8
random partition, b-quark boundary	0.8897, 0.8919	0.9484, 0.9505	33.5, 34.6	73.7, 80.7	0.7932, 0.7996	0.8671, 0.8744	20.6, 21.6	54.6, 59.4	0.7211, 0.7201	0.7979, 0.8018	10.61, 10.78	30.28, 30.65
random partition, random flavor boundary	0.8775, 0.8751	0.9403, 0.9324	27.6, 20.9	65, 41	0.7969, 0.7932	0.8736, 0.8666	22.1, 20.0	66, 54	0.7207, 0.7147	0.8038, 0.7967	11.02, 10.32	33.2, 28.7
randomly initialized network	0.8828, 0.8819 [2 of 3]	0.9395, 0.9385 [2 of 3]	25.42, 24.57 [2 of 3]	59.0, 57.3 [2 of 3]	0.7869, 0.7853 [2 of 3]	0.8628, 0.8610 [2 of 3]	19.68, 19.41 [2 of 3]	57.6, 55.8 [2 of 3]	0.7204, 0.7185 [2 of 3]	0.7969, 0.7955 [2 of 3]	10.66, 10.62 [2 of 3]	31.2, 32.5 [2 of 3]
randomly initialized network, BatchNorm recomputed	0.8818, 0.8798 [2 of 3]	0.9371, 0.9389 [2 of 3]	22.9, 25.6 [2 of 3]	46, 57 [2 of 3]	0.7846, 0.7867 [2 of 3]	0.8609, 0.8611 [2 of 3]	18.83, 18.84 [2 of 3]	53.9, 52.0 [2 of 3]	0.7189, 0.7206 [2 of 3]	0.7971, 0.7985 [2 of 3]	10.36, 10.68 [2 of 3]	30.55, 31.14 [2 of 3]

Table 18: Linear probes against the number of training jets (column headers): JetClass-II AUC averaged over the classes, JetClass accuracy, and R_{30} on top, quark/gluon and Herwig jets with the missing inputs substituted as chosen on validation ([fill fits] for top, [fill fits] for quark/gluon). Mean ± standard deviation over the three pretraining runs. *: a probe that did not converge.

JetClass-II, AUC	1,000	10,000	100,000	1,000,000
188 classes	0.9497 ± 0.0001	0.9701 ± 0.0001	0.9783 ± 0.0001	$0.9801 \pm <0.0001$
162 classes	0.9466 ± 0.0006	0.9692 ± 0.0002	0.9778 ± 0.0001	0.9796 ± 0.0001
43 classes	0.9324 ± 0.0029	0.9633 ± 0.0011	0.9735 ± 0.0003	0.9756 ± 0.0002
17 classes	0.9046 ± 0.0021	0.9508 ± 0.0006	0.9660 ± 0.0004	0.9691 ± 0.0004
162 classes, mass regression	0.9445, 0.9435	0.9684, 0.9684	0.9772, 0.9775	0.9791, 0.9794
17 classes, mass regression	0.8999, 0.9044	0.9483, 0.9518	0.9646, 0.9665	0.9679, 0.9696
randomly initialized network	0.6140, 0.6197 [2 of 3]	0.7140, 0.7171 [2 of 3]	0.7931, 0.7934 [2 of 3]	0.8110, 0.8111 [2 of 3]
randomly initialized network, BatchNorm recomputed	0.637, 0.652 [2 of 3]	0.7905, 0.7958 [2 of 3]	0.8543, 0.8515 [2 of 3]	0.8660, 0.8630 [2 of 3]

JetClass, accuracy	1,000	10,000	100,000	1,000,000
188 classes	0.711 ± 0.006	0.7529 ± 0.0008	0.7686 ± 0.0022	0.7709 ± 0.0023
162 classes	0.714 ± 0.008	0.7537 ± 0.0018	0.7687 ± 0.0002	0.7709 ± 0.0005
43 classes	0.7109 ± 0.0005	0.7521 ± 0.0004	0.7679 ± 0.0012	0.7695 ± 0.0014
17 classes	0.689 ± 0.006	0.742 ± 0.004	0.7586 ± 0.0026	0.7605 ± 0.0027
162 classes, mass regression	0.7083, 0.7093	0.7543, 0.7574	0.7694, 0.7741	0.7718, 0.7760
17 classes, mass regression	0.6863, 0.6947	0.7447, 0.7426	0.7606, 0.7595	0.7624, 0.7612
randomly initialized network	0.3057, 0.3121 [2 of 3]	0.3851, 0.3871 [2 of 3]	0.4031, 0.4061 [2 of 3]	0.4062, 0.4087 [2 of 3]
randomly initialized network, BatchNorm recomputed	0.3579, 0.3585 [2 of 3]	0.4510, 0.4446 [2 of 3]	0.4740, 0.4661 [2 of 3]	0.4764, 0.4691 [2 of 3]

top, R_{30} (missing inputs: zero [choice on validation])	1,000	10,000	100,000	1,200,000
188 classes	39 ± 10	59 ± 11	60 ± 12	60 ± 12
162 classes	46 ± 11	73 ± 18	75 ± 16	76 ± 17
43 classes	42 ± 9	56 ± 13	57 ± 13	57 ± 13
17 classes	37.5 ± 3.2	52 ± 6	54 ± 4	55 ± 5
162 classes, mass regression	43.9, 46.2	75, 86	80, 96	77, 95
17 classes, mass regression	70, 41	103, 60	104, 68	103, 69
randomly initialized network	46.4, 40.0 [2 of 3]	60.3, 57.1 [2 of 3]	59.1, 57.3 [2 of 3]	59.0, 57.3 [2 of 3]
randomly initialized network, BatchNorm recomputed	28, 40 [2 of 3]	47.8, 55.7 [2 of 3]	46, 57 [2 of 3]	46, 57 [2 of 3]

quark/gluon, R_{30} (missing inputs: zero [choice on validation])	1,000	10,000	100,000	1,600,000
188 classes	53 ± 6	63 ± 7	66 ± 7	66 ± 7
162 classes	44 ± 7	57 ± 7	58 ± 9	58 ± 9
43 classes	50 ± 11	59 ± 8	61 ± 9	61 ± 9
17 classes	54 ± 8	59 ± 7	62 ± 6	62 ± 7
162 classes, mass regression	55.8, 46.8	62.7, 58.5	65.6, 60.6	65.4, 61.1
17 classes, mass regression	54.08, 54.20	64.4, 61.7	66.09, 65.57	66.01, 65.53
randomly initialized network	44.40, 44.44 [2 of 3]	55.9, 54.3 [2 of 3]	57.27, 56.37 [2 of 3]	57.6, 55.8 [2 of 3]
randomly initialized network, BatchNorm recomputed	38.7, 43.9 [2 of 3]	50.76, 50.63 [2 of 3]	53.33, 52.47 [2 of 3]	53.9, 52.0 [2 of 3]

quark/gluon, Herwig, R_{30} (missing inputs: zero [choice on validation])	1,000	10,000	100,000	1,600,000
188 classes	28.7 ± 1.9	32.5 ± 1.1	32.9 ± 1.2	32.8 ± 1.0
162 classes	24.5 ± 2.7	29.8 ± 2.6	30.6 ± 3.1	30.4 ± 3.2
43 classes	26 ± 4	29.0 ± 2.4	30.6 ± 1.9	30.8 ± 1.9
17 classes	29.1 ± 3.1	31.6 ± 1.0	33.4 ± 1.5	33.4 ± 1.4
162 classes, mass regression	30.0, 26.1	33.8, 30.5	34.4, 32.2	34.6, 32.1
17 classes, mass regression	27.1, 29.5	32.11, 32.47	32.4, 34.1	32.7, 34.3
randomly initialized network	25.55, 26.09 [2 of 3]	30.62, 31.30 [2 of 3]	31.4, 33.3 [2 of 3]	31.2, 32.5 [2 of 3]
randomly initialized network, BatchNorm recomputed	22.9, 26.8 [2 of 3]	29.30, 30.21 [2 of 3]	30.1, 31.2 [2 of 3]	30.55, 31.14 [2 of 3]

Table 19: Fine-tuning against the number of training jets (column headers), in each task's metric: AUC averaged over the classes on class-balanced JetClass-II (all classes and the QCD classes), accuracy on JetClass, and R_{30} on top, quark/gluon and Herwig jets. Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run. Each pretrained run is fine-tuned once; from scratch: three training seeds on the same jets.

JetClass-II **[not yet read out]**

JetClass-II, QCD classes **[not yet read out]**

JetClass, accuracy	1,000,000
188 classes	0.8491 ± 0.0005
162 classes	0.8495 ± 0.0002
43 classes	0.8480 ± 0.0006
17 classes	0.8434 ± 0.0001
162 classes, mass regression	0.8478, 0.8490
17 classes, mass regression	0.8420, 0.8419
random partition 1, seed 1	0.845
random partition 2, seed 2	0.846
random partition 3, seed 1	0.847
random partition 4, seed 2	0.846
random partition 5, seed 2	0.846
random partition, no flavor boundary	0.8446, 0.8442
random partition, b-quark boundary	0.8451, 0.8469
random partition, random flavor boundary	0.8456, 0.8454
from scratch	0.8269 ± 0.0010

top, R_{30}	1,200,000
188 classes	$1{,}800 \pm 60$
162 classes	$1{,}790 \pm 60$
43 classes	$1{,}780 \pm 150$
17 classes	$1{,}740 \pm 140$
162 classes, mass regression	1,941, 1,590
17 classes, mass regression	1,683, 1,628
random partition 1, seed 1	1,660
random partition 2, seed 2	1,900
random partition 3, seed 1	1,640
random partition 4, seed 2	1,820
random partition 5, seed 2	1,740
random partition, no flavor boundary	1,726, 1,852
random partition, b-quark boundary	1,590, 2,060
random partition, random flavor boundary	1,565, 1,655
from scratch	$1{,}412 \pm 20$

q/g, R_{30}	1,600,000
188 classes	107.3 ± 2.2
162 classes	108.0 ± 1.4
43 classes	108.3 ± 1.0
17 classes	106.5 ± 0.8
162 classes, mass regression	107.0, 106.6
17 classes, mass regression	104.9, 105.6
random partition 1, seed 1	107
random partition 2, seed 2	109

Table 20: Fine-tuning on the full training sets of top and quark/gluon (also scored on Herwig jets), every configuration: accuracy, AUC and $1/\epsilon_B$ at 50% and 30% signal efficiency. Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run. Every rejection rests on at least 98 background jets passing the cut.

pretraining	top				quark/gluon				quark/gluon, Herwig			
	accuracy	AUC	R_{50}	R_{30}	accuracy	AUC	R_{50}	R_{30}	accuracy	AUC	R_{50}	R_{30}
188 classes	0.9400 ± 0.0002	0.9861 ± 0.0001	461 ± 4	$1{,}800 \pm 60$	0.8433 ± 0.0002	0.9148 ± 0.0001	42.56 ± 0.10	107.3 ± 2.2	0.712 ± 0.005	0.8078 ± 0.0023	12.3 ± 0.4	35.2 ± 2.1
162 classes	0.9401 ± 0.0002	0.9861 ± 0.0001	443 ± 17	$1{,}790 \pm 60$	$0.8432 \pm <0.0001$	0.9148 ± 0.0001	42.39 ± 0.22	108.0 ± 1.4	0.7100 ± 0.0017	0.8074 ± 0.0010	12.33 ± 0.28	34.6 ± 2.0
43 classes	0.9402 ± 0.0004	0.9861 ± 0.0001	450 ± 13	$1{,}780 \pm 150$	0.8430 ± 0.0005	0.9146 ± 0.0003	42.2 ± 0.8	108.3 ± 1.0	0.713 ± 0.006	0.809 ± 0.005	12.5 ± 0.6	36.1 ± 2.7
17 classes	0.9398 ± 0.0001	0.9859 ± 0.0001	429 ± 10	$1{,}740 \pm 140$	0.8427 ± 0.0003	$0.9144 \pm <0.0001$	41.97 ± 0.25	106.5 ± 0.8	0.7175 ± 0.0033	0.8103 ± 0.0018	12.79 ± 0.20	36.9 ± 0.9
162 classes, mass regression	0.9406, 0.9402	0.9860, 0.9860	420, 436	1,941, 1,590	0.8425, 0.8427	0.9147, 0.9145	42.46, 42.21	107.0, 106.6	0.7178, 0.7121	0.8109, 0.8074	12.99, 12.21	37.8, 34.4
17 classes, mass regression	0.9397, 0.9399	0.9858, 0.9859	435, 421	1,683, 1,628	0.8425, 0.8424	0.9142, 0.9140	41.77, 41.39	104.9, 105.6	0.7160, 0.7107	0.8101, 0.8052	12.58, 11.96	37.7, 32.9
random partition 1, seed 1	0.940	0.986	432	1,660	0.843	0.915	42.1	107	0.712	0.807	12.2	33.4
random partition 2, seed 2	0.940	0.986	430	1,900	0.843	0.915	42.5	109	0.712	0.808	12.3	35.2
random partition 3, seed 1	0.940	0.986	421	1,640	0.843	0.914	42.1	106	0.711	0.807	12.2	34.8
random partition 4, seed 2	0.940	0.986	458	1,820	0.843	0.915	41.8	107	0.714	0.809	12.7	37.5
random partition 5, seed 2	0.940	0.986	431	1,740	0.843	0.915	42.1	106	0.713	0.808	12.2	35.3
random partition, no flavor boundary	0.9402, 0.9401	0.9860, 0.9861	427, 463	1,726, 1,852	0.8423, 0.8434	0.9144, 0.9146	41.67, 41.95	108.9, 109.2	0.7091, 0.7121	0.8045, 0.8094	12.07, 12.92	34.1, 39.3
random partition, b-quark boundary	0.9401, 0.9405	0.9860, 0.9861	432, 444	1,590, 2,060	0.8430, 0.8431	0.9147, 0.9147	42.43, 42.77	106.5, 106.8	0.7153, 0.7110	0.8109, 0.8066	12.93, 12.19	38.1, 34.7
random partition, random flavor boundary	0.9397, 0.9406	0.9858, 0.9861	394, 447	1,565, 1,655	0.8427, 0.8428	0.9145, 0.9147	42.12, 42.09	108.2, 104.4	0.7114, 0.7171	0.8047, 0.8106	12.18, 12.76	34.9, 37.4
from scratch	0.9387 ± 0.0002	0.9852 ± 0.0001	358 ± 11	$1{,}412 \pm 20$	0.8397 ± 0.0007	0.9121 ± 0.0002	41.47 ± 0.29	104.0 ± 1.0	0.7153 ± 0.0024	0.8061 ± 0.0007	12.18 ± 0.11	35.5 ± 1.2

Table 22: JetClass fine-tuning as table 2, every configuration. Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run. Bold and † as there.

	jets	accuracy	AUC	rejection of QCD								
				$H \to b\bar{b}$ R_{50}	$H \to c\bar{c}$ R_{50}	$H \to gg$ R_{50}	$H \to 4q$ R_{50}	$H \to \ell\nu qq'$ R_{99}	$t \to bqq'$ R_{50}	$t \to b\ell\nu$ $R_{99.5}$	$W \to qq'$ R_{50}	$Z \to q\bar{q}$ R_{50}
from scratch	1 M	0.8269 ± 0.0010	0.9821 ± 0.0001	$5{,}090 \pm 200$	$1{,}540 \pm 50$	92.9 ± 1.0	680 ± 50	$1{,}539 \pm 24$	$5{,}300 \pm 600$	$4{,}150 \pm 270$	282 ± 4	227.0 ± 2.8
188 classes	1 M	$\mathbf{0.8491 \pm 0.0005}$	$\mathbf{0.9859 \pm 0.0001}$	$6{,}520 \pm 120$	$3{,}015 \pm 26$	105.9 ± 3.3	$1{,}205 \pm 33$	$\mathbf{3{,}390 \pm 270}$	$\mathbf{16{,}900 \pm 2{,}200}^{\dagger}$	$\mathbf{11{,}400 \pm 1{,}000}^{\dagger}$	$\mathbf{413 \pm 13}$	$\mathbf{323 \pm 5}$
162 classes	1 M	$\mathbf{0.8495 \pm 0.0002}$	$\mathbf{0.9859 \pm <0.0001}$	$6{,}600 \pm 600$	$3{,}110 \pm 150$	104.3 ± 1.6	$1{,}180 \pm 50$	$\mathbf{3{,}600 \pm 400}$	$\mathbf{14{,}700 \pm 3{,}200}^{\dagger}$	$10{,}400 \pm 1{,}200^{\dagger}$	$\mathbf{424 \pm 13}$	318.7 ± 2.4
43 classes	1 M	0.8480 ± 0.0006	0.9857 ± 0.0001	$\mathbf{6{,}900 \pm 700}$	$2{,}880 \pm 230$	106.6 ± 2.5	$1{,}070 \pm 50$	$3{,}210 \pm 150$	$15{,}100 \pm 1{,}400^{\dagger}$	$10{,}400 \pm 800^{\dagger}$	410 ± 9	$\mathbf{320 \pm 5}$
17 classes	1 M	0.8434 ± 0.0001	0.9849 ± 0.0001	$6{,}600 \pm 600$	$2{,}680 \pm 130$	103.7 ± 2.4	$1{,}017 \pm 22$	$3{,}100 \pm 400$	$11{,}800 \pm 2{,}000^{\dagger}$	$8{,}800 \pm 1{,}100$	382 ± 10	299.3 ± 2.6
162 classes, mass regression	1 M	0.8478, 0.8490	**0.9857, 0.9859**	6,667, 7,143	**3,571, 3,226**	108.1, 109.4	**1,183, 1,258**	3,448, 3,390	$16{,}667,\ 15{,}385^{\dagger}$	$\mathbf{11{,}765,\ 10{,}000}^{\dagger}$	**421.9, 424.6**	319.5, 322.6
17 classes, mass regression	1 M	0.8420, 0.8419	0.9847, 0.9847	6,667, 6,061	2,564, 2,439	107.1, 105.7	985, 1,031	2,703, 2,667	$9{,}091,\ 16{,}667^{\dagger}$	$\mathbf{11{,}765,\ 10{,}526}^{\dagger}$	386.1, 381.0	298.1, 300.3
random partitions (all five)	1 M	0.8461 ± 0.0009	0.9854 ± 0.0001	$6{,}900 \pm 500$	$2{,}860 \pm 250$	106.7 ± 2.6	$1{,}060 \pm 50$	$3{,}120 \pm 150$	$12{,}700 \pm 1{,}600^{\dagger}$	$9{,}300 \pm 800$	405 ± 6	314 ± 7
random partition, no flavor boundary	1 M	0.8446, 0.8442	0.9852, 0.9851	6,250, 6,667	2,564, 3,030	**107.6, 110.3**	1,130, 1,117	2,899, 3,125	$12{,}500,\ 12{,}500^{\dagger}$	8,330, 8,330	397.6, 389.1	314, 302
random partition, b-quark boundary	1 M	0.8451, 0.8469	0.9853, 0.9855	6,250, 6,897	2,500, 2,703	109.2, 107.6	1,070, 1,149	3,030, 2,985	$11{,}765,\ 16{,}667^{\dagger}$	$9{,}524,\ 8{,}696$	390.6, 388.3	306.3, 311.0
random partition, random flavor boundary	1 M	0.8456, 0.8454	0.9853, 0.9853	**7,407, 5,556**	2,817, 2,941	106.3, 109.3	1,064, 1,099	3,448, 2,985	$15{,}385,\ 10{,}526^{\dagger}$	$9{,}091,\ 10{,}000$	398.4, 399.2	306.3, 302.1

Table 23: JetClass linear probes on the frozen features, largest training set, in the columns of table 2 (AUC one against one where the fit records it). Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run. † as there.

pretraining	accuracy	AUC	rejection of QCD								
			$H \to b\bar{b}$	$H \to c\bar{c}$	$H \to gg$	$H \to 4q$	$H \to \ell\nu qq'$	$t \to bqq'$	$t \to b\ell\nu$	$W \to qq'$	$Z \to q\bar{q}$
188 classes	0.7709 ± 0.0023	0.9682 ± 0.0006	840 ± 90	382 ± 34	20.0 ± 1.0	200 ± 15	920 ± 40	$1{,}100 \pm 80$	$2{,}690 \pm 310$	76.1 ± 1.7	63.1 ± 2.5
162 classes	0.7709 ± 0.0005	0.9683 ± 0.0002	800 ± 130	410 ± 50	20.7 ± 0.8	207 ± 15	930 ± 50	$1{,}220 \pm 150$	$3{,}400 \pm 600$	81.6 ± 2.1	65.8 ± 3.2
43 classes	0.7695 ± 0.0014	0.9680 ± 0.0004	880 ± 180	440 ± 31	20.8 ± 0.7	207 ± 13	920 ± 70	$1{,}300 \pm 200$	$3{,}700 \pm 500$	79.1 ± 1.6	69.5 ± 0.9
17 classes	0.7605 ± 0.0027	0.9650 ± 0.0008	831 ± 16	294 ± 20	21.6 ± 0.7	183 ± 12	810 ± 110	$1{,}180 \pm 310$	$3{,}090 \pm 280$	71 ± 5	68.9 ± 2.8
162 classes, mass regression	0.7718, 0.7760	0.9684, 0.9697	823, 806	377, 413	19.9, 22.5	210, 243	901, 957	980, 1,183	5,000, 4,878	77.9, 72.9	69.1, 67.6
17 classes, mass regression	0.7624, 0.7612	0.9655, 0.9651	637, 926	287, 261	21.90, 22.58	169.6, 171.2	781, 890	873, 813	1,923, 2,381	69.7, 65.8	70.9, 68.1
random partition 1, seed 1	0.753	0.964	712	323	18.3	161	557	847	2,250	67.2	56.1
random partition 2, seed 2	0.755	0.964	658	357	18.7	186	667	664	1,550	71.3	57.7
random partition 3, seed 1	0.755	0.964	877	382	18.5	180	617	1,120	2,020	72.4	58.7
random partition 4, seed 2	0.754	0.964	797	348	17.7	154	662	858	2,500	70.1	56.5
random partition 5, seed 2	0.754	0.964	749	353	16.5	164	610	840	2,330	69.0	56.0
random partition, no flavor boundary	0.7535, 0.7486	0.9633, 0.9622	775, 508	233, 244	21.5, 19.2	157.4, 154.2	683, 712	889, 909	1,449, 2,667	65.5, 69.2	69.2, 68.2
random partition, b-quark boundary	0.7615, 0.7644	0.9661, 0.9665	990, 1,099	398, 301	20.6, 18.0	205, 195	722, 690	837, 1,058	1,724, 2,222	75.0, 70.6	67.8, 66.1
random partition, random flavor boundary	0.7593, 0.7564	0.9654, 0.9645	758, 662	332, 318	18.86, 18.80	184.7, 174.7	660, 685	1,117, 758	2,083, 1,818	72.3, 75.0	66.29, 65.64
randomly initialized network	0.4062, 0.4087 [2 of 3]	0.8268, 0.8282 [2 of 3]	26.4, 24.2 [2 of 3]	22.2, 19.5 [2 of 3]	18.7, 17.1 [2 of 3]	17.2, 15.0 [2 of 3]	1.480, 1.413 [2 of 3]	18.72, 19.69 [2 of 3]	1.188, 1.216 [2 of 3]	18.35, 17.80 [2 of 3]	15.3, 14.2 [2 of 3]
randomly initialized network, BatchNorm recomputed	0.4764, 0.4691 [2 of 3]	0.8649, 0.8618 [2 of 3]	48.4, 46.6 [2 of 3]	12.82, 13.28 [2 of 3]	14.80, 15.44 [2 of 3]	12.7, 13.9 [2 of 3]	1.99, 1.61 [2 of 3]	22.1, 24.8 [2 of 3]	2.24, 1.91 [2 of 3]	13.53, 13.72 [2 of 3]	10.48, 10.78 [2 of 3]

Table 24: QCD jets passing each signal class's working point after fine-tuning, the fewest over the runs of each configuration: the counts behind the rejections of table 2.

fine-tuned from	$H \to b\bar{b}$	$H \to c\bar{c}$	$H \to gg$	$H \to 4q$	$H \to \ell\nu qq'$	$t \to bqq'$	$t \to b\ell\nu$	$W \to qq'$	$Z \to q\bar{q}$
from scratch	38	125	2127	274	128	34	45	702	869
188 classes	30	66	1844	162	56	11	16	472	609
162 classes	28	61	1891	163	49	11	17	457	624
43 classes	26	64	1850	179	59	12	18	478	614
17 classes	28	71	1901	192	59	15	20	510	664
162 classes, mass regression	28	56	1828	159	58	12	17	471	620
17 classes, mass regression	30	78	1867	194	74	12	17	518	666
random partitions (all five)	27	64	1829	177	60	14	20	486	617
random partition, no flavor boundary	30	66	1814	177	64	16	24	503	637
random partition, b-quark boundary	29	74	1831	174	66	12	21	512	643
random partition, random flavor boundary	27	68	1830	182	58	13	20	501	653

Table 25: The random partitions on JetClass: which prong-count boundaries each keeps (•: no group holds classes of both prong counts), the groups that mix prong counts, and its fine-tuned rejection of QCD per signal class over that of the 188-class run of its seed. †: fewer than 20 QCD jets pass in one of the two runs.

	prong-count boundary kept				fine-tuned rejection over the 188-class model								
partition	2/3	2/4	3/4	mixed groups	$H \to b\bar{b}$	$H \to c\bar{c}$	$H \to gg$	$H \to 4q$	$H \to \ell\nu qq'$	$t \to bqq'$	$t \to b\ell\nu$	$W \to qq'$	$Z \to q\bar{q}$
random partition 1, seed 1	—	—	—	12	1.15	0.870	1.05	0.864	0.851	0.611†	0.696†	0.955	0.963
random partition 2, seed 2	—	—	—	12	1.00	1.03	1.00	0.915	1.02	0.688†	0.750†	0.966	0.947
random partition 3, seed 1	—	—	—	13	1.03	1.05	1.02	0.877	0.905	0.786†	0.800†	0.967	1.02
random partition 4, seed 2	—	—	—	12	1.15	0.880	1.01	0.885	0.970	0.786†	0.857†	0.986	0.968
random partition 5, seed 2	—	—	—	11	1.03	0.917	0.951	0.835	1.08	0.611†	0.900†	0.950	0.944

Table 26: Binary linear probes between pairs of JetClass-II classes: the AUC and the ratio of $1 - \text{AUC}$ to the 188-class run of the same seed. †: the label set merges the two sides into one class; ‡: it merges the first side with part of the second. Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run. The b-quark boundary and the random flavor boundary lie only in the four-quark decays; every label set but the 188-class one merges all QCD classes.

pretraining	$X \to bc$ vs bq, cs		$X \to b\bar{b}$ vs $c\bar{c}$		$X \to YY \to bbqq$ vs $ccqq$	
	AUC	$\frac{1-\text{AUC}}{(1-\text{AUC})_{188}}$	AUC	$\frac{1-\text{AUC}}{(1-\text{AUC})_{188}}$	AUC	$\frac{1-\text{AUC}}{(1-\text{AUC})_{188}}$
188 classes	0.9409 ± 0.0003	1.00	$0.9977 \pm <0.0001$	1.00	$0.9939 \pm <0.0001$	1.00
162 classes	0.9410 ± 0.0002	0.998 ± 0.007	0.9976 ± 0.0001	1.015 ± 0.027	0.9939 ± 0.0002	1.005 ± 0.032
43 classes	0.9319 ± 0.0014‡	1.151 ± 0.029	$0.9974 \pm <0.0001$	1.102 ± 0.013	$0.9932 \pm <0.0001$	1.117 ± 0.008
17 classes	0.885 ± 0.004†	1.95 ± 0.06	0.9902 ± 0.0007†	4.22 ± 0.29	0.9716 ± 0.0033†	4.7 ± 0.5
162 classes, mass regression	0.9375, 0.9400	1.057, 1.010	0.9973, 0.9975	1.184, 1.087	0.9930, 0.9935	1.164, 1.070
17 classes, mass regression	0.8808, 0.8833†	2.018, 1.964	0.9891, 0.9904†	4.70, 4.13	0.9676, 0.9706†	5.36, 4.82
random partition 1, seed 1	0.931†	1.17	0.996†	1.51	0.991†	1.43
random partition 2, seed 2	0.930‡	1.18	0.996	1.54	0.991	1.46
random partition 3, seed 1	0.937	1.07	0.996†	1.53	0.991†	1.43
random partition 4, seed 2	0.936	1.08	0.997	1.37	0.992	1.32
random partition 5, seed 2	0.927†	1.24	0.996†	1.67	0.991	1.48
random partition, no flavor boundary	0.882, 0.870†	2.00, 2.19	0.986, 0.969†	5.9, 13.4	0.961, 0.943†	6.4, 9.4
random partition, b-quark boundary	0.9103, 0.9091†	1.518, 1.530	0.9957, 0.9959†	1.837, 1.748	0.9909, 0.9912	1.510, 1.447
random partition, random flavor boundary	0.9096, 0.9100†	1.530, 1.515	0.9940, 0.9924†	2.58, 3.25	0.9863, 0.9860†	2.272, 2.281

pretraining	QCD bb vs cc		QCD with b vs light		QCD with c, no b, vs light	
	AUC	$\frac{1-\text{AUC}}{(1-\text{AUC})_{188}}$	AUC	$\frac{1-\text{AUC}}{(1-\text{AUC})_{188}}$	AUC	$\frac{1-\text{AUC}}{(1-\text{AUC})_{188}}$
188 classes	0.9867 ± 0.0001	1.00	0.9880 ± 0.0001	1.00	0.9324 ± 0.0003	1.00
162 classes	0.9848 ± 0.0004†	1.15 ± 0.04	0.9870 ± 0.0001†	1.085 ± 0.004	0.9282 ± 0.0001†	1.061 ± 0.005
43 classes	0.9846 ± 0.0001†	1.163 ± 0.016	0.9870 ± 0.0001†	1.081 ± 0.008	0.9275 ± 0.0003†	1.072 ± 0.008
17 classes	0.9563 ± 0.0003†	3.294 ± 0.033	0.9815 ± 0.0007†	1.54 ± 0.07	0.9048 ± 0.0010†	1.407 ± 0.021
162 classes, mass regression	0.9829, 0.9842†	1.30, 1.19	0.9868, 0.9870†	1.107, 1.078	0.9262, 0.9273†	1.093, 1.070
17 classes, mass regression	0.9481, 0.9531†	3.95, 3.51	0.9804, 0.9800†	1.638, 1.662	0.9046, 0.9002†	1.414, 1.470
random partition 1, seed 1	0.980†	1.56	0.986†	1.20	0.922†	1.16
random partition 2, seed 2	0.979†	1.58	0.985†	1.21	0.921†	1.16
random partition 3, seed 1	0.980†	1.51	0.986†	1.21	0.923†	1.14
random partition 4, seed 2	0.981†	1.43	0.986†	1.16	0.924†	1.12
random partition 5, seed 2	0.979†	1.60	0.986†	1.19	0.922†	1.14
random partition, no flavor boundary	0.944, 0.930†	4.24, 5.21	0.9814, 0.9825†	1.56, 1.45	0.9088, 0.9122†	1.352, 1.293
random partition, b-quark boundary	0.9795, 0.9795†	1.561, 1.532	0.9855, 0.9854†	1.212, 1.213	0.9166, 0.9171†	1.237, 1.220
random partition, random flavor boundary	0.9721, 0.9707†	2.124, 2.189	0.9848, 0.9845†	1.277, 1.286	0.9204, 0.9211†	1.179, 1.162

Table 28: Jet-mass regression in detail, over all test jets and over resonant and QCD jets. Upper part: resolution and median response of the regression output of the models pretrained with the mass regression, and of ridge and multilayer perceptron regressions on frozen features. Lower parts: resolution and median response of the groomed mass and of the regression output against jet p_T and against the generator soft-drop mass; bins with fewer than 100 jets are blank. Mean ± standard deviation over the three pretraining runs; the models with the mass regression have two runs, and both values are given.

		resolution			median response		
estimator	pretraining	all	resonant	QCD	all	resonant	QCD
regression output	162 classes, mass regression	0.1471, 0.1463	0.1332, 0.1320	0.2263, 0.2256	0.995, 1.003	0.996, 1.004	0.9897, 0.9996
regression output	17 classes, mass regression	0.1450, 0.1461	0.1312, 0.1320	0.2239, 0.2240	1.005, 1.005	1.006, 1.006	1.001, 1.000
ridge, frozen features	randomly initialized network	0.312 ± 0.007	0.286 ± 0.006	0.913 ± 0.031	0.9406 ± 0.0035	0.936 ± 0.004	0.9649 ± 0.0035
ridge, frozen features	188 classes	0.3211 ± 0.0034	0.2921 ± 0.0026	0.559 ± 0.013	0.9641 ± 0.0012	0.9754 ± 0.0006	0.905 ± 0.006
ridge, frozen features	162 classes	0.316 ± 0.009	0.288 ± 0.008	0.548 ± 0.021	0.9655 ± 0.0005	0.9768 ± 0.0008	0.909 ± 0.007
ridge, frozen features	43 classes	0.304 ± 0.016	0.280 ± 0.013	0.49 ± 0.06	0.968 ± 0.005	0.977 ± 0.004	0.922 ± 0.012
ridge, frozen features	17 classes	0.293 ± 0.012	0.271 ± 0.012	0.448 ± 0.030	0.973 ± 0.005	0.980 ± 0.004	0.937 ± 0.010
ridge, frozen features	162 classes, mass regression	0.1885, 0.1922	0.1751, 0.1785	0.2674, 0.2701	0.9998, 0.9998	1.003, 1.003	0.9826, 0.9830
ridge, frozen features	17 classes, mass regression	0.1842, 0.1824	0.1696, 0.1687	0.2630, 0.2601	1.002, 1.002	1.005, 1.003	0.9888, 0.9935
MLP, frozen features	randomly initialized network	0.2421 ± 0.0018	0.2214 ± 0.0017	0.502 ± 0.014	0.983 ± 0.005	0.984 ± 0.005	0.980 ± 0.007
MLP, frozen features	188 classes	0.2200 ± 0.0014	0.1990 ± 0.0016	0.3487 ± 0.0029	0.991 ± 0.005	0.994 ± 0.004	0.975 ± 0.007
MLP, frozen features	162 classes	0.2195 ± 0.0017	0.1989 ± 0.0020	0.3425 ± 0.0031	0.9876 ± 0.0014	0.9900 ± 0.0024	0.974 ± 0.006
MLP, frozen features	43 classes	0.2158 ± 0.0029	0.1949 ± 0.0023	0.338 ± 0.007	0.991 ± 0.005	0.993 ± 0.004	0.976 ± 0.010
MLP, frozen features	17 classes	0.2098 ± 0.0031	0.1894 ± 0.0035	0.327 ± 0.006	0.991 ± 0.004	0.993 ± 0.004	0.977 ± 0.008
MLP, frozen features	162 classes, mass regression	0.1591, 0.1586	0.1436, 0.1430	0.2473, 0.2452	1.004, 1.006	1.006, 1.008	0.9883, 0.9930
MLP, frozen features	17 classes, mass regression	0.1583, 0.1588	0.1420, 0.1431	0.2465, 0.2463	1.009, 1.006	1.012, 1.008	0.9942, 0.9932

resolution	jet p_T [GeV]	200–400	400–700	700–1000	1000–2500
groomed mass	all	0.242	0.225	0.231	0.327
162 classes, mass regression	all	0.1720, 0.1703	0.1483, 0.1468	0.1426, 0.1411	0.1396, 0.1395
17 classes, mass regression	all	0.1691, 0.1702	0.1455, 0.1470	0.1403, 0.1414	0.1383, 0.1391
groomed mass	resonant	0.229	0.215	0.216	0.211
162 classes, mass regression	resonant	0.1642, 0.1621	0.1412, 0.1393	0.1337, 0.1315	0.1201, 0.1201
17 classes, mass regression	resonant	0.1610, 0.1620	0.1382, 0.1396	0.1313, 0.1320	0.1188, 0.1200
groomed mass	QCD	0.279	0.333	1.10	3.61
162 classes, mass regression	QCD	0.1945, 0.1928	0.1854, 0.1846	0.2146, 0.2110	0.2801, 0.2831
17 classes, mass regression	QCD	0.1918, 0.1947	0.1816, 0.1830	0.2089, 0.2090	0.2800, 0.2775

median response	jet p_T [GeV]	200–400	400–700	700–1000	1000–2500
groomed mass	all	0.893	0.914	0.917	0.953
162 classes, mass regression	all	0.986, 0.997	0.994, 1.003	0.997, 1.005	0.997, 1.004
17 classes, mass regression	all	0.9930, 0.9974	1.001, 1.003	1.006, 1.007	1.010, 1.007
groomed mass	resonant	0.904	0.917	0.915	0.947
162 classes, mass regression	resonant	0.987, 0.998	0.995, 1.004	0.998, 1.005	0.998, 1.004
17 classes, mass regression	resonant	0.9933, 0.9980	1.002, 1.004	1.006, 1.007	1.010, 1.008
groomed mass	QCD	0.863	0.895	0.931	1.02
162 classes, mass regression	QCD	0.984, 0.995	0.986, 0.998	0.994, 1.003	0.993, 1.001
17 classes, mass regression	QCD	0.9921, 0.9955	0.9965, 0.9973	1.004, 1.004	1.008, 1.002

resolution	generator soft-drop mass [GeV]	20–50	50–100	100–150	150–250	250–500
groomed mass	all	0.281	0.206	0.204	0.210	0.225
162 classes, mass regression	all	0.1751, 0.1756	0.1425, 0.1416	0.1302, 0.1291	0.1233, 0.1213	0.1163, 0.1135
17 classes, mass regression	all	0.1755, 0.1740	0.1418, 0.1411	0.1284, 0.1288	0.1210, 0.1220	0.1134, 0.1145
groomed mass	resonant	0.275	0.194	0.192	0.201	0.216
162 classes, mass regression	resonant	0.1632, 0.1634	0.1366, 0.1359	0.1261, 0.1252	0.1198, 0.1175	0.1133, 0.1107
17 classes, mass regression	resonant	0.1631, 0.1621	0.1357, 0.1351	0.1246, 0.1250	0.1174, 0.1183	0.1107, 0.1118
groomed mass	QCD	0.292	0.239	0.246	0.265	0.278
162 classes, mass regression	QCD	0.2033, 0.2030	0.1653, 0.1639	0.1508, 0.1515	0.1515, 0.1509	0.1511, 0.1470
17 classes, mass regression	QCD	0.2032, 0.2006	0.1630, 0.1631	0.1505, 0.1514	0.1483, 0.1515	0.1434, 0.1482

median response	generator soft-drop mass [GeV]	20–50	50–100	100–150	150–250	250–500
groomed mass	all	1.00	0.936	0.908	0.890	0.852
162 classes, mass regression	all	1.025, 1.037	1.006, 1.014	0.993, 1.001	0.9881, 0.9954	0.9710, 0.9772
17 classes, mass regression	all	1.032, 1.041	1.014, 1.016	1.002, 1.002	0.9986, 0.9970	0.9837, 0.9792
groomed mass	resonant	1.01	0.953	0.922	0.902	0.862
162 classes, mass regression	resonant	1.032, 1.043	1.012, 1.019	0.997, 1.004	0.9905, 0.9977	0.9722, 0.9783
17 classes, mass regression	resonant	1.038, 1.047	1.019, 1.021	1.006, 1.006	1.001, 0.999	0.9847, 0.9804
groomed mass	QCD	0.971	0.864	0.816	0.781	0.738
162 classes, mass regression	QCD	1.006, 1.019	0.9802, 0.9899	0.9670, 0.9766	0.9647, 0.9720	0.9565, 0.9631
17 classes, mass regression	QCD	1.015, 1.021	0.9906, 0.9901	0.9791, 0.9759	0.9787, 0.9743	0.9704, 0.9641

Table 29: Mass sculpting: Jensen–Shannon distance (base 2) $\times 10^2$ between the m_{SD} distributions (24 bins of 20 GeV from 20 to 500 GeV) of all QCD test jets and of those passing a cut on $1 - p_{\mathrm{QCD}}$ at the QCD efficiency given; lower is less sculpting. The last row is the distance expected for a random selection of as many of the QCD test jets (drawn without replacement). Mean ± standard deviation over the three pretraining runs; the models with the mass regression have two runs, and both values are given.

pretraining	$\epsilon_{\mathrm{QCD}} = 10\%$	$\epsilon_{\mathrm{QCD}} = 1\%$
188 classes	5.24 ± 0.34	4.39 ± 0.24
162 classes	5.32 ± 0.09	4.29 ± 0.29
43 classes	5.60 ± 0.27	5.1 ± 0.7
17 classes	5.43 ± 0.10	5.61 ± 0.17
162 classes, mass regression	4.22, 4.64	4.24, 4.42
17 classes, mass regression	5.22, 4.23	4.05, 4.52
random selection of as many QCD jets	0.944	3.13

Table 30: Resonance tagging in the soft-drop mass window, **[samples not in]**–**[samples not in]** GeV: signal efficiency (%) at the QCD efficiency given, mean ± standard deviation over the pretraining runs. Transfer: a classifier on the frozen features, its seeds averaged per run (the randomly initialized network: the linear probe). Direct: Sophon's discriminant from the output scores, where $X \to b\bar{b}$ is a class of its own.

[no result yet]

Table 31: Resonance tagging in the mass window: rejection $1/\epsilon_B$ of QCD jets at 50% and 30% signal efficiency. Transfer: the classifier on frozen features; direct: Sophon's discriminant from the output scores; closest node: the pretrained $X \to bq$ output against QCD. Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run.

[no result yet]

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